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FACULTY OF ENGINEERING**



Blue-Green OptiCom: Design and Demonstration of an Underwater Optical Communication Network

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Committee report

This is to certify that this graduation project has been read by the examining committee, and the student has been examined in its content. In the opinion of the committee, it is adequate as a project document for a B.Sc. in Electronics and Communication Engineering.

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Abstract

One of the leading technologies in underwater optical wireless systems is the so-called blue-green optical link, which overcomes the severe attenuation limitations of sound waves and radio frequencies. By operating within the blue-green wavelength range water channels and media generally exhibit low signal loss in the visible light spectrum, specifically in the 450-570 nm wavelength range. Underwater optical communication enables high-speed and secure transmission of large-capacity data over medium and short distances, making it a prominent area of research in underwater communications.

This project presents a theoretical, practical, and analytical model of several factors affecting the state of an underwater optical communication link. It simulates a real-world channel environment using a method for transmitting and receiving data after it passes through a real water channel, calculating end-to-end signal strength. Finally, it simulates and verifies the system using computer software such as Optisystem and Proteus . Generating a high-speed communication signal for the transmitter is a fundamental technology. In this project, a laser diode array transceiver was implemented, consisting of several LED diodes with a central wavelength. The device utilizes a Manchester-coded OOK modulation mode with a data rate of standard values in this field.

This included developing the necessary microcontroller code and the transceiver protocol for the proposed working model. Ultimately, a simulation and practical demonstration of a point-to-point underwater wireless optical communication link was carried out in an experimental tank measuring 60 meters long, 30 meters wide, and 30 meters deep. The experimental results show that the system functions well.

These preliminary results confirm the viability of this system for high-speed underwater wireless optical communication applications such as unmanned marine vehicles (UUVs), remotely operated vehicles (ROVs), and unmanned aerial vehicles (UAVs), as well as between underwater platforms .Or during the use of ships in civil and environmental application . This project also explored the possibility of improving the performance of the UOWC system using different LED sources within the visible spectrum.

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List of Abbreviations

Abbreviation	Full Meaning / Description
ADC	Analog-to-Digital Converter
APDs	Avalanche Photodiodes
ARM	Advanced RISC Machine
ASK	Amplitude Shift Keying
ATP	Acquisition, Tracking, and Pointing
AUVs	Autonomous Underwater Vehicles
AWS IoT	Amazon Web Services Internet of Things
BER	Bit Error Rate
DPSK	Differential Phase Shift Keying
ELF	Extremely Low Frequency
FEC	Forward Error Correction
FPGA	Field Programmable Gate Array
FSO	Free-Space Optics
GPIO pin	General-Purpose Input/Output pin
IM/DD	Intensity Modulation with Direct Detection
IMUs	Inertial Measurement Units
ISI	Inter-Symbol Interference
KVL	Kirchhoff's Voltage Law
LD	Laser Diode
LEDs	Light Emitting Diodes
Li-Fi	Light Fidelity
LOS	Line-of-Sight
MCU	Microcontroller Unit
MIMO	Multiple-Input Multiple-Output
NRZ	Non-Return-to-Zero
OFDM	Orthogonal Frequency Division Multiplexing
OOK	On-Off Keying
Op-Amp	Operational Amplifier
PAT	Pointing, Acquisition, and Tracking
PIN	Positive-Intrinsic-Negative (Photodiode)
PPM	Pulse Position Modulation
PRBS	Pseudo-Random Bit Sequence
PCB	permanent Printed Circuit Board
QAM	Quadrature Amplitude Modulation
QPSK	Quadrature Phase Shift Keying
RF	Radio Frequency
RMT	Remote Control Peripheral
SNR	Signal-to-Noise Ratio
SSB	Single Sideband

TIA	<i>Transimpedance Amplifier</i>
TTL	<i>Transistor-Transistor Logic</i>
TIR	<i>Total Internal Reflection</i>
UOWC	<i>Underwater Optical Wireless Communication</i>
UVLC	<i>Underwater Visible Light Communication</i>
UWAC	<i>Underwater Acoustic Communication</i>
UWOC	<i>Underwater Wireless Optical Communication</i>
VLC	<i>Visible Light Communication</i>

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CHAPTER 1: *Introduction*

1.1 Background

The increasing need for underwater communication systems stems from the ever expanding range of human undertakings in the maritime environment, including activities like gathering underwater scientific data, monitoring the environment, exploring oil fields, conducting maritime archeology, ensuring port security, and engaging in tactical surveillance. Before delving into wireless technologies, it is essential to acknowledge that wire-line systems, particularly fiber-optic systems, can offer real-time communication solutions for underwater applications and remain a widely used and efficient solution for specific scenarios. However, wire-line systems have limited adaptability and several operational drawbacks that often restrict their practical applicability. These apparent limitations drive the growing requirement for underwater wireless connections. Underwater wireless technology represents a promising and innovative solution to overcome the limitations associated with traditional wired systems used in underwater environments. This approach offers numerous significant advantages, making it an attractive choice for a wide range of applications.

1.2 Wireless Technologies for Underwater Communication

In this subsection, we explore three primary wireless technologies that have revolutionized communication beneath the water’s surface: Radio frequency (RF), acoustic signaling, and optical waves (see Fig1.1) . Each of these technologies offers distinct characteristics and advantages, underscoring the importance of selecting the most suitable option based on the specific requirements and applications of underwater missions. As we progress, we will compare these three technologies, examining their strengths and weaknesses to help make informed decisions in choosing the most suitable technology for high-speed underwater communication.

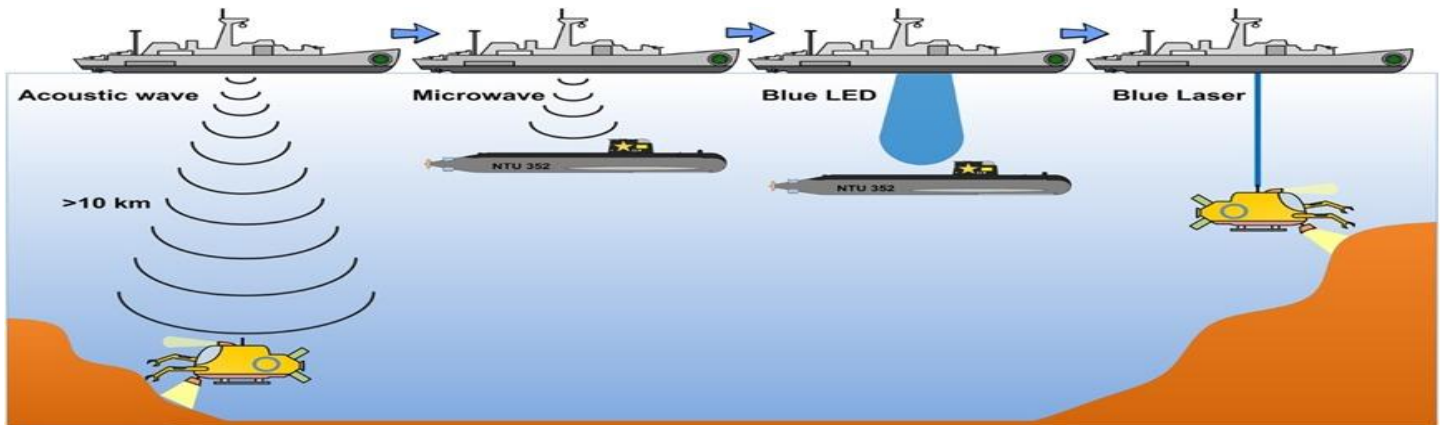


Figure 1.1: Wireless communication technologies

1.2.1 Underwater Acoustic Communications

Acoustic communication has a long history of application in underwater environments and has significantly contributed to various sectors, including marine research, offshore oil and gas exploration, and underwater robotics. Despite its limitations, recent advancements have focused on improving the data rate and reducing latency, making it more suitable for specific real-time applications. For example, in marine research, acoustic communication is utilized for tracking marine life, oceanographic data collection, and monitoring underwater ecosystems. In offshore industries, acoustic modems are used for remote control and monitoring of underwater systems, enhancing efficiency and safety. Ongoing research in this area aims to mitigate the impact of acoustic communication on marine life through the development of low-power transmission methods and the use of advanced signal processing techniques to minimize interference with marine ecosystems.

1.2.2 Underwater RF Communications

RF communication, despite its challenges, has found significant application in various underwater operations, including underwater navigation, remote sensing, and environmental monitoring. The development of compact and energy-efficient transceivers has facilitated the deployment of RF technology in underwater vehicles, enabling real-time data transmission and remote control in challenging underwater environments. Recent advancements in antenna design and signal processing techniques have improved the range and data transfer rates of RF communication, making it a viable option for specific underwater missions. Ongoing research efforts are focused on enhancing the resilience of RF signals to the effects of seawater attenuation and developing adaptive communication protocols for dynamic underwater environments.

1.2.3 Underwater Optical Communications

Optical wireless communication has emerged as a promising solution for high-speed data transmission and real-time monitoring in underwater applications. Recent advancements in underwater optical communication systems have led to the development of compact and robust transceivers, enabling seamless data transmission and imaging capabilities in diverse underwater environments. Applications of optical communication include underwater surveillance, high-definition video streaming, and real-time monitoring of sub-sea infrastructure. Ongoing research in this domain aims to address the challenges associated with signal attenuation and scattering in water by exploring advanced modulation techniques, signal processing algorithms, and adaptive optics for maintaining reliable and high-speed communication links. Additionally, efforts are being made to develop energy-efficient and cost-effective optical communication solutions for long-term underwater deployments.

1.2.4 Comparison of Underwater Wireless Technologies

As discussed above, acoustic, RF, and optical communications are three prominent technologies used for underwater data transmission, each with its unique set of advantages and challenges

Acoustic Communications

- **Advantages:** Underwater acoustic communication offers the advantage of an extensive transmission range, allowing signals to travel over tens of kilometers underwater. This makes it particularly suitable for applications that require long distance communication beneath the water's surface.

- **Disadvantages:** However, acoustic communication does have its drawbacks. It is characterized by a significant delay due to the low propagation speed of sound in water, which may not be suitable for real-time applications. Additionally, the limited available bandwidth results in a low data rate, limiting its capacity for data-intensive tasks. Moreover, acoustic communication experiences frequency dependent attenuation, with different frequencies experiencing varying levels of signal loss due to factors such as water composition and physical characteristics. Furthermore, the use of acoustic waves can have adverse effects on marine life, posing environmental concerns.

RF Communications

- **Advantages:** RF communication presents several advantages for underwater applications. RF signals are relatively tolerant of water turbulence and turbidity, making them suitable for challenging underwater environments. They can seamlessly transition between air and water, making it possible to create unified communication systems under and above water.

- **Disadvantages:** However, RF communication faces its own set of challenges. In water environments, it may require large antenna sizes, which can be cumbersome for certain applications. Additionally, RF signals experience high attenuation in seawater, which limits their range. This technology is also constrained by relatively short link ranges, which can be a limiting factor for certain underwater missions.

Optical Communications

- **Advantages:** Underwater optical wireless communication (UOWC) is known for its low delay, high security, high data rates, and cost-effectiveness. Its low delay makes it ideal for real-time applications such as underwater imaging and video transmission. Line-of-sight (LOS) configurations in optical communication enhance security by preventing eavesdropping. UOWC offers high data rates, facilitating the efficient transfer of large volumes of data. Moreover, it is cost effective and energy-efficient, making it an attractive choice for various underwater communication needs.

- **Disadvantages:** Despite its advantages, UOWC also faces challenges. Optical signaling suffers from severe absorption and scattering in water, which can limit its effective range. LOS optical links require precise transceiver alignment, which may be difficult to maintain in dynamic underwater environments. Additionally, UOWC is sensitive to environmental factors such as water temperature and salinity, which can influence its performance.

Table 1.1: Comparison of wireless communication technologies underwater[1].

Technology	Advantages	Disadvantages
Acoustic communications	Extensive transmission range (tens of kilometers) Widely used technolog. More tolerant to water turbulence and turbidity	Significant delay Limited bandwidth and low data rate Frequency-dependent attenuation Adverse effects on marine life
RF communications	Tolerance of water turbulence and turbidity Relatively smooth transition to cross air/water boundary Moderate data rate (up to 100 Mb/s at close distances)	Large antenna sizes in underwater environments High attenuation in seawater, limiting range Relatively short link ranges Bulky, costly, and energy-consuming transceivers
Optical communications	Low delay and high security High data rates for efficient data transfer Ultra-high data rate (in the order of Gb/s) Low cost and small volume transceivers	Severe absorption and scattering in water, limiting effective range Can't cross air/water boundary easily Suffer from severe absorption and scattering. Short link range (up to tens of meters) LOS optical links require precise transceiver alignment Sensitivity to environmental factors like water temperature and salinity

1.3 Problem Statements

Despite the continuous development in underwater communication, traditional wireless technologies face inherent physical limitations that hinder their performance in modern high-speed applications. These drawbacks constitute the core problem that necessitates a more efficient alternative.

The fundamental problem lies in the technical constraints of conventional underwater wireless methods. Acoustic communication, while capable of long-range transmission, suffers from extremely low data rates (bandwidth bottleneck), high latency due to the slow speed of sound in water, and significant multipath fading. On the other hand, Radio Frequency (RF) technology is severely limited by the high conductivity of seawater, which causes rapid signal attenuation and requires large, impractical antennas for even short-distance links. Consequently, neither technology can support the real-time, high-definition data transfer required for modern subsea missions, creating a critical gap in underwater communication infrastructure.

1.4 Motivation

The motivation for this project arises from the urgent need to overcome the aforementioned limitations by adopting Underwater Visible Light Communication (UVLC). UVLC provides a transformative solution that addresses the 'bandwidth bottleneck' of acoustic waves by offering gigabit-per-second data rates and ultra-low latency. Furthermore, unlike RF waves, optical signals in the blue-green spectrum experience a 'low-attenuation window' in water, allowing for high-speed transmission over short to medium ranges with much smaller and more energy-efficient hardware (LEDs and Laser Diodes). By exploring UVLC, this project aims to provide a reliable and high-capacity wireless link that empowers autonomous underwater vehicles (AUVs) and underwater sensor networks to operate with unprecedented efficiency and speed.

1.5 Project Objectives & Scope

1.5.1 Objectives

The main objectives of this project are:

- *To design and implement an underwater optical transmitter integrated with an environmental sensor*
- *To develop a surface receiver station capable of reliably extracting data from the optical link*
- *To evaluate the performance of the underwater optical communication system, including link efficiency and signal stability*
- *To integrate the system with a local server for data storage and visualization*
- *Implementation of a cloud-based server connected to the ESP module, enabling real-time data uploading from the environmental sensor*
- *Remote monitoring and visualization of collected underwater data through the cloud server interface*

1.5.2 Project Scope

The project encompasses three main deliverables:

- *UWOC Transmitter Development: Designing and constructing the underwater optical transmitter module, including the integration of an environmental sensor and the implementation of efficient signal modulation circuitry.*
- *Surface Receiver and Data Recovery: Developing the surface receiving station equipped with a high-sensitivity photodetector and implementing robust digital signal processing techniques for reliable data extraction.*

- *Link Budget Analysis: Evaluating the performance of the underwater optical communication channel by analyzing the link budget, including transmission losses, received power, and system efficiency under various underwater conditions.*
- *System Integration and Data Archiving: Integrating the transmitter and receiver into a complete operational system, culminating in the automated forwarding of recovered sensor data to a local data server.*

1.6 Methodology

- *Literature Review and Feasibility Study.*
- *Selection of Optical Wavelength and Modulation Scheme.*
- *Design of Optical Communication System.*
- *Environmental Adaptation and Signal Optimization.*
- *Performance Testing and Optimization.*
- *Comparison with Acoustic Communication Systems.*
- *Security and Data Integrity.*
- *Scalability and Future Enhancement.*
- *Final Testing and Deployment*

CHAPTER 2: *Literature Review*

Underwater communication has become a crucial domain for marine exploration, environmental monitoring, and tactical operations. Traditionally, acoustic waves have been the dominant approach due to their long transmission ranges. However, acoustic communication suffers from low bandwidth, high latency (due to the slow speed of sound in water, $\approx 1500\text{m/s}$), and susceptibility to Doppler shifts. Radio Frequency (RF) waves, while offering higher bandwidth, experience severe attenuation in seawater due to high electrical conductivity.

To overcome these limitations, Underwater Optical Wireless Communication (UOWC) has emerged as a high-potential alternative, offering high data rates (Gbps), ultra-low latency, and enhanced security over short-to-medium ranges.

2.1 Related Work

In recent years, significant research has been dedicated to improving the data rate, link distance, and reliability of UOWC systems under various water conditions:

Research 1: High Data Rate using Discrete Multi-Tone and Laser Diode

- **What they did:** Investigators experimentally demonstrated a high-speed UWOC system using a single 450 nm blue Laser Diode (LD). By employing adaptive bit-power loading Discrete Multi-Tone (DMT) modulation combined with a Volterra series-based post-nonlinear equalization technique, they successfully achieved a record data rate of 16.6 Gbps over a 5-meter tap water channel. Furthermore, the system maintained robust performance by achieving 13.2 Gbps over 35 meters and 6.6 Gbps over an extended distance of 55 meters, with bit error rates (BER) falling safely below the standard hard-decision forward error correction (HD-FEC) limit.

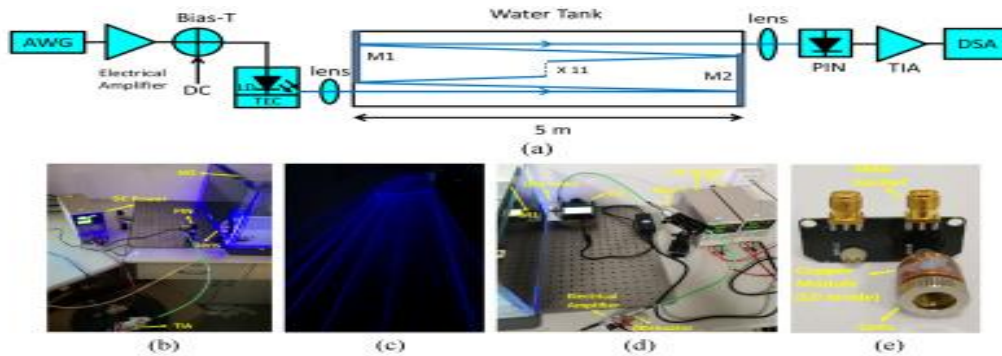


Figure 2.1: (a) The schematic of the experimental setup and (b) - (e) sections of the UWOC system. (b) Receiver side; (c) Water tank; (d) Transmitter side; (e) Thermal copper module (LD inside) and PCB. AWG: arbitrary waveform generator; M1(2): mirror 1(2); TIA: trans-impedance amplifier; DSA: digital serial analyzer; TEC: thermo-electric cooler [2].

Research 2: Gb/s UWOC Using Series-Connected GaN Micro-LED Arrays

- **What they did:** Investigators demonstrated a high-speed UWOC system utilizing a novel series-connected Gallium Nitride (GaN) micro-LED array source. To evaluate performance across extended distances, they constructed an experimental setup (as illustrated in Fig. 4) using a 160 L water tank with physical dimensions of $1.5\text{ m} \times 0.35\text{ m} \times 0.35\text{ m}$. By strategically placing two mirrors (M1 and M2) within the channel, they successfully extended the effective optical link length from 1.5 meters up to 3 meters and 4.5 meters through multi-reflection. The transmitter integrated an Arbitrary Waveform Generator (AWG), an amplifier, and a Bias Tee to combine AC data signals with DC bias, while the receiver utilized a PIN photodiode, a Fresnel lens, and a post-amplifier connected to an oscilloscope for real-time Gb/s data rate verification.

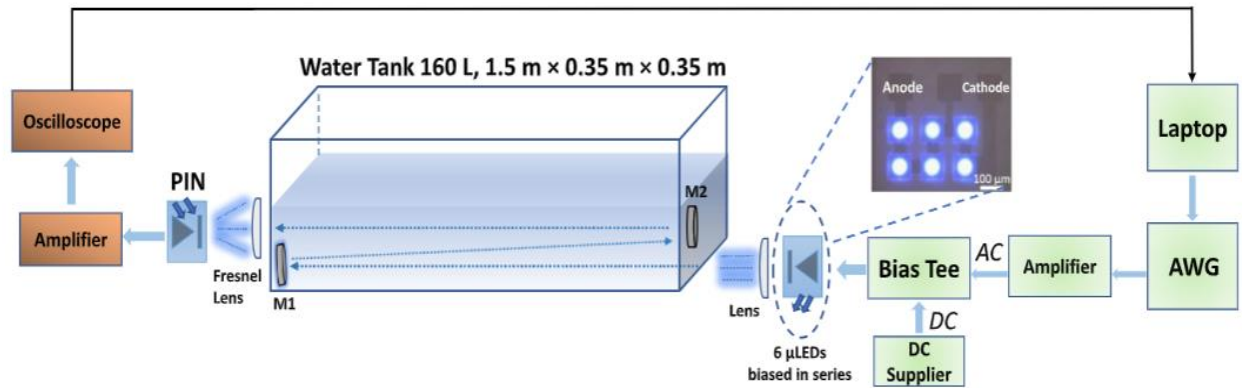


Figure 2.2: Illustration of the experimental set-up used for the UWOC measurements of this work. Two mirrors (M1 and M2) were placed appropriately to increase the optical link from 1.5 m up to 3 m and 4.5 m [3].

Research 3: High Data Rate using Laser Diodes

- **What they did:** Investigators demonstrated a high-speed UWOC system using a blue Laser Diode (LD) operating at a wavelength of 450 nm. By employing On-Off Keying (OOK) modulation, they achieved a data rate of 2.5 Gbps over a 20-meter tank filled with clear ocean water. This work highlighted that blue-green lasers fall perfectly within the minimum attenuation window of seawater.

Research 5: High-Capacity OCDMA Coding via Green Spectrum

- **What they did:** This study introduced a multi-user, high-capacity UOWC architecture that integrates Optical Code Division Multiple Access (OCDMA) utilizing Inverted Coding Stacking Method (ICSM) codes operating within the green optical spectrum. Through extensive numerical and performance modeling across different water types, the investigators demonstrated that the system can achieve an aggregate capacity of 30 Gbps. The effective communication range extended up to 50 meters in pure sea water (characterized by a low extinction coefficient), while maintaining a localized transmission distance of 5.2 meters in highly turbid Harbor II water environments.

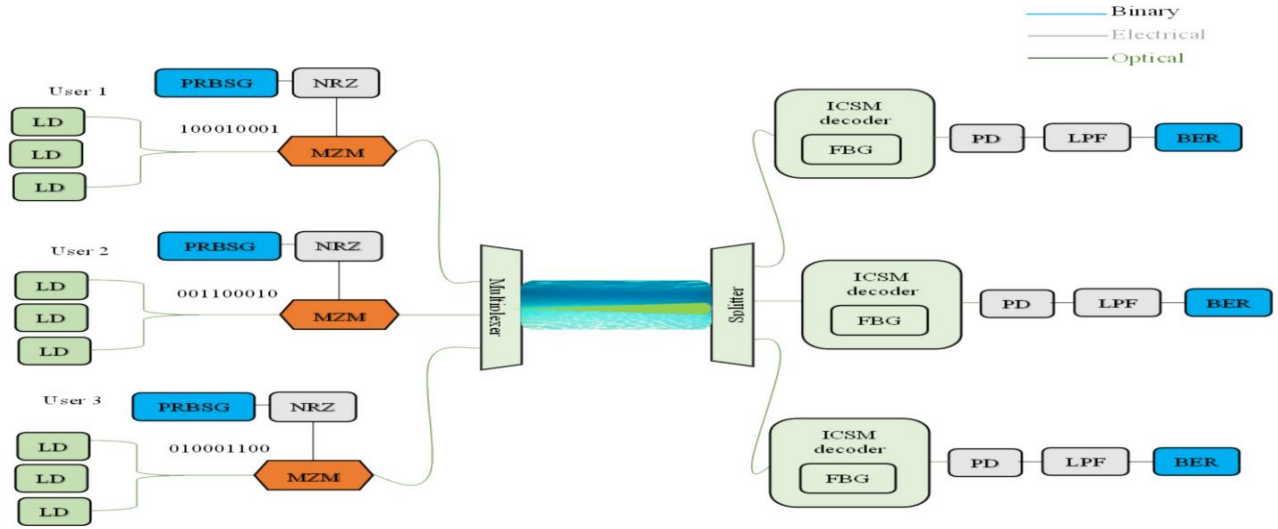


Figure 2.5: Layout of proposed UOWC with ICSM based OCDMA system[6].

CHAPTER 3: *Underwater Optical Link : Analysis & Components*

3.1 *Background*

To establish a robust underwater wireless network, three primary physical carriers are traditionally considered: acoustic, radio frequency (RF), and optical waves. Each carrier is governed by distinct physical laws that dictate its bandwidth, operational range, and reliability across different aquatic environments.

3.1.1 *Underwater Acoustic Communication (UWAC)*

Historically, acoustic waves have been the primary carrier for underwater wireless communication due to their relatively low absorption in water, which enables long-range propagation spanning several kilometers. Early acoustic systems, developed using single sideband (SSB) suppressed carrier amplitude modulation, often suffered from poor signal quality and required significant human cognitive processing to decode distorted speech.

However, as digital communication advanced in the 1960s, improvements in data rates were achieved through multipath compensation.

Modern acoustic systems now utilize sophisticated digital techniques, such as Orthogonal Frequency Division Multiplexing (OFDM), to achieve higher throughput without the need for complex equalizers. Despite these advancements, Underwater Acoustic Communication (UWAC) faces critical limitations.

The available bandwidth is extremely limited and strictly distance-dependent, with data rates ranging from a few hundred bps for long-distance links to approximately 500 kbps for very short ranges. Furthermore, sound travels at a relatively slow speed (≈ 1500 m/s), leading to high latency and significant propagation delays. The performance is further hindered by time-varying multipath reflections from the sea surface and bottom, which cause delay spreads that result in severe Inter-Symbol Interference (ISI). Additionally, acoustic channels are highly unpredictable; a system designed for shallow water may fail in deep-ocean environments due to shifting refractive indices and noise profiles.

3.1.2 *Underwater Radio Frequency (RF) Communication*

In an effort to overcome the bandwidth limitations of acoustics, Underwater Radio Frequency (RF) communication has been explored, offering higher carrier frequencies and significantly faster propagation speeds of approximately 2.25×10^8 m/s. RF system architectures generally fall into two categories: buoyant RF systems, which link submerged units to terrestrial stations via MHz to GHz frequencies, and direct RF systems, which utilize Extremely Low Frequencies (ELF: 30–300 Hz) to penetrate the conductive medium between submerged

transceivers. However, RF signals suffer from extreme attenuation in seawater due to their high conductivity, which is nearly 100 times higher than that of fresh water. This conductivity creates high absorption losses that increase proportionally with frequency. Consequently, achieving reasonable distances requires massive antenna sizes and high transmission power, often exceeding 100W for a mere 100-meter range, making RF impractical for compact, battery-powered underwater autonomous vehicles (AUVs).



Figure 3.1 Direct RF communication System[6].

3.1.3 Underwater Optical Wireless Communication (UOWC)

As a solution to the "bandwidth bottleneck" of acoustics and the "attenuation/size" hurdles of RF, Underwater Optical Wireless Communication (UOWC) offers a high-capacity, low-latency alternative for moderate-range applications. Utilizing the high frequency of the optical carrier, UOWC is capable of exceeding data rates of 1 Gbps.

While early experiments in the 1990s achieved 50 Mbps over 9 meters, modern advancements in LED arrays and sophisticated modulation schemes have pushed operational ranges to over 50 meters. The technology capitalizes on the specific "blue-green" spectral window (400–550 nm) where seawater exhibits its minimum absorption coefficient, allowing optical beams to propagate with sufficient energy for detection. Practically, UOWC offers near-instantaneous transmission and significantly lower power consumption—approximately 30,000 bits/Joule compared to acoustic systems.

Unlike RF, UOWC utilizes compact and cost-effective transceivers, such as LEDs and photodiodes, allowing for easy integration into small underwater sensors. Despite these advantages, UOWC remains sensitive to turbidity and scattering caused by suspended particles, and high-speed laser-based links require strict Pointing, Acquisition, and Tracking (PAT) systems due to their narrow beam widths.

TABLE 3.1 Comparison of different wireless underwater technologies[6].

Parameters	Acoustic	RF	Optical
Attenuation	Distance and frequency dependent (0.1 - 4 dB/km) [12]	Frequency and conductivity dependent (3.5 - 5 dB/m) [13]	0.39 dB/m (ocean) - 11 dB/m (turbid) [14]
Speed (m/s)	1500 m/s	$\approx 2.255 \times 10^8$	$\approx 2.255 \times 10^8$
Data rate	~ kbps	~ Mbps	~ Gbps
Latency	High	Moderate	Low
Distance	up to kms	up to ≈ 10 meters	$\approx 10 - 100$ meters
Bandwidth	Distance dependent [8]: 1000 km < 1kHz 1 - 10 km ≈ 10 kHz < 100 m ≈ 100 kHz	$\approx$ MHz	10 - 150 MHz
Frequency band	10 - 15 kHz	30 - 300 Hz (ELF) (for direct underwater communication system) or MHz (for buoyant communication system)	$10^{12} - 10^{15}$ Hz
Transmission power	tens of Watts (typical value)	few mW to hundreds of Watts (distance dependent)	Few Watts
Antenna size	0.1 m	0.5 m	0.1 m
Efficiency	≈ 100 bits/Joules		$\approx 30,000$ bits/Joules
Performance parameters	Temperature, salinity and pressure	Conductivity and permittivity	Absorption, scattering/turbidity, organic matter

3.1.4 Rationale for Selecting Optical Carriers in Underwater Networks

Based on the comparative analysis, Underwater Optical Wireless Communication (UOWC) is selected as the primary technology for this project due to its unique ability to combine high-speed data transfer with low-latency performance.

By utilizing the vast THz bandwidth of the optical carrier, the system can achieve data rates in the Gbps range, far exceeding the kilobit-per-second limitations of acoustic waves. Furthermore, UOWC offers exceptional energy efficiency, approximately 30,000 bits per Joule and a compact form factor that is ideal for battery-constrained underwater nodes. By leveraging the speed of light in water (2.25×10^8 m/s), the system ensures the near-instantaneous communication required for real-time control. Ultimately, the use of optical components provides a cost-effective and scalable solution that effectively addresses the classical bottlenecks of underwater exploration.

3.2 The general system of Underwater Optical Communications

3.2.1 UWOC System

A typical UWOC system's components are shown in **Figure (3.2)**. It consists of a source for data generation that is modulated in an optical transmitter and transmitted over long distances at a high rate of data. The transmitter has a projection lens and beams steering components to focus the optical wave on the position of the receiver. The data carrying wave can then propagate along the water channel. An optical wave at the receiving end is converted from optical to electric by the photodetector. A demodulator and a signal processor can then process the electric signal to recover the original signal.

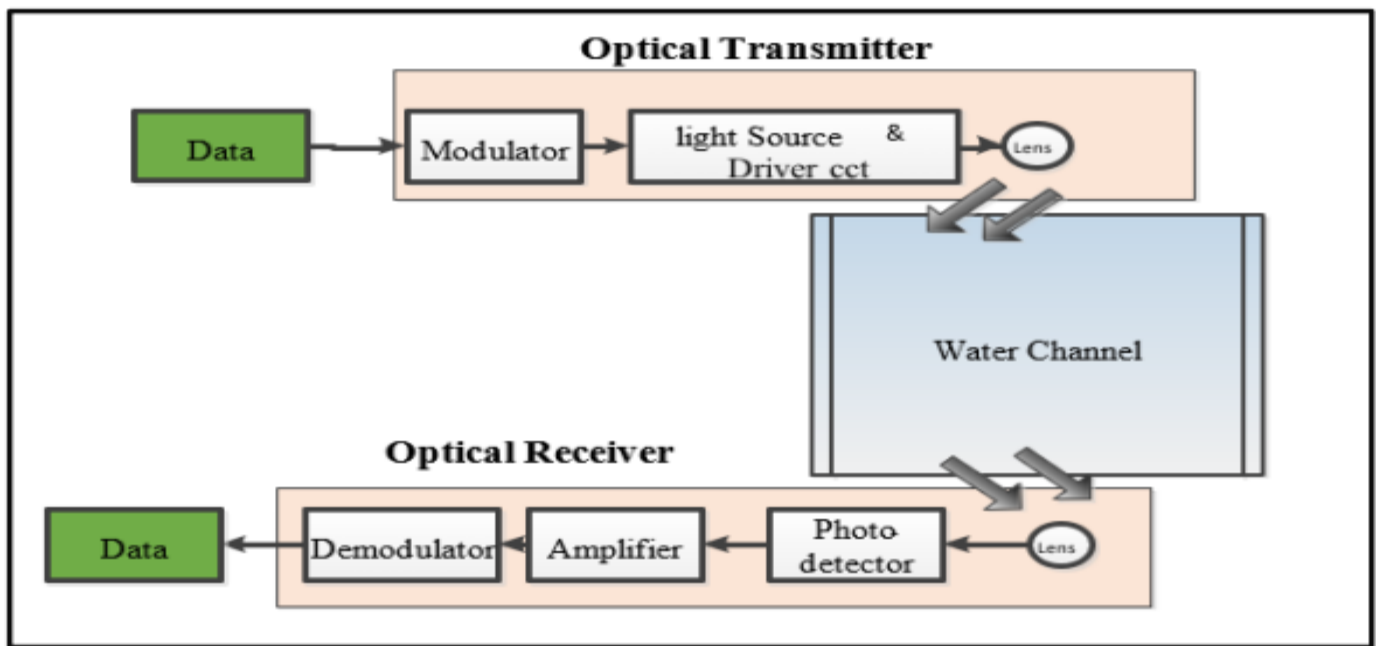


Fig.3.2: Block Diagram of UWOC System[9].

3.2.2 Transmitter of UWOC System

The electrical data signal is transformed into an optical signal and projects the optical signal into a transmission channel. **Figure (3.3)** shows a standard optical transmitter with an original wave, an optical driver system, a source of light, and additional optics components for the focusing and conditioning of light beams (lenses and reflectors). The role of the optical source is to convert an electrical input signal into the corresponding optical signal. There are two available types of optical transmitter device based on photon source: LEDs and LDs

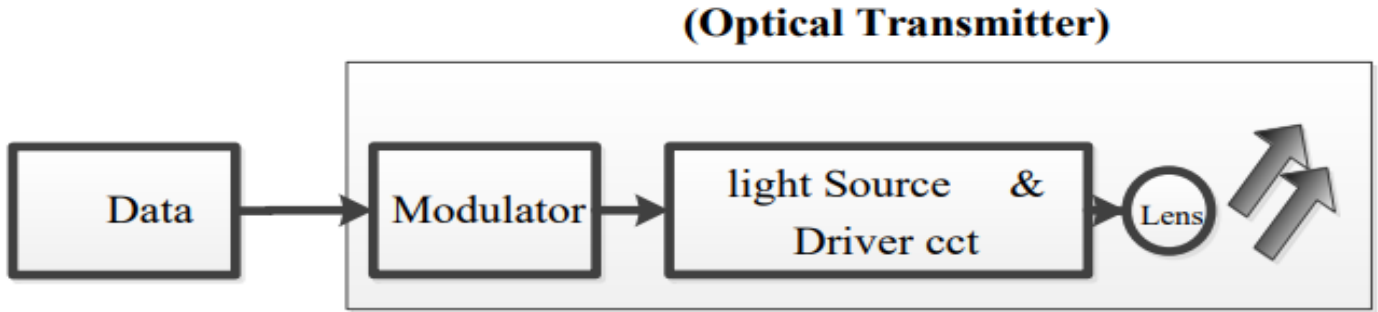


Fig.3.3: A Typical Optical Transmitter[9].

3.2.3 Optical Receiver

The role of an optical receiver is to convert the optical signal back into electrical form and recover the data transmitted through the light wave system. Its main component is the photodetector that converts light into electricity through the photoelectric effect. It consists of an optical signal detector that transform into an electric current. **Figure (3.4)** shows the system-level receiver design. Two popular examples of photodetector currently in use include p-i-n photodiodes (PIN) and avalanche photodiodes (APD).

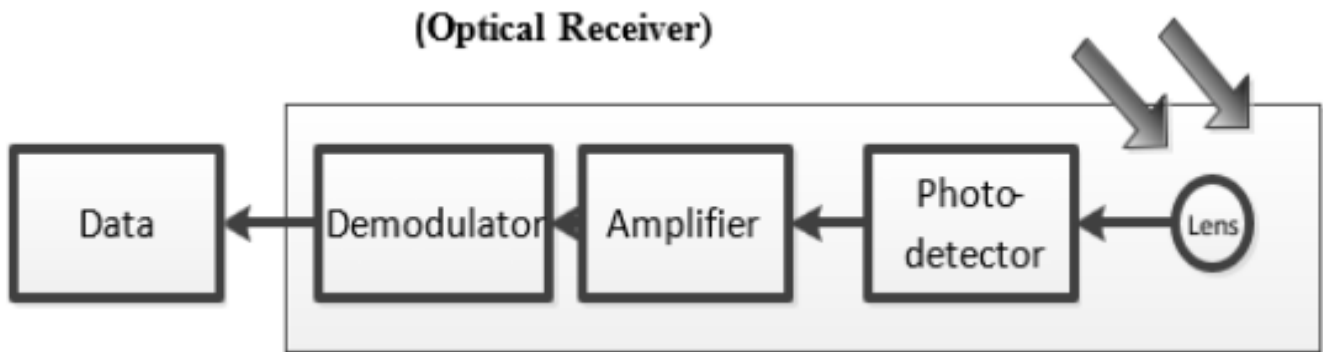


Fig 3.4: A Typical Optical receiver[9]

3.3 Transmitter Architecture and Light Source Selection

3.3.1 Overview of Transmitter Components

The transmitter subsystem is responsible for converting the digital information source into a modulated optical signal capable of propagating through the restrictive underwater channel. To achieve high-speed and power-efficient transmission, the transmitter architecture consists of several cascading functional blocks, as illustrated in the system block diagram below:

1. Information Source & Microcontroller Block:

This is the initial stage where the input data (such as sensor readings, text, or commands) is generated. A microcontroller unit (MCU) processes the raw data and prepares the bitstream for transmission by organizing it into structured packets.

2. Modulator Circuit (Modulation Stage):

The digital bitstream from the microcontroller is fed into the modulator. Based on the selected signaling scheme—such as On-Off Keying (OOK) or Pulse Position Modulation (PPM)—the modulator switches the state of the signal to represent logical '1's and '0's. This stage determines the time-domain characteristics of the pulses.

3. Driver Circuit (Amplification & Switching Stage):

Since the output pins of a microcontroller cannot provide sufficient electrical current to drive high-power optical sources, a dedicated LED/Laser Driver Circuit is strictly required. This circuit acts as an electronic switch (typically utilizing high-speed MOSFETs or BJTs) that translates the low-voltage modulation signal into the high-current pulses needed to toggle the light source at high frequencies.

4. Optical Source (Emitter Block):

The final component of the transmitter is the electro-optical transducer, which converts electrical current into optical power. Depending on the design constraints:

- **Light Emitting Diodes (LEDs):** *Are utilized for short-range, wide-divergence applications where strict alignment is difficult.*
- **Laser Diodes (LDs):** *Are selected for high-speed, long-range directional links.*

The emitter is strictly chosen to operate within the optimal blue-green visible light spectrum (450 nm – 530 nm) to ensure minimal attenuation in the aquatic medium.

3.3.2 Comparison: Laser Diodes (LDs) vs. Light Emitting Diodes (LEDs)

The selection of an appropriate optical transmitter is a fundamental design decision that directly influences the communication range, alignment complexity, and overall system reliability. In the field of Underwater Optical Wireless Communication (UOWC), two primary light sources are traditionally considered: Laser Diodes (LDs) and Light Emitting Diodes (LEDs). While both technologies utilize the blue-green spectral window for transmission, they differ significantly in their physical characteristics and operational requirements.

Laser Diodes are characterized by high optical power density and narrow spectral widths, enabling them to achieve exceptional data rates in the Gbps range and extend communication over longer distances. However, the highly collimated nature of laser beams introduces a significant technical challenge: the requirement for precise Pointing, Acquisition, and Tracking (PAT) systems. In dynamic underwater environments, even minor turbulence,

water currents, or platform vibrations can cause a total link failure due to misalignment. Furthermore, LDs are sensitive to temperature fluctuations and require complex feedback circuitry to maintain stable optical output.

In contrast, Light Emitting Diodes (LEDs) offer a much wider beam divergence, which inherently simplifies the alignment process between the transmitter and the receiver. This characteristic makes LED-based systems far more resilient to the "pointing errors" common in underwater scenarios. While LEDs typically offer lower modulation bandwidths compared to lasers, they are sufficient for a wide range of practical applications, including sensor data collection and environmental monitoring.

TABLE 3.2 :Advantages and disadvantages of LASER and LED light source of UOWC systems[7].

Light Source	Advantages	Drawbacks
LASER	High directionality. Lasers produce a low-divergence beam. This allows for long-distance transmission with low dispersion. High data rate. Due to the coherent nature of the light emitted by lasers, systems using this type of light source can support high-speed data transmission.	Alignment sensitivity. The narrow beam of light, typical of lasers, requires precise alignment between the transmitter and receiver. This makes lasers difficult to use in dynamic underwater environments. Dispersion issues. Particles suspended in the water can cause dispersion, reducing signal quality. High cost. Laser systems tend to be more expensive than LED-based solutions.
LED	Wide Beam. LEDs emit a relatively wide beam of light. This reduces the need for precise alignment between the transmitter and receiver, making it easier to implement systems that can be used in dynamic underwater environments. Cost-effective. LED-based systems are generally cheaper to manufacture and maintain. Resistance to water turbulence. The wider beam is less affected by small movements or disturbances in the water.	Lower data rates. LEDs cannot be modulated at high frequencies. In addition, LEDs are poorly suited to complex modulation schemes. Data rates are generally in the Mbps range. Low communication distance. Due to the lower optical power density (divergent beam), the effective communication range is in the order of 10 m.

The fundamental difference in how these two sources emit light is a primary factor in system design. As illustrated in **Figure 3.5**, the Laser Diode produces a highly directional, narrow beam, whereas the LED emits light in a wider, conical shape. This geometric difference directly impacts the ease of establishing a stable communication link in turbulent waters.

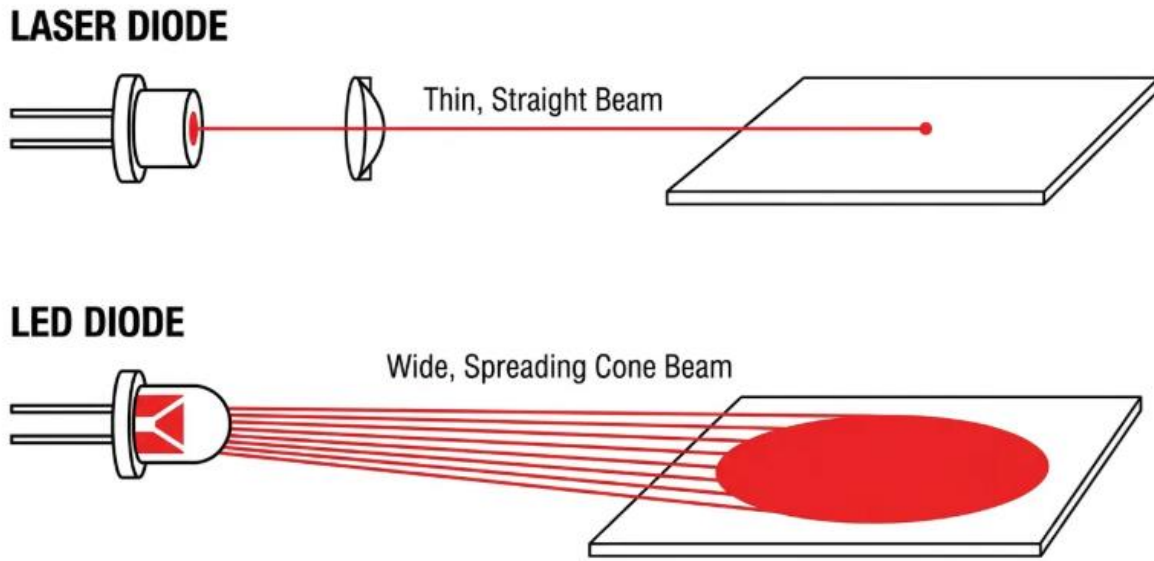


Figure 3.5 Comparison of beam profiles for LD and LED sources, illustrating the wider coverage area provided by LED-based transmitters [Adapted from online open-source repository].

3.3.3 Comparative Analysis: Single LED vs. LED Array Configuration

The choice between a single LED and an LED array is pivotal for optimizing the performance of Underwater Wireless Optical Communication (UWOC) systems. While a single LED inherently provides a wider emission angle compared to laser sources—thereby offering better tolerance for non-line-of-sight configurations. it often falls short in providing sufficient optical power and spatial uniformity for practical underwater deployment. In dynamic environments, the light intensity from a single source degrades significantly as the receiver moves away from the central axis, making the link susceptible to disruptions caused by water turbulence and currents.

To further enhance this advantage, as investigated in the study by Liu et al, the implementation of an LED array is proposed to mitigate the severe attenuation and scattering effects of seawater. The primary strength of the array configuration lies in its ability to amplify the total transmitted power while enabling "spatial diversity." By optimizing the arrangement and the specific orientation (pitch and deflection angles) of multiple LEDs, the system can expand the effective communication envelope far beyond what a single source can achieve. This creates a highly uniform spatial coverage field that maintains a stable Signal-to-Noise Ratio (SNR) across a broader area. Consequently, the LED array approach transforms the transmitter from a localized light source into a robust wide-coverage system, which is essential for the reliable operation of sensor nodes

3.3.4 Overview of Optical Modulation Techniques

In underwater optical wireless communication (UWOC) systems, several modulation techniques can be used to encode information onto the optical signal. Common schemes include On-Off Keying (OOK), Pulse Position Modulation (PPM), Differential Phase Shift Keying (DPSK), Quadrature Phase Shift Keying (QPSK), and higher-order techniques such as 32-PSK and 64-QAM.

Lower-order modulation schemes, such as OOK and PPM, are simple to implement and require lower power for reliable transmission. They are generally more robust to channel impairments such as noise, absorption, and scattering. However, they offer lower spectral efficiency compared to advanced modulation techniques.

On the other hand, higher-order modulation schemes like QPSK and QAM provide higher data rates and better bandwidth utilization. Despite this advantage, they require a higher signal-to-noise ratio (SNR) and are more sensitive to channel degradation. In underwater environments, where attenuation and scattering are significant, these complex schemes often suffer from performance limitations.

3.3.5 On-Off Keying (OOK) Modulation

On-Off Keying (OOK) is one of the simplest forms of digital modulation and is widely used in optical communication systems. It is considered a special case of amplitude shift keying (ASK), where the carrier signal is switched between two levels: a non-zero amplitude to represent binary '1' and zero amplitude to represent binary '0' (Haykin, 2009; Proakis, 2001).

In optical systems, this corresponds to the presence or absence of light, making OOK highly suitable for intensity modulation and direct detection (IM/DD) schemes commonly employed in underwater optical wireless communication (UWOC) systems.

Mathematically, the OOK signal can be expressed as a time-domain waveform where the transmitted signal is present during a binary '1' and absent during a binary '0'. This simple representation allows efficient implementation using basic switching mechanisms at the transmitter and threshold-based detection at the receiver (Agrawal, 2010). In practical systems, the optical source, such as an LED or laser diode, is directly modulated to generate the required signal pattern.

The operation of an OOK system involves three main stages: modulation, transmission, and demodulation. During modulation, the binary input data is used to control the optical source, turning it on and off according to the data sequence. The modulated signal then propagates through the underwater channel, where it is affected by attenuation, scattering, and noise. At the receiver, a photodetector converts the received optical signal into an electrical signal, and a decision circuit determines whether the received signal corresponds to a binary '1' or '0' based on a predefined threshold (Ghassemlooy et al., 2017).

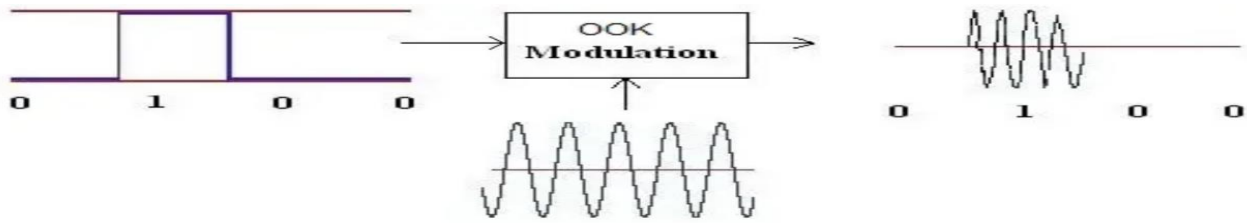


Figure 3.6 Block diagram of On-Off Keying (OOK) modulation showing binary input, carrier signal, and modulated output [9].

One of the key advantages of OOK is its simplicity. It requires minimal hardware complexity and low computational resources, making it attractive for practical implementations. Additionally, OOK is energy-efficient, as no power is transmitted during the transmission of binary '0'. These characteristics make it particularly suitable for systems operating under strict power and complexity constraints, such as UWOC systems (Agrawal, 2010).

However, OOK also has several limitations. It is sensitive to noise and interference, as errors may occur when noise causes false detection of signal presence or absence. Furthermore, OOK has relatively low spectral efficiency compared to higher-order modulation techniques such as QPSK and QAM, which can transmit more bits per symbol. Despite these drawbacks, OOK remains a widely adopted modulation scheme due to its robustness and ease of implementation (Proakis, 2001; Haykin, 2009).

In the context of optical wireless communication, including visible light communication (VLC) and Li-Fi systems, OOK continues to play a significant role due to its compatibility with intensity modulation and its ability to operate effectively in channels with high attenuation and noise (Ghassemlooy et al., 2017; Haigh et al., 2019).

3.3.6 Justification for Selecting OOK

The selection of a suitable modulation technique in underwater optical wireless communication (UWOC) systems is highly dependent on trade-offs among communication range, data rate, power efficiency, and environmental conditions. A comprehensive study conducted by Zayed and Shokair (2025) evaluated multiple modulation techniques, including OOK, PPM, DPSK, QPSK, 32-PSK, and 64-QAM, under different water types such as pure seawater, clear ocean, coastal water, and turbid harbor conditions, using both LED and laser diode (LD) photo-sources.

The results demonstrated that OOK consistently achieves the longest communication distance among all modulation schemes across all water environments. In contrast, higher-order modulation techniques such as QPSK and 64-QAM.

Although capable of delivering higher data rates, they suffer from significantly reduced transmission ranges due to their higher sensitivity to attenuation and noise.

Table 3.3 Communication distances for various modulation techniques, water types, and photo-sources at receiver sensitivity – 53.4 dBm [11].

Water Type	Modulation Type	LED-PS Distance (m) at $P_S = -53.4$ dBm	LD-PS Distance (m) at $P_S = -53.4$ dBm
Pure Sea Water	OOK	27.37	53.87
	PPM	25.52	51.19
	DPSK	24.45	49.62
	QPSK	21.90	45.75
	32-PSK	18.37	40.18
	64-QAM	15.86	36.03
Clear Ocean	OOK	19.44	33.97
	PPM	18.35	32.56
	DPSK	17.71	31.73
	QPSK	16.16	29.68
	32-PSK	13.96	26.67
	64-QAM	12.34	24.38
Coastal Ocean	OOK	10.14	15.50
	PPM	9.71	15.00
	DPSK	9.46	14.70
	QPSK	8.83	13.97
	32-PSK	7.91	12.87
	64-QAM	7.22	12.03
Turbid Harbor	OOK	3.33	4.51
	PPM	3.23	4.40
	DPSK	3.17	4.34
	QPSK	3.02	4.18
	32-PSK	2.80	3.94
	64-QAM	2.63	3.75

Furthermore, the study highlights a fundamental trade-off between signal-to-noise ratio (SNR) and communication distance. Higher-order modulation schemes require higher SNR values to maintain reliable performance, which limits their achievable range in underwater environments. Conversely, simpler schemes such as OOK can operate at lower SNR levels while maintaining longer transmission distances. This trade-off is particularly evident in the simulation results, where increasing modulation complexity leads to shorter communication ranges but improved spectral efficiency.

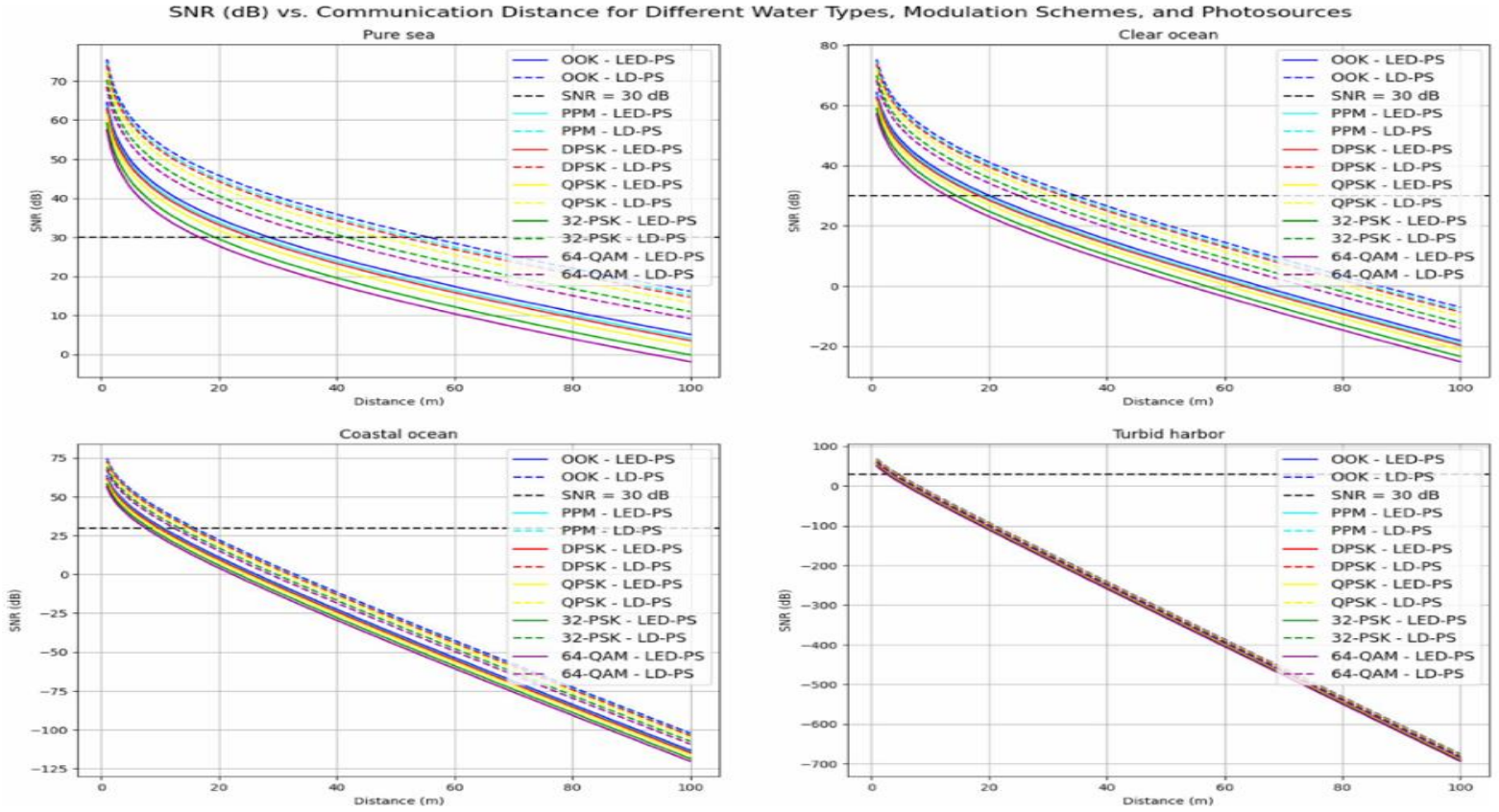


Figure 3.7 SNR vs. communication distance for various water types, modulation techniques, and transmitter types[10].

Even at a fixed SNR threshold of 30 dB, OOK continues to outperform higher-order modulation techniques in terms of achievable communication distance across all water types. This confirms the robustness of OOK in underwater environments characterized by high attenuation and scattering.

On the other hand, higher-order modulation schemes such as 64-QAM achieve significantly higher channel capacity. For example, in clear water conditions, 64-QAM can reach channel capacities exceeding 50 bps/Hz, whereas OOK provides much lower spectral efficiency. However, this increase in capacity comes at the cost of higher power requirements and reduced communication range.

Additionally, the study shows that environmental conditions play a critical role in system performance. As water turbidity increases, the communication distance decreases drastically for all modulation schemes. In highly turbid environments, even robust techniques such as OOK experience significant range limitations, highlighting the strong impact of absorption and scattering on UWOC systems.

Based on these findings, the choice of modulation technique must align with the intended application. For long-range and energy-efficient applications, such as environmental monitoring or military underwater communication, OOK combined with laser diode sources provides superior performance, achieving distances exceeding 120 m in clear seawater conditions at low power and SNR requirements.

In this project, the primary objective is to maximize communication distance between the transmitter and receiver in underwater environments, particularly for potential military applications. Therefore, OOK modulation is selected due to its simplicity, robustness, lower power requirements, and ability to achieve the longest transmission range under various channel conditions (Zayed & Shokair, 2025).

3.4 Theoretical Background and Channel Environment

3.4.1 Optical Propagation Fundamentals in Water

Optical wireless propagation in an underwater environment is fundamentally governed by the interaction of light photons with water molecules and suspended particulate matter. Unlike terrestrial free-space optics (FSO), the aquatic medium introduces severe deterministic degradation to the propagating beam. When an optical beam travels through water, it loses energy and changes direction due to two primary physical phenomena: absorption and scattering. The combination of these two processes dictates the overall degradation of the channel, severely constraining the achievable transmission distance and the vital signal-to-noise ratio (SNR) at the receiver end.

3.4.2 Absorption and Scattering Effects

Absorption and scattering are the main factors that influence the attenuation and propagation of light in water (Spagnolo et al., 2025). As an optical beam moves through the aqueous medium, photons engage with water molecules, dissolved substances, and suspended particles, leading to a reduction in the radiation intensity that reaches the receiver. **Figure 3.8** provides a schematic representation of light absorption and scattering.

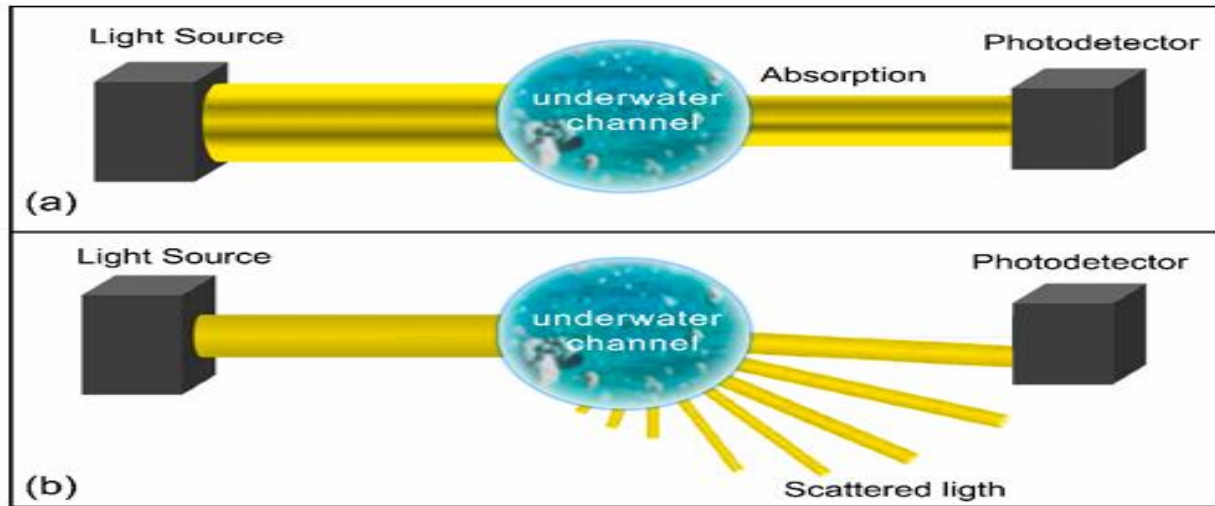


Figure 3.8, A schematic overview of light absorption and scattering in water: (a) the light experiences a reduction in intensity as a result of absorption by the molecules found in the transmission channel; (b) the light intensity decreases due to scattering induced by particles that are dispersed throughout the transmission channel [7].

The propagation of an optical beam through an aqueous medium is primarily hindered by the complex physical interactions between photons and the water's constituents. These interactions result in a continuous reduction of optical power, a phenomenon collectively known as attenuation. Absorption and scattering are the main factors that influence the attenuation and propagation of light in water.

As an optical beam moves through the aqueous medium, photons engage with water molecules, dissolved substances, and suspended particles, leading to a reduction in the radiation intensity that reaches the receiver. Specifically, the process of absorption involves the irreversible conversion of photon energy into other forms, such as thermal or chemical energy, quantified by the coefficient $a(\lambda)$. In contrast, scattering, measured by $b(\lambda)$, occurs when photons are deflected from their original path due to collisions with particles like plankton and micro-bubbles. The combined impact of these phenomena gives rise to the total wavelength-dependent attenuation coefficient $c(\lambda)$.

As highlighted in recent studies (Spagnolo et al., 2025), the behavior of these coefficients varies significantly with the water's purity. In pure sea water, as demonstrated in **Figure 3.9**, the energy loss due to scattering for visible light is practically negligible, making attenuation almost entirely driven by absorption. However, in natural seawaters, the presence of organic matter significantly alters this balance.

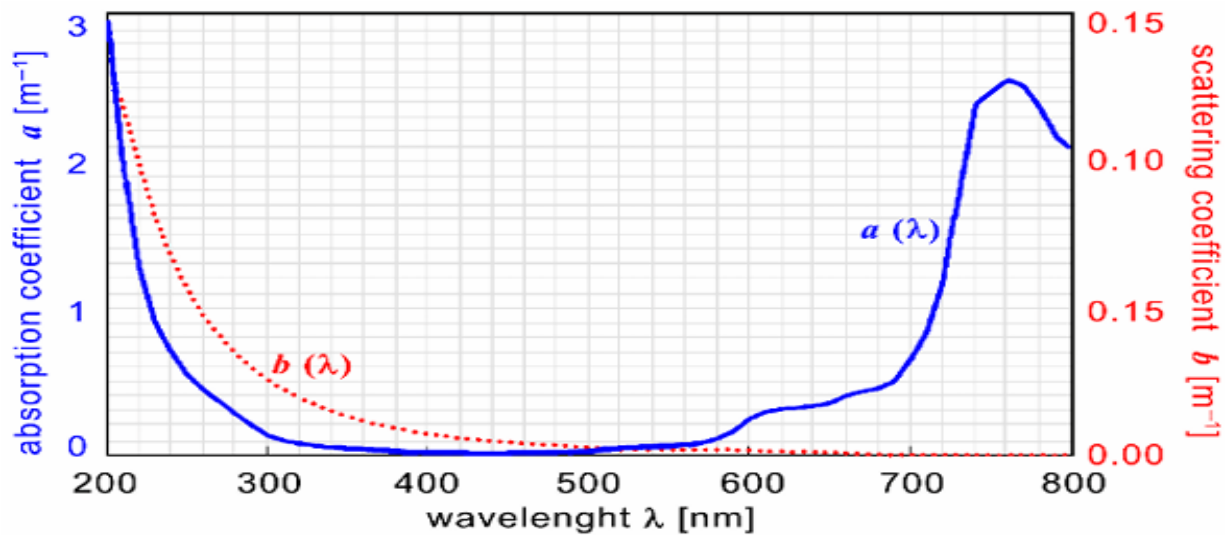


Figure 3.9: Absorption $a(\lambda)$ and scattering $b(\lambda)$ coefficients of pure sea water., showing the dominance of absorption in the visible spectrum [7].

Figure 3.10 illustrates the absorption spectrum of chlorophyll, revealing how biological components create a dynamic environment where the "optical window" shifts across the spectrum. This shift directly influences the efficiency of the communication link, necessitating a careful selection of the operating wavelength based on the concentration of these biological factors.

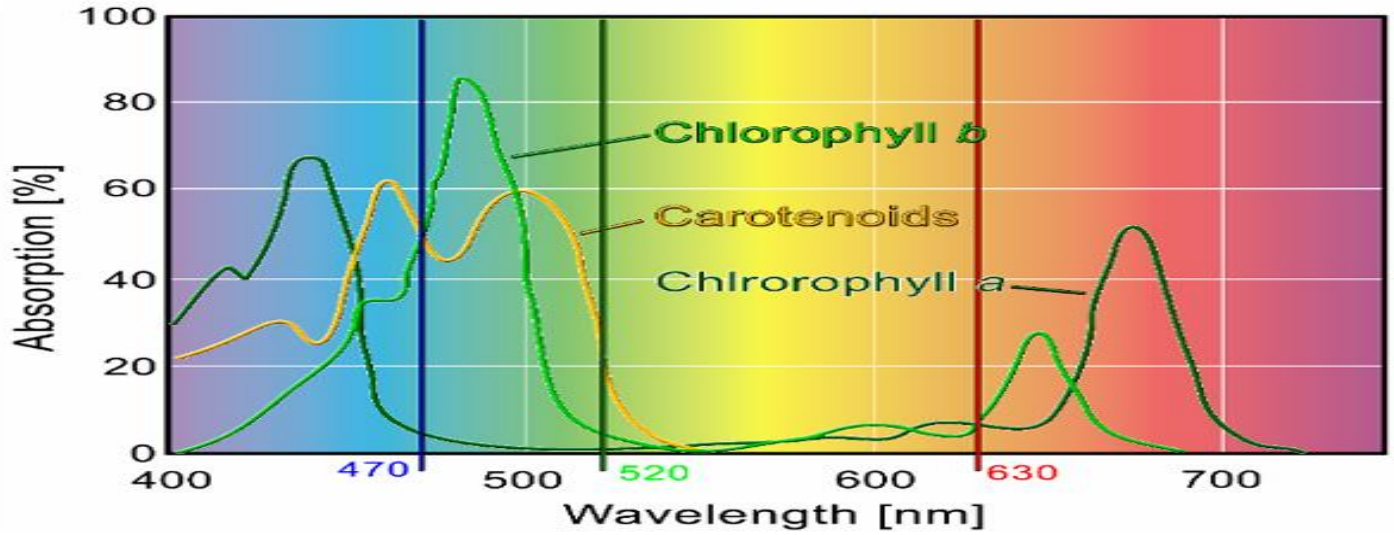


Figure 3.10: The absorption spectrum of chlorophyll, demonstrating its significant impact on the total attenuation and the shifting of the transmission window in natural waters [7].

3.4.3 Beer-Lambert Law for Optical Attenuation

The total loss of optical intensity as light propagates through a water channel is mathematically modeled using the classical Beer–Lambert Law. This law assumes a linear relationship between the concentration of attenuating species in the medium and the exponential decay of optical power over distance.

The total extinction (attenuation) coefficient of the water channel, $c(\lambda)$, is defined as the sum of the absorption coefficient, $a(\lambda)$, and the scattering coefficient, $b(\lambda)$:

$$c(\lambda) = a(\lambda) + b(\lambda) \quad (3.1)$$

Where:

- $c(\lambda)$ is the total attenuation coefficient (m^{-1}).
- $a(\lambda)$ is the absorption coefficient (m^{-1}).
- $b(\lambda)$ is the scattering coefficient (m^{-1}).
- λ represents the operational optical wavelength (nm).

According to the Beer–Lambert Law, the transmitted optical intensity decreases exponentially with propagation distance z , expressed as:

$$I(z) = I_0 e^{-c(\lambda)z} \quad (3.2)$$

Where:

- $I(z)$ is the remaining optical intensity at distance z .
- I_0 is the initial transmitted optical intensity at $z = 0$.
- z is the physical link range or propagation distance (m).

3.4.4 Jerlov Water Classification and Optimal Optical Window

To standardize the study of light propagation, the oceanographer Nils Jerlov developed a globally recognized classification system that categorizes water bodies based on their optical clarity and attenuation properties. As summarized in **Table 3.4**, this system distinguishes between open ocean waters (Types I, IA, IB, II, and III) and coastal waters (Types 1C through 9C).

TABLE 3.4 Jerlov water types vs. turbidity [7].

Open ocean	Jerlov I	
	Jerlov IA	
	Jerlov IB	
	Jerlov II	
	Jerlov III	
Coastal	Jerlov IA	
	Jerlov 3C	
	Jerlov 5C	
	Jerlov 7C	
	Jerlov 9C	

This categorization is fundamental for UOWC design because it predicts how the "optical transmission window" shifts as the water composition changes from clear to turbid. In open ocean environments (Jerlov Type I), the water is exceptionally clear, and the minimum attenuation occurs at wavelengths around 470 nm, which corresponds to blue light. However, as the water type transitions toward coastal categories (e.g., Jerlov Type 9C), the concentration of chlorophyll and suspended sediments increases significantly.

According to the data provided in **Table 3.4**, the attenuation coefficient $c(\lambda)$ in these turbid environments can be several orders of magnitude higher than in pure water. More importantly, the peak transparency window shifts away from blue and toward the green spectrum ($\approx 520\text{--}550\text{ nm}$). This shift is a critical factor in the selection of the light source for the current project. While blue light is often considered the standard for deep-sea exploration, it suffers from extreme absorption in coastal and harbor waters due to the presence of dissolved organic matter. Consequently, by analyzing the Jerlov attenuation profiles, it is evident that Green LEDs offer the most robust and versatile performance for a wide range of underwater conditions, providing the lowest possible signal loss in the coastal environments.

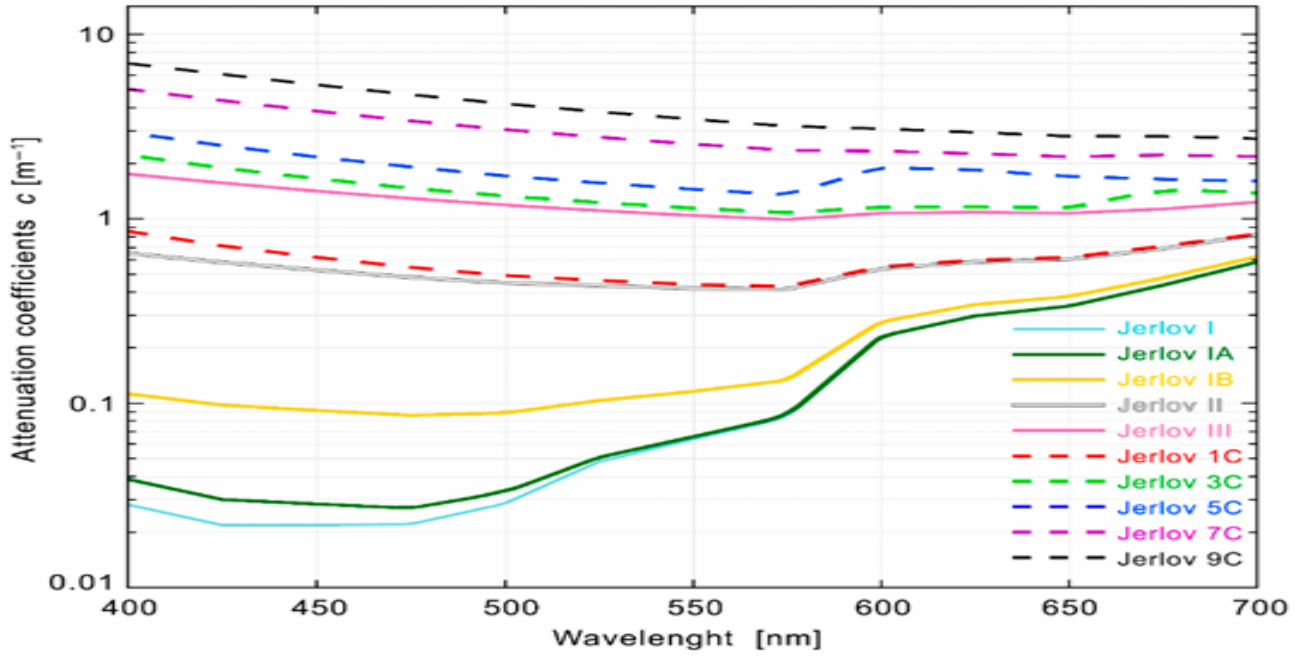


Figure 3.11: Attenuation coefficient for Jerlov water types [7].

3.4.5 Channel Loss and Power Budget Equations

To design a reliable physical link, a precise power budget must be established. The total channel loss factor, L_{channel} , accounts for the geometric spreading of the beam and the medium attenuation. For a line-of-sight (LOS) configuration, the optical power arriving at the receiver photodetector, P_{RX} is governed by the following geometric and physical constraints:

$$P_{RX} = P_{Tx} \cdot \eta_{Tx} \cdot \eta_{Rx} \cdot \frac{(A_{Rx} \cdot \cos(\theta))}{(\pi \cdot (z \cdot \tan(\phi))^2)} \cdot e^{-c(\lambda)z} \quad (3.3)$$

Where:

- P_{Tx} is the average transmitted optical power (W).
- η_{Tx} and η_{Rx} are the optical efficiencies of the transmitter and receiver optics, respectively.
- A_{Rx} is the active surface area of the photodetector (m^2).
- θ is the angle between the optical beam axis and the receiver normal vector (pointing error angle).
- ϕ is the transmitter beam divergence semi-angle (rad).

The corresponding link power budget in decibels (dB) can be expressed as:

$$P_{RX(dBm)} = P_{Tx(dBm)} - L_{\text{geometric(dB)}} - L_{\text{attenuation(dB)}}$$

(3.4)

Where the medium attenuation loss is given by:

$$L_{\text{attenuation(dB)}} = 4.343 \cdot c(\lambda) \cdot Z$$

(3.5)

3.4.6 Signal-to-Noise Ratio (SNR) in UWOC Systems

At the receiver, the optical power is converted into an electrical current by a photodetector (typically a PIN diode or Avalanche Photodiode - APD). The electrical Signal-to-Noise Ratio (SNR) determines the ultimate Bit Error Rate (BER) performance of the link. Assuming intensity modulation with direct detection (IM/DD), the SNR is modeled as:

$$SNR = \frac{(R \cdot P_{Rx})^2}{(\sigma_{\text{shot}}^2 + \sigma_{\text{thermal}}^2 + \sigma_{\text{background}}^2)}$$

(3.6)

Where:

- R is the responsivity of the photodetector (A/W).
- M is the internal multiplication gain multiplier (for standard PIN diodes, $M = 1$; for APDs, $M > 1$).
- σ_{shot}^2 is the shot noise variance, given by $2qM^2F(RP_{Rx})B$
- $\sigma_{\text{thermal}}^2$ is the thermal noise variance, given by $\frac{4k_BTB}{R_L}$
- q is the electron charge ($1.6 \cdot 10^{-19}C$), B is the electrical bandwidth (Hz), k_B is the Boltzmann constant, T is the absolute temperature (K), R_L is the load resistance, and F is the APD excess noise factor

3.5 Receiver Architecture and Components

3.5.1 Receiver Overview

The optical receiver system detects the optical signal and transforms it into an electrical signal. The photon packets that propagate to the receiver plane are terminated and their data is recorded, including the transmitter altitude relative to sea level and the ambient temperature

It consists of a photon detector, which converts an optical signal into an electrical current, as well as additional electronics to convert the photon detector's electrical signal into TTL voltage. **Figure 3.12** shows this the system-level receiver design.

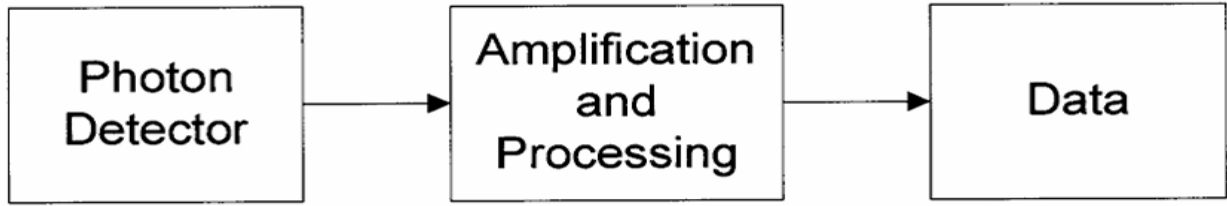


Figure 3.12 Receiver system diagram

3.5.2 Receiver Design

The receiver design consists of several stages starting from the photodetector, which converts the incoming optical signal into a proportional electrical current. This current is then converted into a voltage signal and amplified using a transimpedance amplifier (TIA). The resulting analog signal is subsequently processed by a comparator, which performs threshold detection to generate a digital output in the form of a TTL-compatible logic signal. Finally, this digital signal is fed into a microcontroller for further processing and system-level interpretation. **Figure 3.13** shows the receiver system architecture.

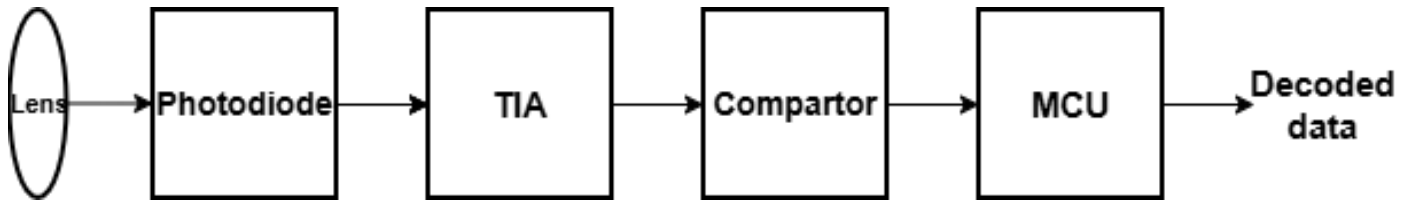


Figure 3.13 Optical receiver block diagram

The subsequent sections describe each stage in detail.

1. Lens

The optical lens is the first component of the receiver, responsible for collecting and focusing the incoming optical signal onto the active area of the photodiode. This increases the amount of received optical power, improving the receiver sensitivity and enhancing the overall performance of the underwater optical communication system. Proper lens selection also helps minimize optical losses caused by beam divergence and misalignment.

2. Photodiode

Photodiode are opto-electronic devices used to convert incident optical signals into electrical signals and the relationship between the incident optical power and the generated photocurrent in a photodiode can be expressed as:

$$i_{sig} = \frac{q P_{rx} G_{det} QE}{h\nu}$$

(3.7)

where each term is defined as follows:

- i_{sig} : the generated signal photocurrent.
- q : the electron charge.
- P_{rx} the received optical power incident on the photodiode.
- G_{det} : the detector gain (if applicable, depending on the photodiode type).
- QE : the quantum efficiency of the photodiode, representing the conversion efficiency of incident photons into electrons.
- h : Planck's constant.
- ν : the optical frequency of the incoming light.

An ideal photodetector should respond rapidly to incoming photons without adding significant noise, while also being compact, cost-effective, power-efficient, and robust. However, in practice, these requirements are often conflicting, and design trade-offs must be made depending on the application.

For optical communication systems, particularly underwater optical wireless communication, the primary requirements are high switching speed and sufficient light sensitivity, provided that constraints on size and power consumption are satisfied.

Several types of photodetectors are considered below with respect to these requirements:

- **Photoresistors**

Photoresistors, also known as light dependent resistors (LDRs) or cadmium sulfide (CdS) cells, are photoconductive devices whose resistance changes when exposed to light. In darkness, their resistance is very high (in the mega-ohm range), while under illumination it decreases significantly, sometimes to a few hundred ohms.

Despite their high sensitivity to light, photoresistors suffer from extremely slow response times, typically in the millisecond range, with recovery times that can extend to several seconds. This makes them unsuitable for high-speed communication applications, especially those requiring bandwidths above 1 MHz.

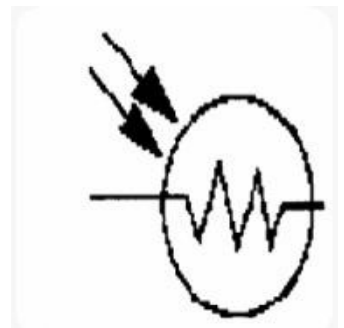


Figure3.14:photoresistor

- **Photothyristors**

Photothyristors are light-activated thyristors that behave similarly to diodes with an additional gate terminal controlling conduction. Once triggered by sufficient optical excitation, the device remains in the conducting state even after the light is removed, until the circuit is reset. Due to this latching behavior, photothyristors are not suitable for optical communication systems where continuous and time-vary signal detection is required.

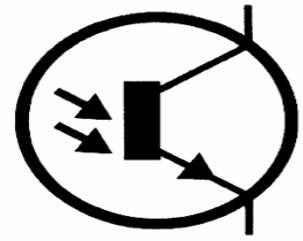


Figure 3.15 photothyristor

- **Phototransistors**

Phototransistors are similar in structure to conventional bipolar junction transistors or field-effect transistors, with the base or gate region exposed to light. Incident photons generate a base current (or gate voltage), which is then amplified internally by the transistor. This internal gain, typically ranging from 10^2 to 10^3 , makes phototransistors highly sensitive to low light levels. However, their switching speed is limited, generally up to a few hundred kilohertz (around 250 kHz), which restricts their use in high-speed optical communication systems.



Figure 3.16 phototransistor

- **p-n Photodiodes**

Photodiodes operate by converting incident photons into electrical current through the generation of electron-hole pairs within a p-n junction. When photons with sufficient energy strike the depletion region, electron-hole pairs are generated and separated by the built-in electric field, resulting in a photocurrent.

The device is often operated under reverse bias to increase the depletion region width, which enhances quantum efficiency and improves bandwidth. However, increasing the depletion width introduces a trade-off between sensitivity and response speed due to carrier transit time and junction capacitance effects.

Photodiodes do not provide internal gain, but they offer excellent linearity, high speed (up to tens of picoseconds rise time), compact size, and strong noise immunity. These characteristics make them well-suited for optical communication systems.

- **Avalanche Photodiodes (APDs)**

Avalanche photodiodes are similar to p-n photodiodes but operate under high reverse bias, enabling internal signal multiplication through avalanche breakdown. A single photon can generate multiple electron-hole pairs, resulting in internal gains typically ranging from 10^2 to 10^4 .

While this improves sensitivity, APDs introduce higher noise levels, nonlinear behavior, and increased system complexity. They also require precise bias control and high operating voltages (30–300 V), making them more difficult to integrate into simple and low-power systems.

Photodetector Selection

A comparison of different photodetectors shows clear trade-offs between speed, sensitivity, complexity, and power requirements. While photomultiplier tubes and avalanche photodiodes offer high sensitivity, they are unsuitable for compact and low-power communication systems. Phototransistors and photoresistors, although simple and sensitive, do not meet the speed requirements of optical communication.

Therefore, p-n photodiodes were selected as the most appropriate choice for the receiver design. This decision is based on their high switching speed, good linearity, compact size, robustness, and ease of integration with electronic circuitry, making them well suited for underwater optical wireless communication applications.

3. Transimpedance Amplifier (TIA)

A Transimpedance Amplifier (TIA) is used in optical receiver systems to convert the small current generated by a photodiode into a usable voltage signal.

- **Principle of Operation**

The TIA is typically implemented using an operational amplifier (op-amp) in an inverting configuration. The photodiode is connected to the inverting input, while the non-inverting input is grounded.

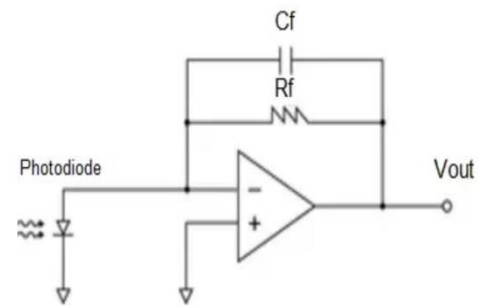


Figure 3.17 Transimpedance circuit

Due to negative feedback, the op-amp maintains the inverting input at approximately 0 V (virtual ground). This forces the photodiode current to flow through the feedback resistor, producing an output voltage:

$$V_{out} = -I_{in} \times R_f \quad (3.8)$$

where:

- I_{in} : is photodiode current
- R_f : is the feedback resistor

- **Main Components**

1. Operational Amplifier (Op-Amp)

The op-amp keeps the input at virtual ground, which helps:

- Improve response speed
- Reduce the effect of the photodiode capacitance

2. Feedback Resistor (R_f)

The feedback resistor converts the input current into a voltage.

- *A larger (R_f) increases the output voltage (higher sensitivity)*
- *A smaller (R_f) increases the speed (higher bandwidth)*

3. Feedback Capacitor (C_f)

A small capacitor is placed in parallel with (R_f) to stabilize the circuit.

Its functions:

- *Prevent oscillations*
- *Reduce high-frequency noise*
- *Improve signal stability*

An approximate Value Can be estimated by:

$$C_f \approx \frac{1}{2\pi R_f f_{BW}}$$

(3.9)

Where:

- *f : is the required bandwidth.*

4. comparator

A comparator is an electronic device or circuit that compares two input voltages and produces a digital output based on the comparison. It has two input terminals: the non-inverting input (+) and the inverting input (-). If the voltage at the non-inverting input is higher than the voltage at the inverting input, the output becomes HIGH. Otherwise, the output becomes LOW. Comparators are widely used in electronic systems for signal detection, threshold sensing, analog-to-digital conversion, and waveform generation because they provide fast and accurate switching between two output states

5. Microcontroller

The microcontroller serves as the final processing stage in the receiver system. It is responsible for receiving the conditioned digital signal from the comparator (or Schmitt Trigger) and performing the required data decoding and interpretation.

In this system, the microcontroller ensures reliable signal processing by converting the incoming electrical pulses into meaningful digital data that can be further used for communication or display purposes. It also provides flexibility for implementing different decoding algorithms depending on the application requirements.

Additionally, the microcontroller operates as an interface between the analog front-end (photodetector and signal conditioning stages) and the digital output, enabling efficient integration of the overall system.

CHAPTER 4: *System Design & Realization*

4.1 *Optical Transmitter*

In Visible Light Communication (VLC) systems, the optical transmitter is designated as the frontline stage responsible for converting electrical data signals into modulated optical intensity fields suitable for propagation through a physical channel (such as air or water). Rather than utilizing conventional Radio Frequency (RF) antennas, solid-state optoelectronic light sources specifically Light Emitting Diodes (LEDs) are employed to carry information across the visible light spectrum.

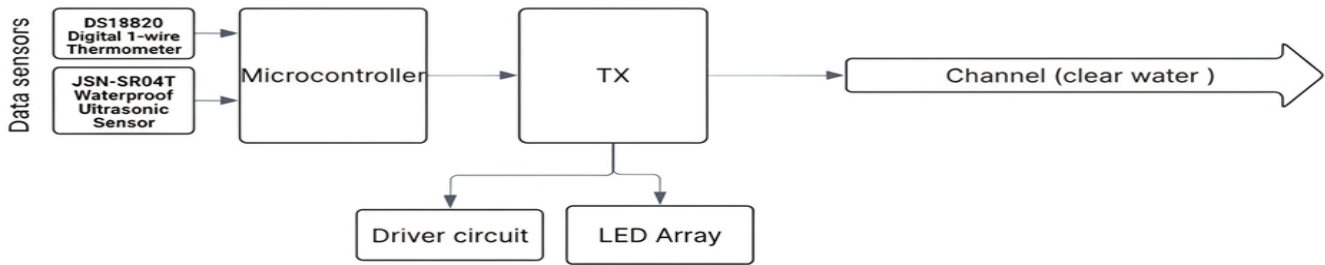


Figure 4.1: Block diagram of the optical transmitter system architecture.

As a fundamental architectural layout, the general pipeline of the developed optical transmitter node is structurally divided into four interconnected sequential blocks:

- 1) Data Acquisition Layer: Designated for collecting raw physical environmental variables from the integrated sensor modules.*
- 2) Microcontroller Unit (MCU): Utilized as the central processing core to manage system states, execute embedded firmware, and coordinate data streams.*
- 3) Data Encoding and Modulation (Manchester Encoding): Implemented from scratch within the firmware to translate binary data into a DC-balanced, flicker-free optical pulse stream.*
- 4) Transmitter Driver Circuit and LED Array: Engineered to translate low-power digital logic into high-current electrical drives, executing the final electro-optical conversion via the high-power blue LED configuration.*

Each of these foundational stages is comprehensively detailed and analyzed in the subsequent subsections.

4.1.1 System Design and Simulation Setup

1) Data Acquisition Layer:

The transmitter node's data acquisition layer integrates two primary environmental sensors with the microcontroller. These sensors are specifically selected to gather environmental metrics required for processing and subsequent optical transmission.

A. Temperature Sensor (DS18B20):

The DS18B20 is a digital, waterproof temperature sensor that provides 9-bit to 12-bit Celsius temperature measurements. Key Features: It communicates over a 1-Wire bus, which means it requires only one data line (and ground) to communicate with the central microcontroller. It has an operating range of -55 to +125. Hardware Connection: As illustrated in DS18B20.png, the sensor is interfaced with the ESP32 microcontroller. The VDD pin is connected to the 3.3V power rail, GND is grounded, and the Data-line is connected to a GPIO pin (e.g., GPIO15). A 4.7kOm pull-up resistor is connected between the data line and the VDD line to ensure stable data transmission over the 1-Wire protocol.

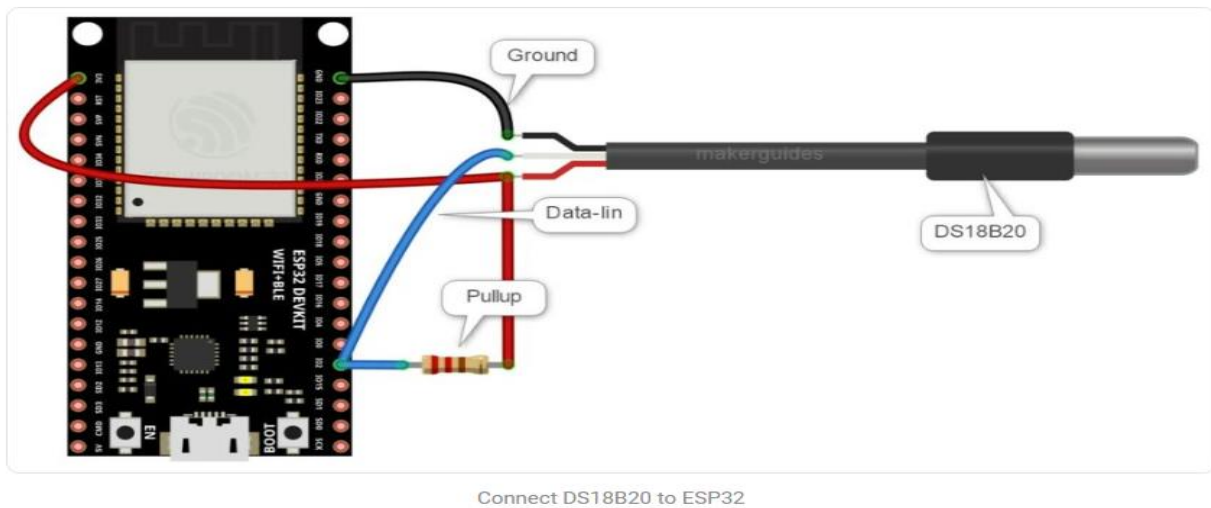
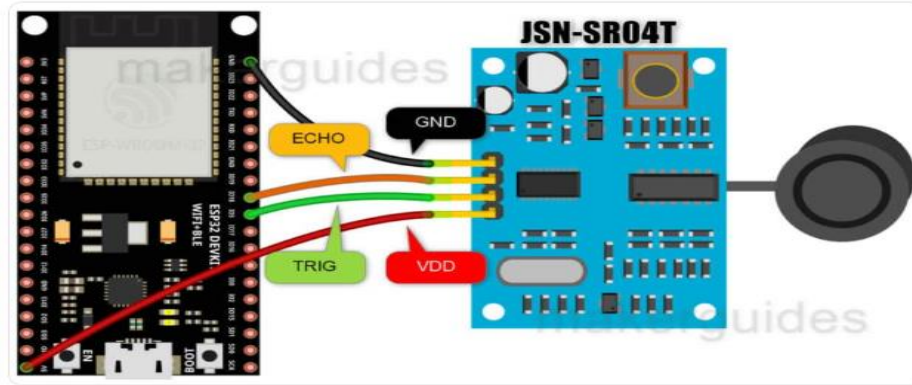


Figure 4.2: Wiring and hardware interfacing of the DS18B20 temperature sensor with ESP32.

B. Ultrasonic Distance Sensor (JSN-SR04T-2.0):

The JSN-SR04T-2.0 is an industrial-grade, waterproof ultrasonic distance measurement sensor used to detect obstacle proximity. Key Features: Unlike standard ultrasonic sensors (like HC-SR04), this module features a separate integrated probe that is entirely waterproof, making it ideal for underwater or high-moisture sensing. It operates within a range of 20 cm to 600cm. Hardware Connection: The module requires a 5V power supply to operate effectively. As shown in Figure 4.3 and its configuration overview in SN-SR04T1.jpg, it interfaces with the ESP32 using two main control pins: TRIG (Trigger): Connected to an ESP32 output pin to initiate the ultrasonic pulse. ECHO: Connected to an ESP32 input pin to measure the duration of the reflected pulse, which is then used to calculate the distance.



Connecting JSN-SR04T to ESP32

Figure 4.3: Schematic connection of the waterproof JSN-SR04T ultrasonic sensor to the ESP32 microcontroller.

2) Microcontroller Unit (MCU):

The central processing core of the optical transmitter node is the ESP32 DevKit microcontroller. The MCU acts as the bridge between the data acquisition layer (sensors) and the physical transmission layer Visible Light Communication (VLC).

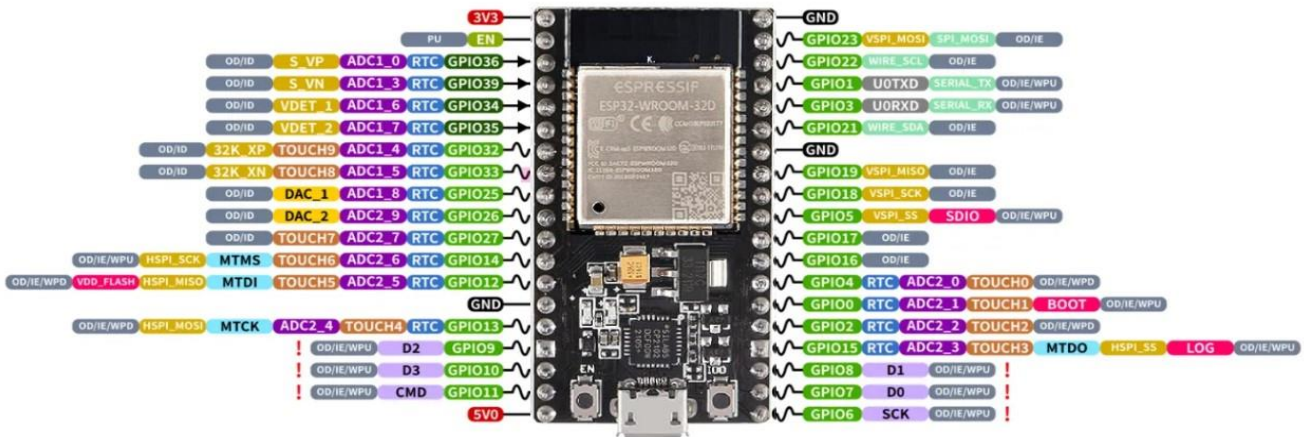


Figure 4.4: Pinout configuration and peripheral mapping of the ESP32- DevKit.

A. Technical Justification:

The ESP32 was selected for this project due to several advanced hardware features that make it superior to traditional 8-bit microcontrollers:

Dual-Core Processor: It features a Tensilica Xtensa 32-bit LX6 dual-core microprocessor operating at up to 240MHz, providing high computational power for real-time sensor polling and signal modulation.

Peripheral Richness: It includes built-in hardware peripherals essential for this system, such as Analog-to-Digital Converters (ADC), Timers, and the Remote Control (RMT) peripheral, which is highly efficient for generating precise Manchester-encoded signals.

B. Functional Role in the Transmitter:

The ESP32 executes three primary sequential tasks within the transmitter node: Data Acquisition & Filtering: The MCU continuously polls the DS18B20 via the 1-Wire protocol and triggers the JSN-SR04T to measure distance.

It processes the raw digital signals and converts them into structured telemetry data (temperature in C and distance in cm).

Data Packaging: The processed sensor readings are structured into a light data packet (payload) to prepare them for transmission.

Signal Modulation & Encoding: Since standard digital data streams (NRZ) are prone to synchronization errors and DC-bias in Visible Light Communication (VLC), the ESP32 performs Manchester Encoding on the data packet. It then outputs this high-frequency modulated signal through a designated GPIO pin to drive the switching circuit.

3) Data Encoding and Modulation (Manchester Encoding):

To ensure reliable and error-free communication over the Visible Light Communication (VLC) channel, a custom Manchester Encoding protocol was developed and implemented from scratch on the ESP32, bypassing high-overhead external libraries.

A. Protocol Architecture and Packet Structure:

*As illustrated in **Figure 4.5**, the developed protocol enforces a strict structure to maintain synchronization and data integrity over the optical link:*

The Frame Layout: The data packet begins with a Header (Start of Frame) to alert the receiver of incoming data, followed by the Payload containing the dynamic sensor readings (Temperature and Distance), and ends with a Footer (End of Frame) or error-checking bits. Bit Transition Mapping: In this implementation, a logical '1' is mapped to a High-to-Low transition, and a logical '0' is mapped to a Low-to-High transition. Each bit period (T_b) is precisely bisected into two halves (T_{half}), guaranteeing a constant 50% DC balance. This eliminates LED flickering and embedded clock recovery at the receiver stage.

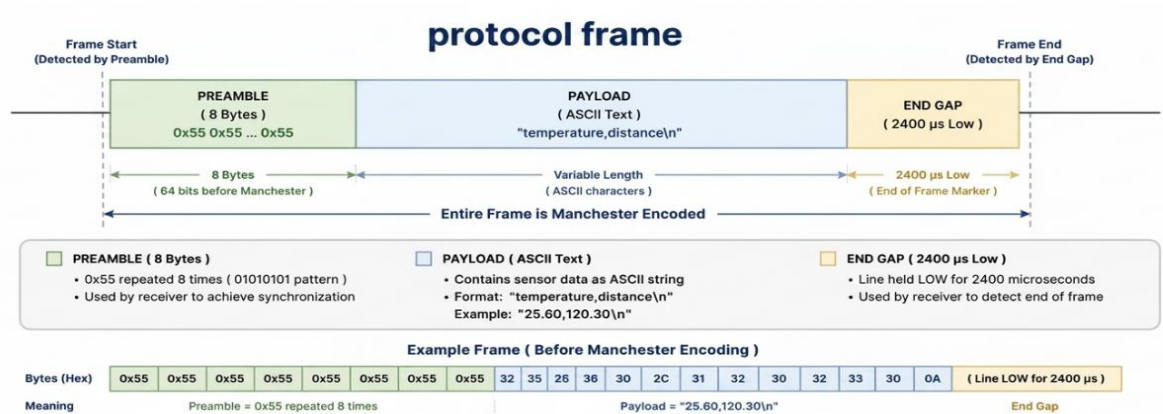


Figure 4.5: Custom Manchester encoding protocol, and packet structure.

B. Algorithmic Logic and Adaptive Transmission Rates:

Adaptive Transmission Rate Selection: A standout feature of this custom protocol is its scalability. The firmware is designed to dynamically adjust the data transmission speed within a broad operating range, spanning from 2.5 kbps up to 200 kbps. This is achieved by varying the half-bit duration (T_{half}) programmatically (e.g., $T_{half}=200\mu s$ for 2.5 kbps, down to $T_{half}=200\mu s$ for 200 kbps). This adaptability allows the system to scale down its speed to overcome high scattering/attenuation in turbid water or scale up for high-throughput environments. *Sequential Transmission:* The ESP32 processes each byte bit-by-bit. It checks the logical state of the bit, applies the corresponding transition at the exact mid-point (T_{half}) using high-precision hardware timing, and sequentially drives the output GPIO pin to trigger the transmitter driver circuit.

4) Transmitter Driver Circuit and LED Array

The primary function of the transmitter driver circuit is to translate the low-power, high-frequency Manchester-encoded digital signal from the ESP32 GPIO pin into high-current pulses capable of driving the optical source (LED array) at high switching speeds.

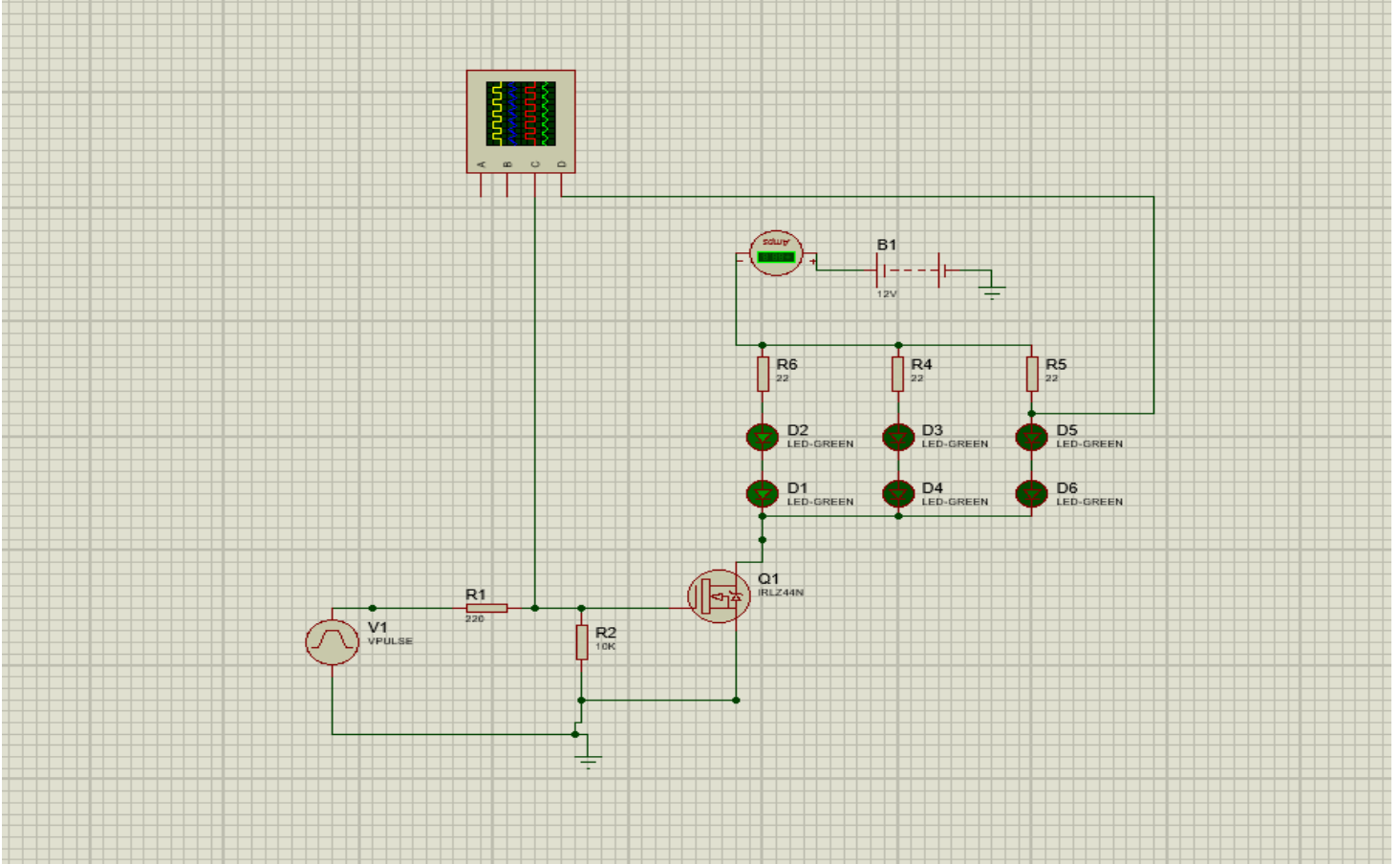


Figure 4.6: Schematic diagram of the high-speed N-channel MOSFET driver circuit for the LED transmitter array by proteus.

A. Circuit Design and Components:

As illustrated in the schematic diagram (**Figure 4.6**), the driver circuit is designed around a fast-switching N-Channel Power MOSFET. This configuration is highly efficient for Visible Light Communication (VLC) due to the following structural connections:

The Gate Control Input: The modulated digital signal from the ESP32 GPIO pin is connected directly to the Gate (G) terminal of the MOSFET. When the GPIO outputs a logic HIGH (3.3 V), it charges the internal gate capacitance, turning the MOSFET channel fully ON (saturation). When the GPIO outputs a logic LOW (0V), the channel turns fully OFF (cutoff). *The Low-Side Switching Configuration:* The Source (S) terminal of the MOSFET is tied directly to the common ground (GND). The Drain (D) terminal is connected to the cathode side of the LED array. This "low-side switch" topology ensures that the MOSFET can handle high load currents and voltages while being controlled by a low-voltage microcontroller.

B. LED Array Topology and Current Regulation:

The optical radiation source consists of a high-power LED Array arranged in a hybrid series-parallel topology to optimize power distribution and brightness: *Series-Parallel Layout:*

The array is divided into three parallel branches. Each individual branch consists of Two LEDs connected in series. This series connection ensures that the exact same forward current flows through all LEDs in that branch, maintaining uniform light intensity and wavelength emission.

Current-Limiting Resistors (R_4, R_5, R_6): Each branch is equipped with its own dedicated power resistor (R_4, R_5, R_6) connected in series with the LEDs. These resistors are critical to limit the maximum forward current (I_F) flowing through the LEDs when the MOSFET is ON, preventing thermal runaway and protecting the optical components from burning out under the 12V main power supply (VCC).

4.1.2 Design Calculations and Component:

To validate the hardware architecture and ensure the long-term operational reliability of the transmitter node, precise engineering calculations were executed regarding electrical power constraints, voltage drops, and temporal high-frequency limits.

A. Current-Limiting Resistor and Voltage Calculations:

The optical array is powered by a regulated $V_{supply} = 12\text{ V}$ source and consists of three parallel branches. According to the updated schematic configuration, each individual branch contains two high-power Green LEDs connected in series with a dedicated current-limiting resistor ($R = 22\text{ }\Omega$). Based on the manufacturer's datasheet, the forward voltage drop (V_f) across a single Green LED under typical operating conditions is 3.2 V. The total voltage drop across the series LEDs in a single branch (V_{LED_Total}) is determined by:

$$V_{LED_Total} = 2 \times V_F = 2 \times 3.2\text{ V} = 6.4\text{ V} \quad (4.1)$$

Applying Kirchhoff's Voltage Law (KVL) across the branch loop, and assuming the saturation voltage drop across the IRLZ44N power MOSFET (V_{DS}) is negligible (0 V), the remaining voltage drop designated across the current-limiting resistor (V_R) is calculated as:

$$V_R = V_{supply} - V_{LED_Total} = 12\text{ V} - 6.4\text{ V} = 5.6\text{ V} \quad (4.2)$$

Using Ohm's Law, the continuous forward current flowing through each individual branch (I_{branch}) is regulated by the 22 Ω resistor:

$$I_{branch} = \frac{V_R}{R} = \frac{5.6\text{ V}}{22\text{ }\Omega} \approx 0.2545\text{ A} \text{ (254.5 mA)} \quad (4.3)$$

B. Total Transmitter Power Consumption:

The total electrical current (I_{total}) drawn by the entire optical transmitter structure from the 12 V source is the sum of the currents passing through the three matched parallel branches:

$$I_{total} = 3 \times I_{branch} = 3 \times 254.5 \text{ mA} \approx 763.5 \text{ mA} \text{ (0.7635 A)} \quad (4.4)$$

The total electrical power consumed by the transmitter during a continuous logic 'HIGH' transmission state is:

$$P_{total} = V_{supply} \times I_{total} = 12 \text{ V} \times 0.7635 \text{ A} \approx 9.162 \text{ W} \quad (4.5)$$

C. Bandwidth and Switching Constraints

The bandwidth limits of the physical driver circuit are fundamentally dictated by the switching speeds of the IRLZ44N N-Channel Power MOSFET, which is governed by the RC time constant of the gate driver stage. The input gate resistor used to damp switching ringing from the ESP32 is $R_1 = 220 \text{ Ohm}$. According to the official datasheet of the IRLZ44N, the typical input capacitance (C_{iss}) is approximately 1700 pF.

The RC charging time constant of the gate (τ) is calculated as:

$$\tau = R_1 \times C_{iss} = 220 \text{ } \Omega \times (1700 \times 10^{-12} \text{ F}) \approx 374 \text{ ns} \quad (4.6)$$

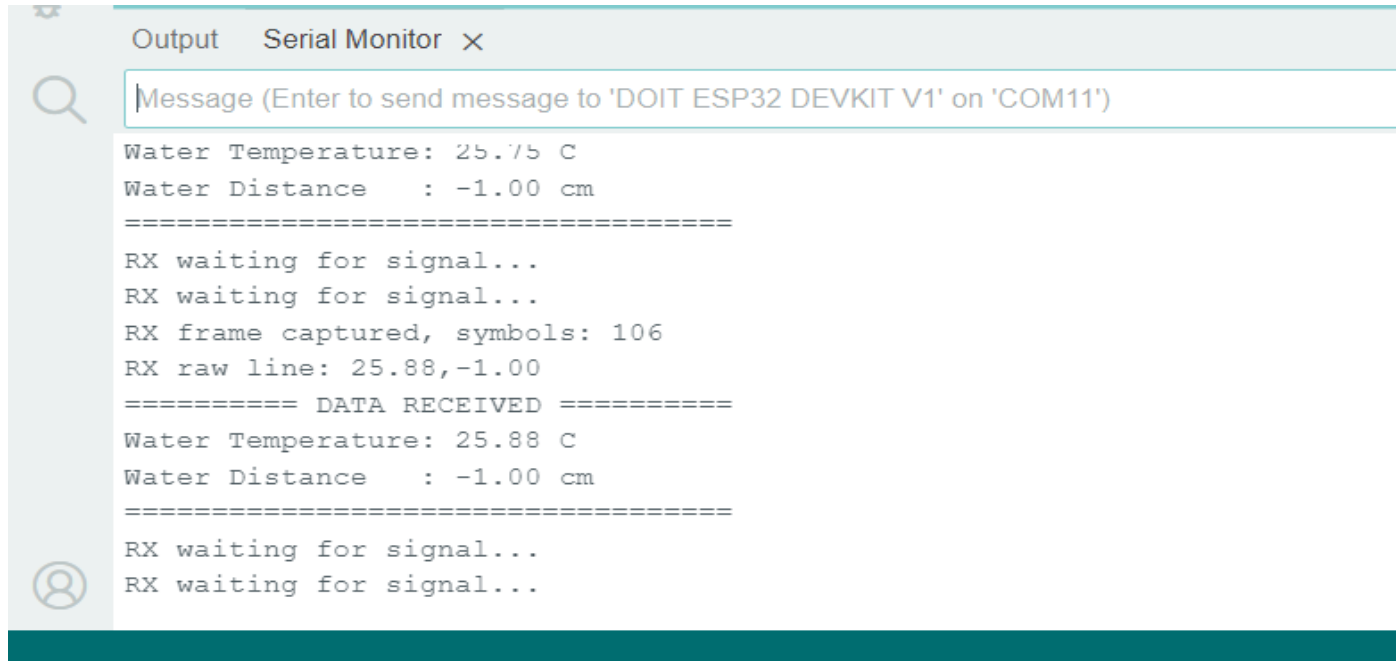
The corresponding theoretical electrical cutoff frequency (-3dB bandwidth) of the driver circuit interface (f_{-3dB}) is defined by:

$$f_{-3dB} = \frac{1}{2\pi\tau} = \frac{1}{2 \times \pi \times 374 \times 10^{-9} \text{ s}} \approx 425.5 \text{ kHz} \quad (4.7)$$

4.1.3 Simulation Results and Performance Analysis

1) Wired Protocol Validation and Logical Verification:

Before deploying the system onto physical optical hardware, a wired simulation phase was executed to strictly validate the software logic of the Manchester protocol. As demonstrated in the serial debugging logs:



```
Output Serial Monitor x
Message (Enter to send message to 'DOIT ESP32 DEVKIT V1' on 'COM11')
Water Temperature: 25.75 C
Water Distance : -1.00 cm
=====
RX waiting for signal...
RX waiting for signal...
RX frame captured, symbols: 106
RX raw line: 25.88,-1.00
===== DATA RECEIVED =====
Water Temperature: 25.88 C
Water Distance : -1.00 cm
=====
RX waiting for signal...
RX waiting for signal...
```

Figure 4.7: Arduino IDE Serial Monitor output validating the software logic, packet capture, and successful frame extraction of the custom Manchester protocol over a wired simulation link.

Frame Detection and Synchronization: The firmware successfully recognized the start of the frame, captured the continuous incoming bit stream, and decoded the transitions in real-time.

Data Integrity: The dynamic payload string containing sensor data was extracted with zero character corruption or bit displacement, mathematically proving that the custom state-machine logic handles bit transitions flawlessly over the digital domain.

2) Frequency Transient Waveform Analysis:

To stress-test the switching capability of the IRLZ44N MOSFET driver and verify pulse shape fidelity, virtual oscilloscope Probes were analyzed across a scalable frequency spectrum, ranging from 1 kHz up to 20:

Low-Frequency Performance (1 kHz and 5 kHz): The output square waves exhibit perfectly symmetrical transitions. The bit boundaries are sharp, and the flat-top phases of the pulses maintain a rock-solid steady voltage, ensuring safe constant illumination. **High-Frequency Performance (10 kHz and 20 kHz):** As the modulation rate increases up to 20 kHz, the physical switching edges remain crisp and strictly vertical. The simulation shows zero trace of parasitic capacitive rounding or signal dampening.

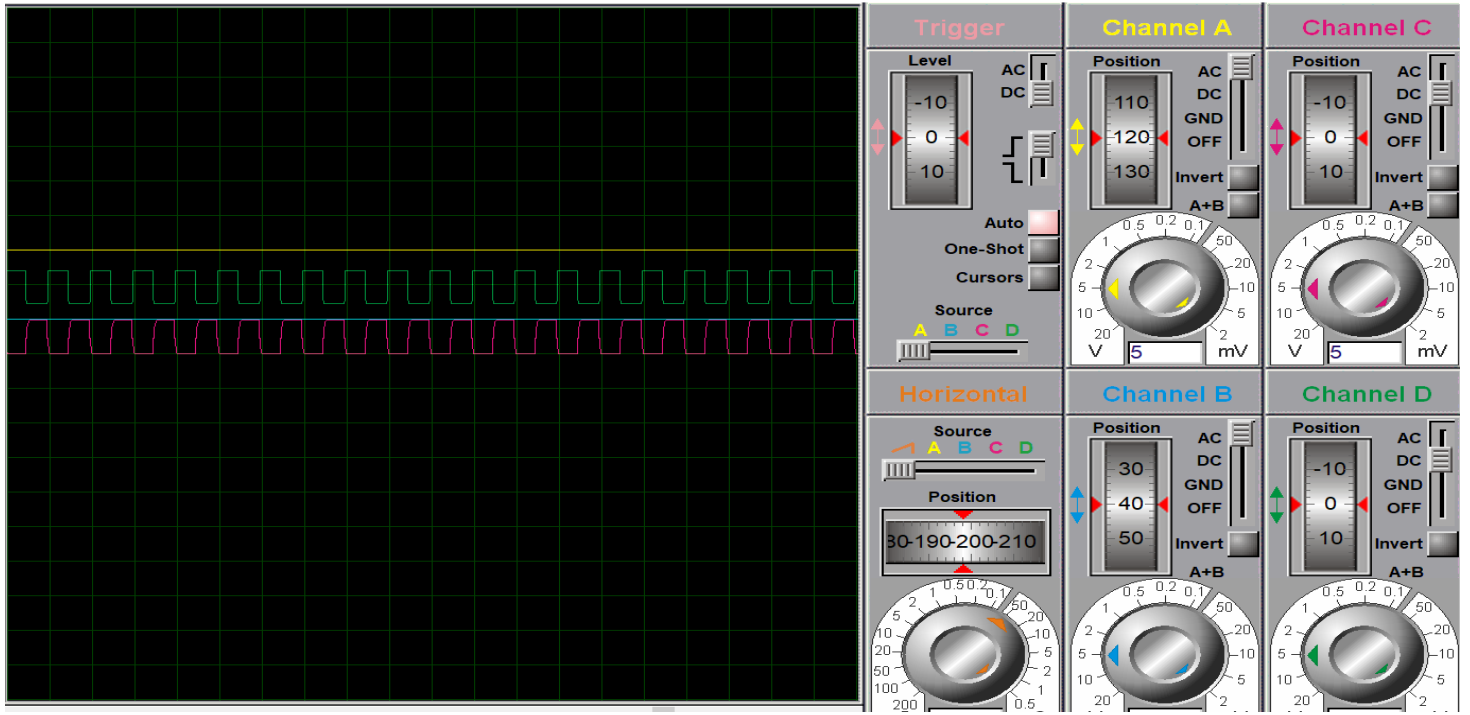


Figure 4.8: Proteus virtual oscilloscope waveform showcasing Manchester encoded pulse stream stability across testing frequency (20 kHz).

The consolidated simulation results conclusively confirm that:

No Inter-Symbol Interference (ISI): The temporal transition margins are clean across all tested rates, leaving adequate headroom for data scaling up to the physical channel limit.

Firmware Reliability: Developing the protocol from scratch provided high deterministic execution timing on the microcontroller architecture.

4.2 Proposed Optical Beam Collimation Using a 15° PMMA TIR Lens

To improve the performance of the proposed underwater optical wireless communication (UOWC) system and mitigate the optical spreading produced by the LED transmitter, a 15° Total Internal Reflection (TIR) collimator lens is proposed as an optical beam-shaping element. The primary objective of this lens is to reduce beam divergence, increase the transmitted optical power density, and improve the amount of optical energy reaching the photodiode receiver. Beam collimation is considered one of the most effective optical techniques for enhancing underwater optical communication performance, particularly over short and medium transmission distances (Kaushal & Kaddoum, 2016; Arnon, 2010).

4.2.1 Lens Type and Specifications

The proposed optical beam shaping element is a 13 mm PMMA Total Internal Reflection (TIR) collimator lens designed for use with high-power LEDs. This type of lens is widely used in optical communication and LED illumination applications because it efficiently collects and redirects the emitted light into a controlled beam with a specified divergence angle. Compared with conventional LED emission, the TIR lens significantly reduces beam spreading and improves the optical power directed toward the receiver (Saleh & Teich, 2007; Ghassemlooy et al., 2019).

PMMA (Polymethyl Methacrylate) was selected as the lens material due to its high optical transparency, lightweight structure, excellent mechanical properties, and low optical absorption within the visible spectrum. These characteristics make PMMA an appropriate material for underwater optical communication applications (Saleh & Teich, 2007).

Table 4.1: Proposed Lens Specifications

Parameter	Value
Lens Type	PMMA TIR Collimator
Material	Optical PMMA
Beam Angle	15°
Diameter	13 mm
Height	8.5 mm
Compatible LED	1–5 W High-Power LED
Optical Transmittance	>92%



Figure 4.9: Actual PMMA 15° Total Internal Reflection (TIR) collimator lens proposed for the optical transmitter.

Working Principle of the Proposed TIR Lens

point (F) of the lens. Light rays emitted from the LED diverge naturally in different directions. However, after passing through the TIR lens, the rays are redirected by refraction and total internal reflection, producing a nearly parallel optical beam with significantly reduced divergence. This optical configuration increases the transmitted optical power density and improves the propagation distance of the underwater optical communication system

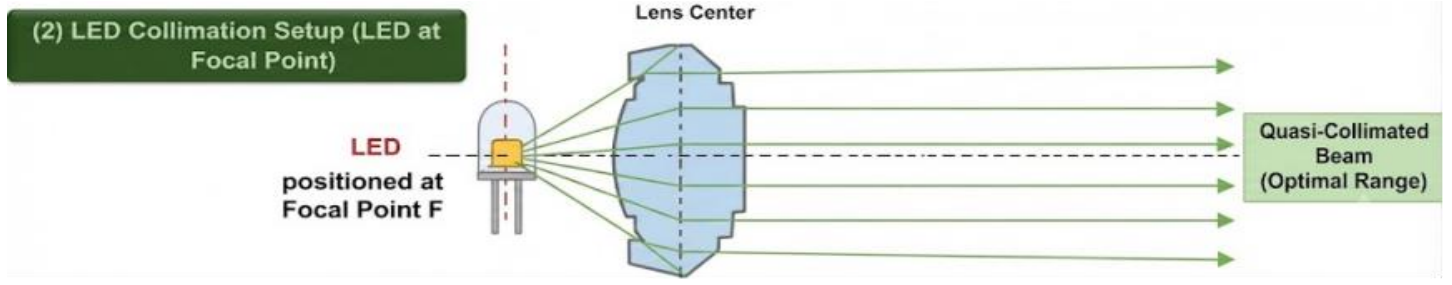


Figure 4.10: Operating principle of the proposed PMMA TIR collimator lens showing the LED positioned at the focal point (F) and the formation of a nearly parallel optical beam.

4.2.2 Selection of the 15° Beam Angle

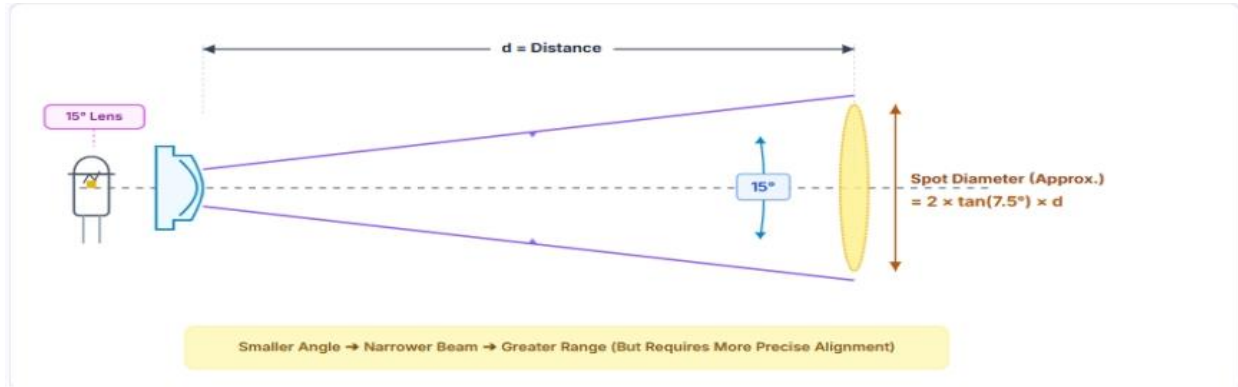


Figure 4.11: Beam Geometry of the Proposed 15° TIR Collimator Lens

PMMA Total Internal Reflection (TIR) collimator lens. The lens is positioned directly in front of the LED to reduce the natural beam divergence and reshape the emitted light into a narrow optical beam.

The full beam angle produced by the lens is 15°, which corresponds to a half-angle of 7.5° on each side of the optical axis. As the optical beam propagates through the transmission medium over a distance d , the beam gradually expands, forming a circular illuminated spot at the receiver plane. The spot diameter depends on both the propagation distance and the beam divergence angle and can be estimated using the following geometrical relationship (Hecht, 2017):

$$D = 2 d \tan(\theta) \quad (4.8)$$

where:

- D is the projected spot diameter.
- d is the transmission distance.
- θ is the beam half-angle (7.5°).

For the proposed transmission distance of 20 cm, the projected optical spot diameter is approximately 5.3 cm, providing a concentrated optical footprint while maintaining sufficient alignment tolerance between the transmitter and the receiver.

Using a narrow 15° beam angle significantly increases the optical power density directed toward the photodiode compared with wider beam angles. Consequently, more optical energy reaches the receiver, reducing geometric spreading losses and improving the expected communication performance in underwater optical wireless communication systems (Kaushal & Kaddoum, 2016; Arnon, 2010; Saleh & Teich, 2007).

4.2.3 Expected Advantages of the Proposed 15° TIR Lens

The proposed 15° PMMA Total Internal Reflection (TIR) collimator lens is expected to provide several improvements to the performance of the underwater optical wireless communication system. By reducing the natural divergence of the LED beam, the lens concentrates the emitted optical power into a narrower optical path, allowing a larger portion of the transmitted energy to reach the photodiode receiver. Consequently, the received optical power density is expected to increase, improving the efficiency of the optical communication link (Kaushal & Kaddoum, 2016; Ghassemlooy et al., 2019).

In addition, the reduced beam divergence is expected to minimize geometric spreading losses, which become more significant as the transmission distance increases. The concentrated optical beam may also improve the received Signal-to-Noise Ratio (SNR) by directing more optical energy toward the receiver while reducing unnecessary optical dispersion (Arnon, 2010; Saleh & Teich, 2007).

Furthermore, the calculated beam footprint of approximately 5.3 cm at a transmission distance of 20 cm provides a reasonable alignment tolerance between the transmitter and the receiver. This beam geometry is expected to simplify the alignment process during laboratory demonstrations while maintaining sufficient optical coverage over the receiver area (Hecht, 2017).

Overall, integrating a 15° PMMA TIR collimator lens represents a promising enhancement for future versions of the proposed system, particularly for extending the communication range and improving the optical efficiency of underwater wireless communication links.

4.3 Under Water Optical Channel

4.3.1 System Design and Simulation Setup

In this section, a UWOC system is modeled and simulated using OptiSystem software to evaluate the performance of optical signal transmission under various channel conditions. The primary objective of the simulation is to determine the maximum achievable communication range while maintaining acceptable performance in terms of Bit Error Rate (BER) and Q-factor. To achieve this, the system is analyzed under different underwater environments characterized by varying attenuation and scattering levels, representing multiple water

types. The simulation further investigates the effect of transmission distance on key performance metrics, including received power, BER, and signal quality. A comparative analysis is conducted to assess system behavior across these different conditions, providing insight into the trade-offs between communication range, channel characteristics, and signal reliability.

- **Simulation Model Architecture and Simulation Setup**

The simulated UWOC system consists of three main blocks: a transmitter, an underwater optical channel, and a receiver. The transmitter generates the optical signal using a green LED source and employs On-Off Keying (OOK) modulation.

The modulated signal propagates through the underwater channel, where attenuation and scattering effects are incorporated into the model. At the receiver, a photodetector converts the received optical signal into an electrical signal, which is then processed using a low-pass filter to reduce noise and enhance signal integrity prior to performance evaluation.

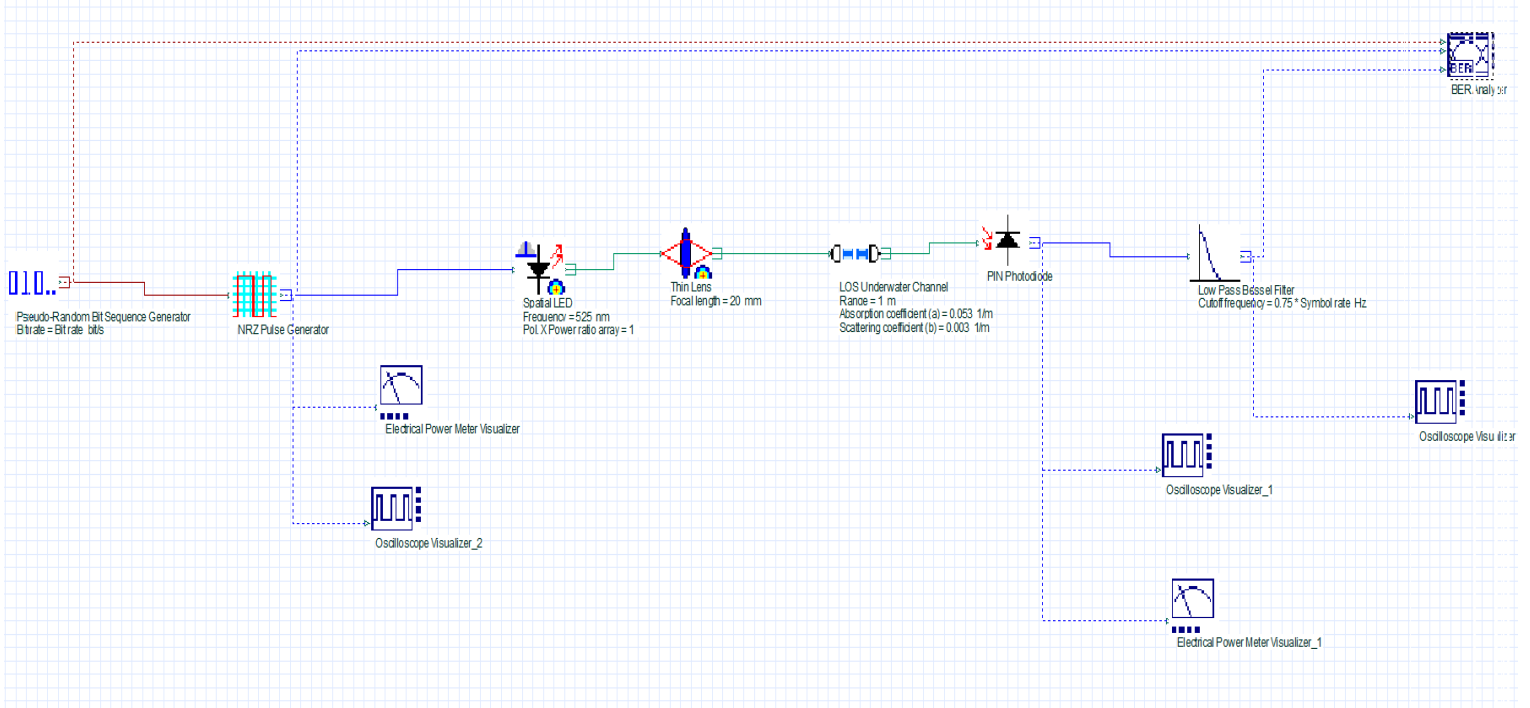


Figure 4.12. OptiSystem simulation of the UWOC system.

Following the system architecture shown in **Figure 4.12**, each simulation component is selected to represent a practical UWOC system. A pseudo-random bit sequence (PRBS) generator is used to produce the input binary data, simulating the output of sensing devices. The generated data is then passed through a Non-Return-to-Zero (NRZ) pulse generator to convert the binary sequence into an electrical signal suitable for modulation.

A directly modulated LED source is employed as the optical transmitter, eliminating the need for an external modulator, as intensity modulation is inherently achieved in LED-based systems. The LED is configured to operate at a wavelength of 525 nm within the green region, which corresponds to the minimum attenuation

window in underwater environments. To improve beam directivity, an optical lens is introduced to reduce the LED divergence angle from approximately 120° to about 15° , using a focal length of 20 mm and a diameter of 5 mm, thereby enhancing the effective transmission range.

The underwater optical channel is modeled by incorporating attenuation and scattering effects based on clear water conditions, with parameters selected according to values reported in the literature. At the receiver side, a photodetector is used to convert the incoming optical signal into an electrical signal. The output is then passed through a low-pass filter to suppress noise and improve signal quality before further performance evaluation.

Table 4.2: Simulation parameters

<i>Simulation parameters</i>		
<i>Bit rate</i>		<i>0.01e+006 bps</i>
<i>Sequence length</i>		<i>128</i>
<i>Samples per bit</i>		<i>1024</i>
<i>Guard bits</i>		<i>0</i>
<i>Symbol rate</i>		<i>0.01e+006 symbols/sec</i>
<i>Number of samples</i>		<i>131072</i>
<i>Frequency</i>		<i>525 nm</i>
<i>Focal length</i>		<i>20 mm</i>
<i>Aperture effects</i>		<i>1</i>
<i>Lens diameter</i>		<i>5 mm</i>
<i>Geometrical loss and gain</i>		<i>NO</i>
<i>Line of sight loss</i>		<i>YES</i>
<i>Shot noise</i>		<i>NO</i>
<i>Thermal noise</i>		<i>NO</i>
<i>Responsivity</i>		<i>0.5 A/W</i>
<i>Dark current</i>		<i>2 nA</i>

Table 4.3: components used for simulation

Components used for simulation	
Pseudo random bit sequence generator	To evaluate the performance of the proposed UWOC system, a series of simulation scenarios is developed by systematically varying key channel parameters. In particular, the transmission distance and attenuation coefficient are adjusted to emulate different underwater environments with varying levels of absorption and scattering. These variations allow the modeling of realistic conditions ranging from clearer water to more turbid environments, thereby capturing the impact of channel impairments on optical signal propagation.
NRZ pulse generator	
Spatial LED	
Thin lens	
LOS underwater channel	
PIN photodiode	
Low pass Bessel filter	

For each simulation scenario, the system performance is analyzed using multiple performance metrics, including received optical power, Bit Error Rate (BER), and Q-factor. The received power provides insight into signal attenuation over distance, while BER and

Q-factor are used to evaluate the quality and reliability of the communication link. By observing how these parameters change with increasing distance and channel loss, it becomes possible to identify the operational limits of the system.

This methodology enables a comprehensive assessment of the trade-offs between communication range and signal quality. It also provides a clear understanding of how different underwater channel conditions influence system performance, thereby supporting the design and optimization of UWOC systems for reliable long-distance communication.

4.3.2 Simulation Results and Performance Analysis

In this study, the proposed UWOC system was simulated under different underwater channel conditions corresponding to several Jerlov water types. The system performance was evaluated using received power, Signal-to-Noise Ratio (SNR), Bit Error Rate (BER), and Q-factor. The effect of attenuation coefficient variation on communication performance was analyzed to evaluate the reliability of the proposed optical communication link.

- **Evaluation Parameters and Performance Indicators**

The performance of the proposed Underwater Optical Wireless Communication (UWOC) system was evaluated using several key performance indicators, including received optical power, Signal-to-Noise Ratio (SNR), Bit Error Rate (BER), and Q-factor. These parameters provide a comprehensive assessment of the communication link quality, transmission reliability, and overall system efficiency under different underwater channel conditions.

As discussed previously in the channel modeling section, underwater optical propagation is significantly influenced by absorption and scattering effects, which increase with water turbidity and propagation distance. Consequently, the received optical power decreases exponentially according to the Beer–Lambert attenuation model. Since received power directly affects the signal quality at the receiver side, it is considered one of the most

important parameters for evaluating UWOC system performance. Similar observations were reported in previous underwater optical communication studies, where attenuation was identified as a dominant factor affecting communication reliability.

The Signal-to-Noise Ratio (SNR), previously defined, was also used to evaluate the quality of the received signal under different channel conditions. In UWOC systems, lower received optical power leads to reduced SNR values due to the increased influence of noise and channel distortion. Therefore, higher SNR values indicate improved signal quality and more reliable data detection at the receiver side.

In addition, the Bit Error Rate (BER) was used as a primary performance metric to evaluate transmission reliability. BER represents the probability of incorrect bit detection and is directly related to the received SNR. Lower BER values correspond to better communication performance and lower transmission errors. Since the proposed system employs On-Off Keying (OOK) modulation, BER performance was analyzed under different attenuation conditions to investigate the effect of underwater channel degradation on signal detection accuracy.

Furthermore, the Q -factor was utilized to evaluate signal quality and eye diagram opening characteristics. Higher Q -factor values indicate better signal distinguishability and lower intersymbol interference (ISI), while lower Q -factor values reflect increased signal distortion and degraded communication performance. The combined analysis of received power, SNR, BER, and Q -factor provides a clear evaluation of the proposed UWOC system under different underwater environments and attenuation conditions (Sahu, P. K., & Ghosh, D 2025).

• Findings and Discussion

The proposed UWOC system was evaluated using OOK modulation at a wavelength of 525 nm under different underwater channel conditions. The analysis was performed for clear seawater at distances of 1 m, 50 m, and 200 m, coastal water at 1 m, 20 m, and 40 m, and turbid water at 1 m and 20 m. In all simulation scenarios, the transmitted optical power was fixed at 471 mW. The received power results under different water conditions are presented in **Table 4.4**

Table 4.4 Received Power Performance under Different Water Types and Transmission Distances

Type of water	Attenuation c (m^{-1})	LED-PS Distance (m)	P_t mW	P_r dBm
Clear Sea water	0.056	1	471	66.7
		50		-25
		200		-100
Coastal Water	0.305	1	471	64.8
		20		-37.3
		40		-100
Turbid Water	2.170	1	471	62.3
		20		-89

As shown in **Table 4.4**, the received optical power decreases with increasing transmission distance due to absorption and scattering losses in the underwater channel. According to the Beer–Lambert attenuation relation, the received power is inversely affected by both propagation distance and the attenuation coefficient of the water medium. Consequently, clearer water conditions enabled longer communication distances than in highly turbid environments.

For clear seawater ($c = 0.056 \text{ m}^{-1}$), the proposed UWOC system maintained signal propagation up to 200 m before the received power reached approximately -100 dBm . In contrast, coastal water and turbid water experienced faster signal degradation because of their higher attenuation coefficients, limiting the achievable communication range to shorter distances. The results clearly demonstrate that increasing water turbidity significantly reduces the received optical power and deteriorates channel performance.

Once the received optical power approaches the minimum detectable level of approximately -100 dBm , reliable signal detection becomes difficult, and further communication is no longer achievable under the considered simulation conditions.

These results confirm the strong dependence of the UWOC system performance on water turbidity and attenuation characteristics.

• BER and Q-Factor Performance Analysis

In addition to the received power analysis, the communication link quality was evaluated using BER and Q-factor measurements. These parameters provide a direct indication of transmission reliability and signal integrity under different underwater attenuation conditions.

Table 4.5: BER and Q-Factor Performance under Different Underwater Conditions

Type of water	LED-PS Distance (m)	Q factor	BER
Clear Sea water	1	41	0
	50	32.9	10^{-239}
	200	11.9	10^{-33}
Coastal Water	1	36	10^{-288}
	20	31.7	10^{-221}
	40	31.6	10^{-220}
Turbid Water	1	31	10^{-218}
	20	25	10^{-144}

As shown in **Table 4.5**, the proposed UWOC system achieved excellent performance under clear seawater conditions due to the low attenuation coefficient. At a distance of 1 m, the Q -factor reached 41 while the BER was approximately zero, indicating an almost ideal communication link. As the transmission distance increased, the Q -factor gradually decreased and the BER increased due to underwater attenuation and propagation losses. However, the system still maintained good communication performance at 200 m with a Q -factor of 11.9 and a BER of (10^{-33}) .

In coastal water, the achievable communication distance decreased significantly because of the higher attenuation coefficient, limiting the signal propagation to approximately 40 m. Despite the reduced range, the Q -factor and BER values still indicated reliable communication performance within the supported distances.

For turbid water conditions, the communication range decreased further due to severe absorption and scattering effects. Nevertheless, the system maintained a Q -factor value of 25 at 20 m, demonstrating acceptable signal quality for short-range underwater optical communication.

Eye Diagram Analysis

The eye diagrams obtained under different underwater channel conditions were used to visually evaluate signal quality, intersymbol interference (ISI), and noise effects at different transmission distances. In general, wider eye openings indicate better communication performance and lower signal distortion, while reduced eye openings reflect increased attenuation and channel degradation

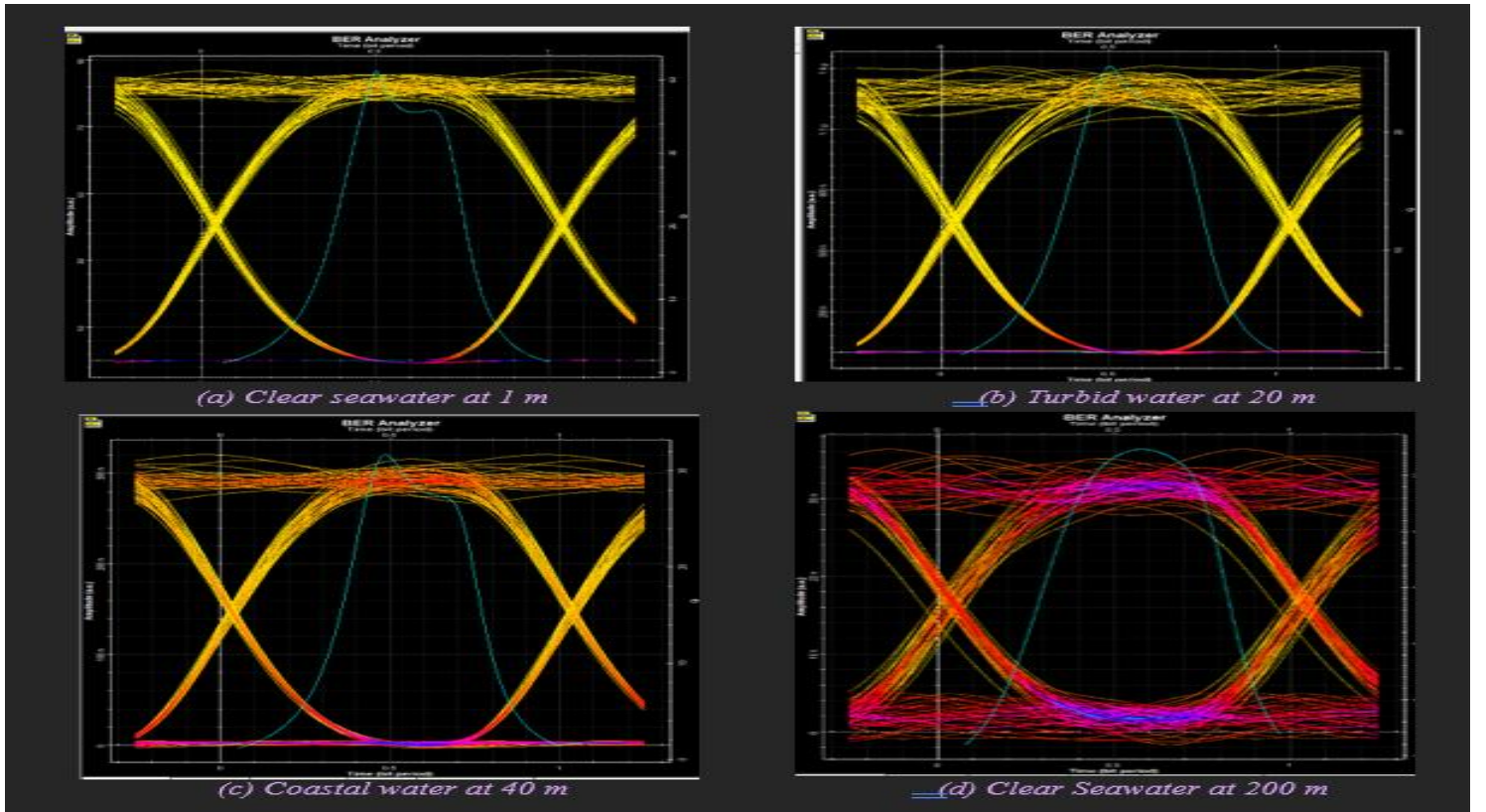


Figure 4.13: Eye Diagram Performance under Different Underwater Channel Conditions and Transmission Distances

shown in **Figure 4.13**, the eye diagrams remained open under all investigated underwater conditions, indicating successful signal detection and acceptable communication performance. The clearest eye-opening was observed for clear seawater at short transmission distances due to the low attenuation and reduced signal distortion. As the transmission distance increased and the water turbidity became higher, the eye opening gradually decreased because of stronger absorption and scattering effects within the underwater channel. Nevertheless, the proposed UWOC system maintained sufficiently open eye patterns even at longer distances, confirming reliable signal transmission under the considered simulation conditions.

• Input and Output Signal Analysis

The transmitted and received signals were analyzed using the Oscilloscope Visualizer components in OptiSystem. The input transmitted signal was observed using Oscilloscope Visualizer 2, while the received filtered signal was monitored using Oscilloscope Visualizer 0. As shown in **Figure 4.14**, the output signal maintained approximately the same amplitude level as the transmitted signal, indicating successful signal transmission through the underwater optical channel. However, slight waveform distortion and noise effects can be observed in the received signal due to underwater attenuation, channel noise, and non-ideal characteristics of the low-pass filter used at the receiver side. Despite these minor distortions, the received signal preserved a clear waveform shape, confirming reliable signal recovery and acceptable communication performance for the proposed UWOC system.

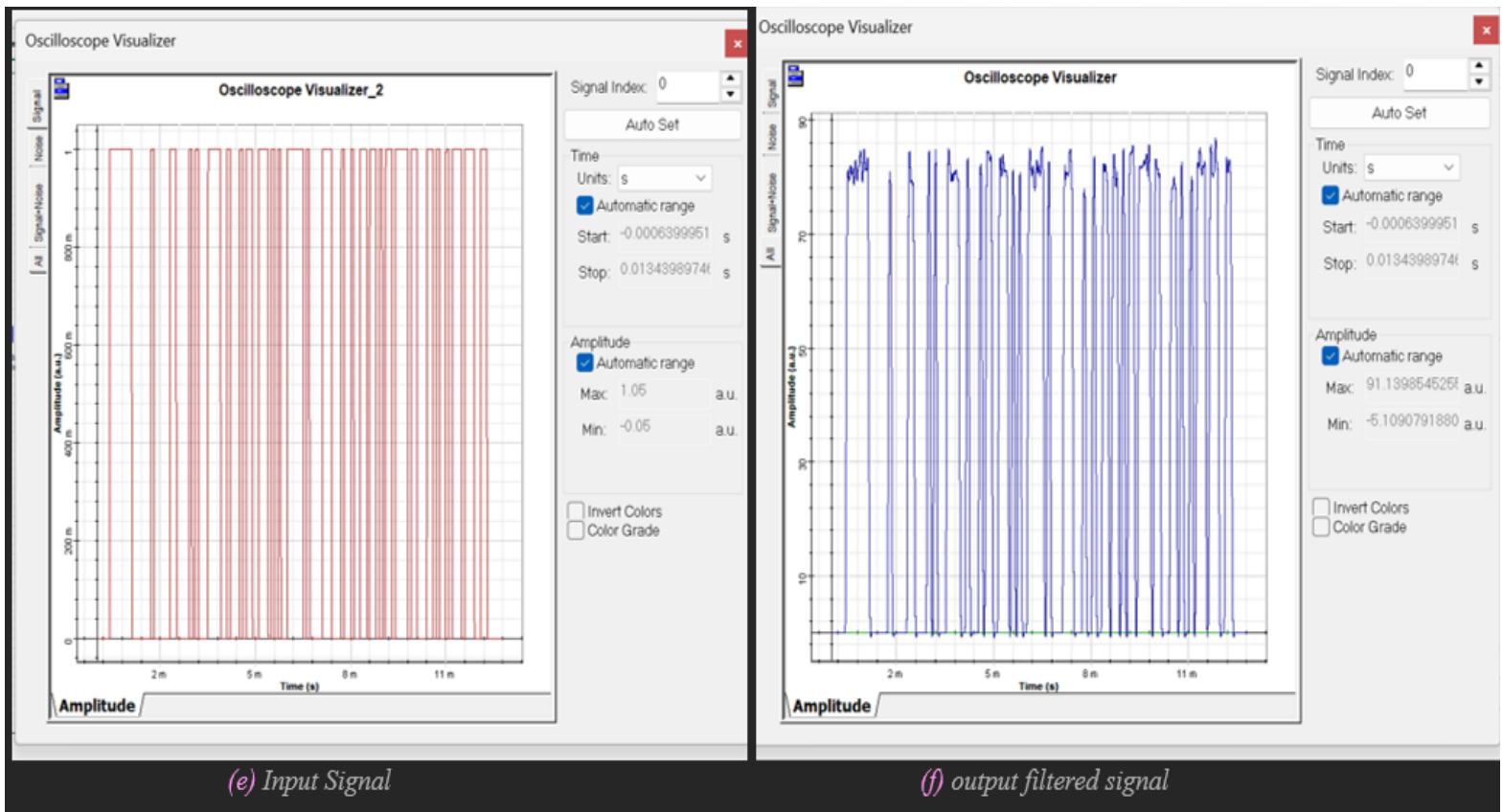


Figure 4.14: Input and Output Filtered Signal Waveforms of the Proposed UWOC System

The obtained simulation results confirm that the performance of the proposed UWOC system is strongly affected by underwater attenuation and propagation distance. Clear seawater achieved the best communication performance due to its low attenuation coefficient, while coastal and turbid water conditions reduced the achievable transmission range because of stronger absorption and scattering effects. Nevertheless, the proposed OOK-based UWOC system maintained acceptable BER, Q-factor, and eye diagram performance within the supported communication distances, demonstrating the feasibility of underwater optical wireless communication under different channel conditions.

4.4 Optical Receiver

4.4.1 System Design and Simulation Setup

- **Receiver implementation**

The receiver was implemented to detect the optical signal transmitted through the underwater optical communication channel and recover the original transmitted data. **Figure 4.15** illustrates the block diagram of the implemented receiver.

The received optical signal is first collected by a lens, which concentrates the incoming light onto the BPW34 photodiode to improve the received optical power. The photodiode converts the incident optical signal into a photocurrent proportional to the received light intensity. Since this photocurrent is very small, it is converted into a voltage and amplified using a Transimpedance Amplifier (TIA) based on the LM358N operational amplifier. The amplified analog signal is then applied to an LM393 comparator, which regenerates a clean digital waveform suitable for digital processing. Finally, the regenerated signal is received by the ESP32 microcontroller, where the transmitted OOK data is detected and processed.

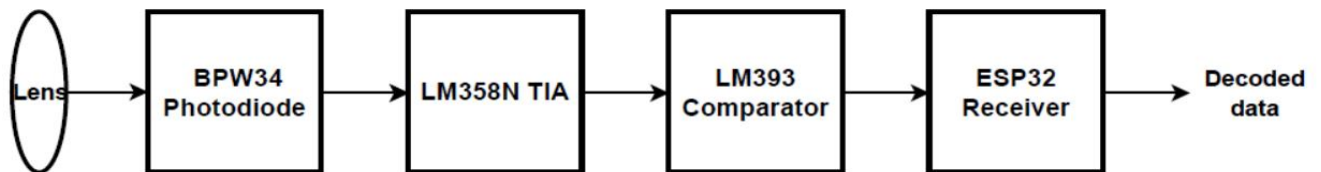


Figure 4.15: the block diagram of the implemented optical receiver.

- **Circuit implementation and Simulation Setup**

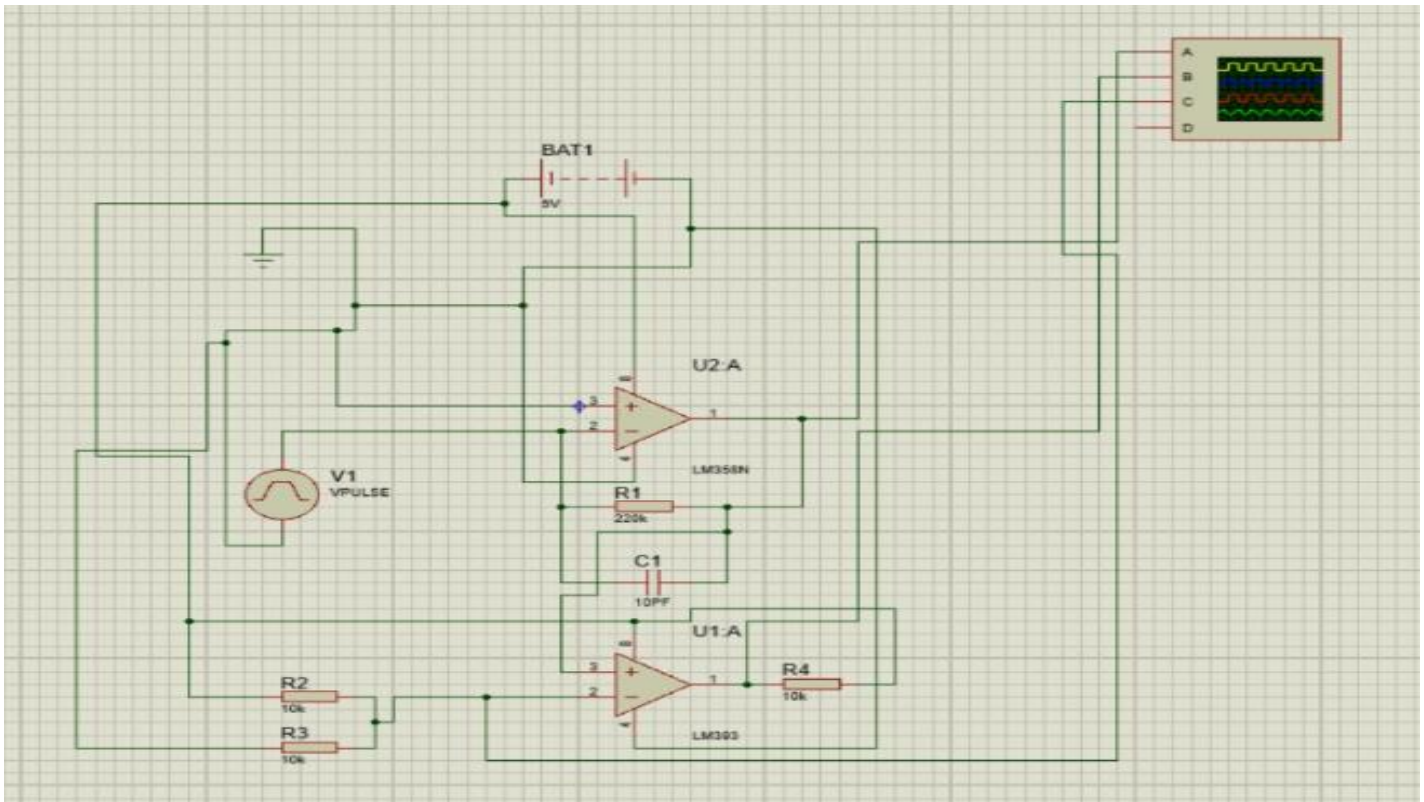


Figure 4.16: Simulation of optical receiver by proteus

In this simulation, the VPULSE source is used instead of the photodiode. The VPULSE generates pulse signals that represent the photocurrent produced by the photodiode.

The pulse signal is applied to the Transimpedance Amplifier (TIA), which consists of an LM358N operational amplifier, a 220 k Ω feedback resistor (R1), and a 10 pF feedback capacitor (C1).

The TIA converts the input current into a voltage signal and amplifies it.

The output of the TIA is connected to the positive (+) input of the LM393 comparator.

The negative (–) input of the comparator is connected to a voltage divider made of two 10 k Ω resistors (R2 and R3) supplied with 5 V, creating a 2.5 V reference voltage.

Since the LM393 has an open-collector output, a 10 k Ω pull-up resistor (R4) is connected to +5 V.

Finally, the comparator converts the amplified analog signal into a clean digital output, producing pulses that switch between 0 V and 5 V.

- **component used**

1. Photodiode

The BPW34 is used to convert the received optical signal into a photocurrent. It is well suited for underwater visible light communication due to its large active area, fast response, and good sensitivity to blue and green wavelengths.

Key Features

- PIN silicon photodiode.
- Large active area.
- High sensitivity to blue/green light.
- Fast response for real-time data transmission.
- Low cost and widely available.

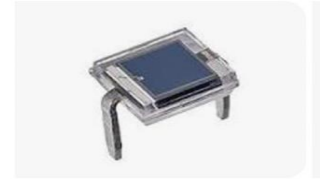


Figure 4.17: Photodiode

2. Transimpedance Amplifier (TIA)

Since the photocurrent generated by the photodiode is very small, a Transimpedance Amplifier (TIA) is used to convert it into a voltage signal suitable for further processing.

The TIA consists of:

- **Operational Amplifier (LM358N):** A dual operational amplifier used to amplify the photodiode output and perform the current-to-voltage conversion. It operates from a single power supply and provides a simple, low-cost solution for this application.
- **Feedback Resistor (R_f):** Determines the amplifier gain. A resistance has value $220\text{ k}\Omega$.
- **Feedback Capacitor (C_f) (Optional):** Connected in parallel with the feedback resistor to improve stability and reduce high-frequency noise.



Figure 4.18: Transimpedance Amplifier

3. Comparator

The LM393 comparator converts the amplified analog signal into a digital HIGH/LOW output, enabling reliable detection of the transmitted OOK data.

Key Features:

- Dual voltage comparator.
- Fast switching response.
- Open-collector output.
- Single-supply operation.

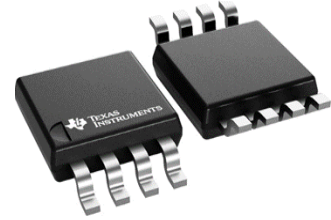


Figure 4.19: comparator

4. . Microcontroller

The ESP32 microcontroller is used to process the digital signal received from the comparator. It detects the transmitted OOK data, reconstructs the received bit stream, and performs the required data processing. The ESP32 was selected for its high processing capability, low power consumption, and ease of programming.

Key Features

- High-performance dual-core microcontroller.
- High processing speed for real-time data reception.
- Multiple GPIO interfaces for easy hardware integration.
- Integrated Wi-Fi and Bluetooth.

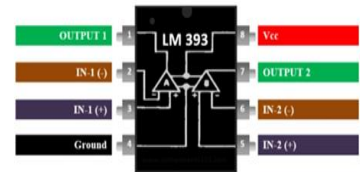


Figure 4.20: ESP32 microcontroller

4.4.2 Simulation Results

After completing the receiver design, Proteus simulation was performed to verify the functionality of each receiver stage. The waveform at the output of the transimpedance amplifier confirms successful current-to-voltage conversion, while the comparator output demonstrates proper regeneration of the digital OOK signal.

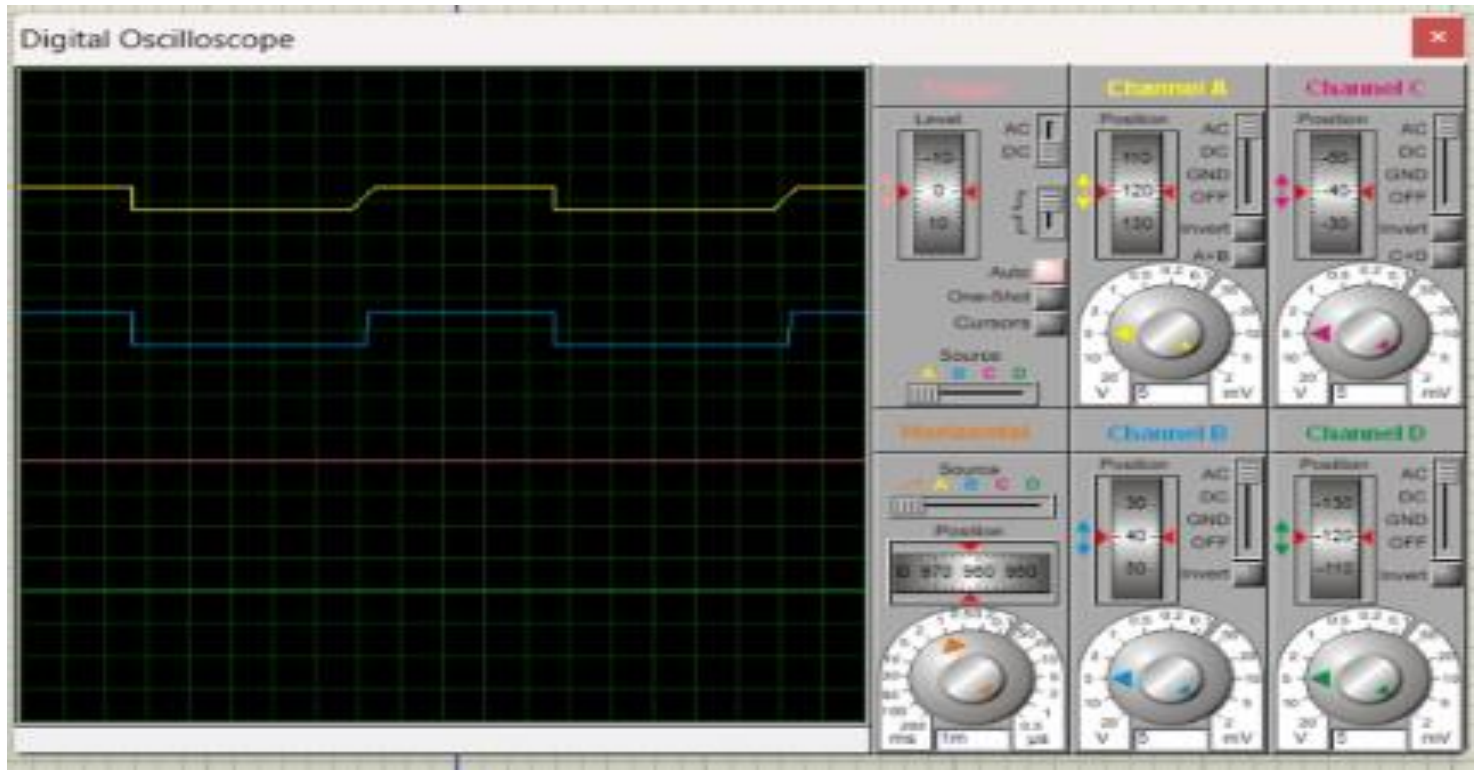


Figure 4.21: the simulation results of optical receiver in proteus

The obtained simulation results indicate that the receiver successfully converts the input pulse signal into a stable digital waveform suitable for processing by the ESP32 microcontroller.

CHAPTER 5: *Hardware Implementation & Results*

5.1 Introduction

In practical optical communication systems, hardware validation typically proceeds from the transmitter to the receiver. However, due to the lack of specialized optical testing devices (such as optical power meters and high-speed optical detectors) within the college laboratory environment, a reverse testing strategy was strategically adopted.

To evaluate the performance of our customized transmitter, we first needed a fully verified, calibrated, and reliable baseline receiver. Therefore, the hardware implementation began with the development and characterization of the receiver module. Once the receiver's operational limits, frequency response, and transimpedance gains were empirically established, it was utilized as the benchmark diagnostic tool to measure and characterize the transmitter's optical performance and channel response.

5.2 Receiver Circuit Development and Characterization

The receiver architecture consists of a photodiode stage, a Transimpedance Amplifier (TIA) to convert the photodiode's weak current into a measurable voltage signal, and a voltage comparator stage to restore the analog signal into a clean digital pulse.

5.2.1 Component Assembly and Breadboard Testing

The receiver circuit was initially prototyped on a standard digital lab breadboard to analyze its core sub-systems.

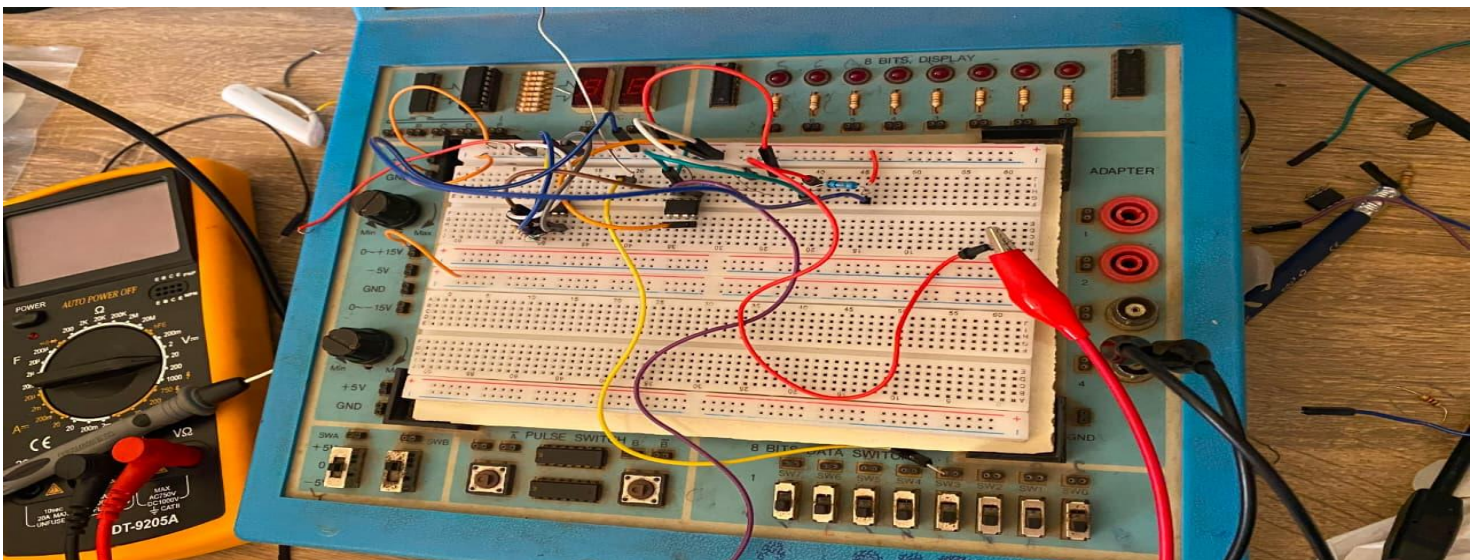


Figure 5.1: Receiver Prototyping and Testing Circuit on a Breadboard

5.2.2 TIA and Comparator Frequency Response Analysis

The mathematical models and theoretical calculations predicted a -3dB cutoff frequency of approximately **60 kHz** for the receiver stage. However, real-world components and parasitic breadboard capacitances led to different empirical results.

During laboratory testing, the frequency of the input signal was gradually swept while observing the output on a digital storage oscilloscope. The circuit exhibited a clean, un-distorted output up to **20 kHz**. Beyond 20 kHz, severe signal attenuation and wave distortion began to manifest. Consequently, the practical -3dB cutoff frequency of the receiver was determined to be **20 kHz**, which served as our verified operational boundary for subsequent tests.

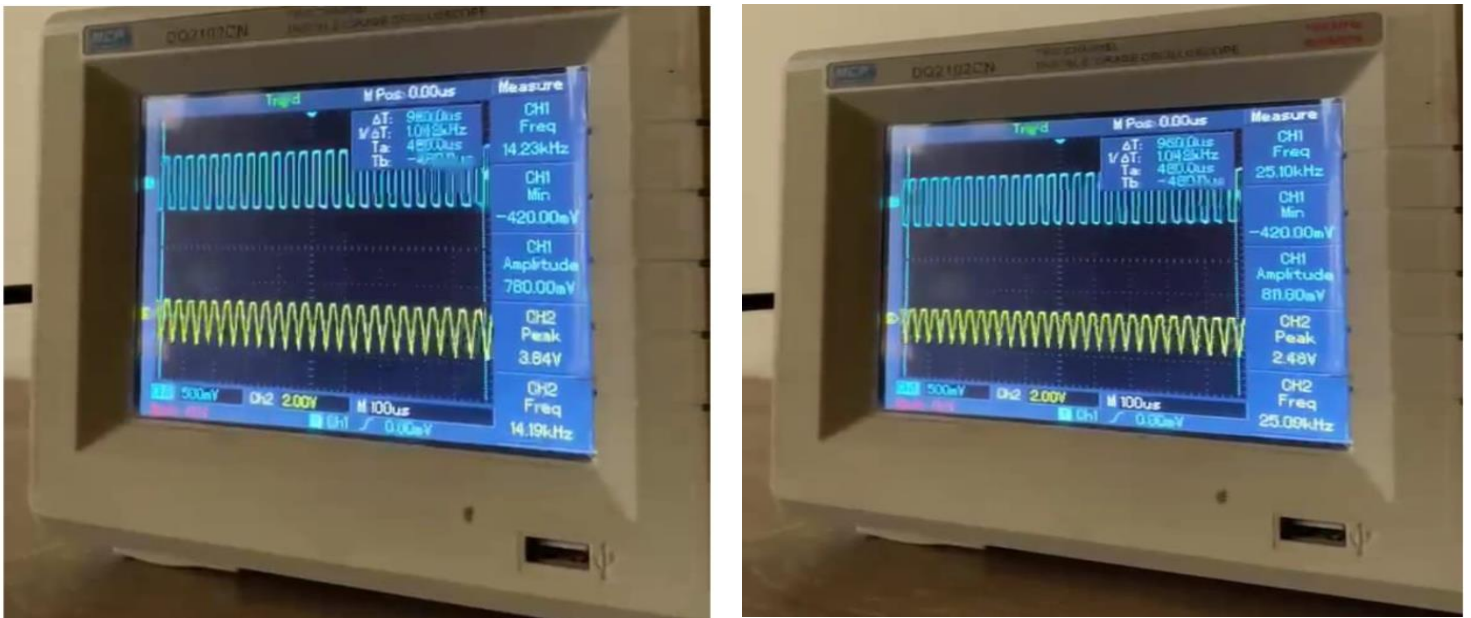


Figure 5.2: Oscilloscope Captured Response Proving Reliable Performance up to 20 kHz

5.2.3 Hardware Optimization and Design Modifications

- **Unavailability of Pico-Farad Feedback Capacitors**

The initial simulation design incorporated a feedback capacitor (C_{FB}) in the pico-Farad (pF) range to stabilize the TIA and mitigate high-frequency noise. However, neither the university laboratory nor the local electronic component stores supplied capacitors in the pF scale; the absolute minimum available value was in the micro-Farad (μF) range.

- **Design Modification:**

An attempt was made to deploy a μF -scale capacitor in the feedback loop. This substitution introduced catastrophic damping and heavy distortion, severely suffocating the signal bandwidth. To resolve this, the feedback capacitor was **completely removed** from the physical hardware setup. Testing confirmed that eliminating the capacitor restored the expected voltage response without causing unstable oscillations, marking our first successful divergence from the simulation design.

5.2.4 Photodiode DC Current Response Test

Before proceeding to high-frequency tests, the fundamental light-to-current conversion of the photodiode needed verification. The photodiode was integrated into the TIA circuit, and a smartphone flashlight was used as a stable DC light source.

When the light beam was perfectly aligned with the photodiode at a distance of 10 cm to 15 cm, the TIA successfully generated a sharp voltage swing, stabilizing at an output of approximately **3.6V**. This confirmed that the photodiode bias and the transimpedance stage were operating with excellent sensitivity under DC conditions.



Figure 5.3: Photodiode Voltage Shift Verification Under a Mobile Flashlight DC Source

5.3 PCB Fabrication and Spatial Alignment Considerations

- **Breadboard Looseness and Optical Alignment Constraints**

While testing the discrete components, a massive practical issue arose during the integration of the transmitter LEDs with the receiver photodiode. Because both sections were constructed on separate breadboards with loose jumper wires, maintaining a rigid, straight-line optical path (Line-of-Sight Alignment) proved nearly impossible. Any slight vibration shifted the components, breaking the optical link and preventing consistent data collection.

- **Solution:** To eliminate structural instability and enforce rigid positioning, the circuit schematics were transferred to permanent Printed Circuit Boards (PCBs). Both the transmitter and receiver modules were successfully etched, drilled, and soldered onto independent PCBs, providing the structural integrity needed for stable alignment.

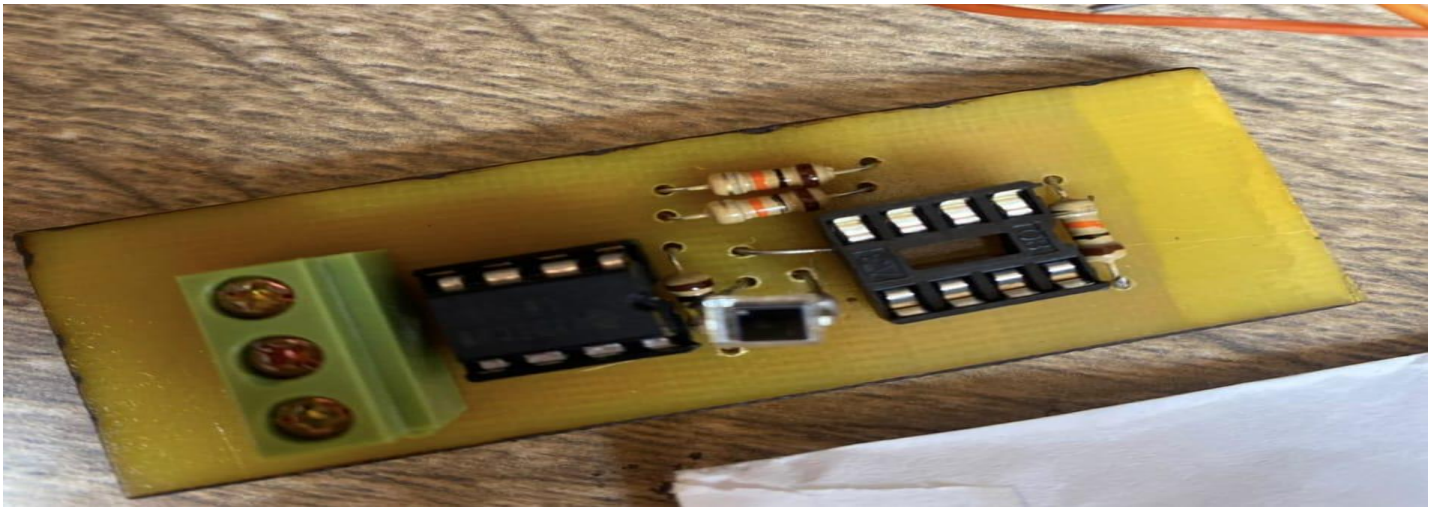


Figure 5.4: Fabricated and Soldered Receiver PCB Modul



Figure 5.5: Fabricated Transmitter PCB Featuring the Central LED Array

5.4 Transmitter Integration and System Optimization

With a fully verified receiver PCB and a structurally stable transmitter PCB, complete end-to-end testing was conducted.

5.4.1 Initial Transmitter Evaluation and Frequency Constraints

Initial trials utilized a static DC drive current to test the collective brightness of the LED array. The initial optical link yielded an extremely weak response at the receiver TIA, measuring a mere 1.0V at a highly restricted physical separation of less than 5 cm.

To understand the system limitations, the distance issue was temporarily set aside to map out the frequency behavior under an AC signal. An alternating signal was fed into the transmitter while measuring the TIA output. The frequency response characteristics revealed the following:

- **From 0 Hz to 1 kHz:** *The output signal was exceptionally clean and highly stable.*
- **From 1 kHz to 2 kHz:** *Minor signal degradation and distortion emerged, but the signal remained readable and acceptable.*
- **Above 2 kHz:** *The output voltage decayed drastically, rendering the signal completely unusable.*

Throughout this sweep, the maximum peak-to-peak output voltage hovering around 1.0V was insufficient for reliable medium-range deployment.

5.4.2 Design Modification: Feedback Resistor Optimization

To amplify the received signal voltage without rebuilding the physical optical channel, we turned to the fundamental Transimpedance relation:

$$V_{\text{out}} = I_{\text{photodiode}} \times R_{\text{FB}}$$

(5.1)

- **Design Modification:**

*The original simulation layout utilized a 220 kΩ feedback resistor, which had been scaled down to 100 kΩ during the very first breadboard assembly. To force a higher voltage gain out of the low photodiode current, the 100 kΩ resistor was replaced with a much larger 560 kΩ **feedback resistor**.*

5.4.3 Final System Integration Results

Upgrading the feedback resistor yielded an immediate and substantial performance boost. Upon re-testing the unified system, the peak output voltage at the receiver TIA increased from 1.0V to 3.2V.

More importantly, this elevated voltage allowed the link to overcome background thermal noise, expanding the operational physical distance between the transmitter and receiver to a stable **15 cm**. The complete system achieved highly reliable, un-distorted data transmission up to a maximum frequency of **1.5 kHz**.



Figure 5.6: Oscilloscope Waveforms of the Transmitted vs. Successfully Received AC Signal at Peak Operating Frequency

5.5 Engineering Discussion

The hardware implementation concluded with a fully functional point-to-point Underwater Wireless Optical Communication link operating reliably at **1.5 kHz** over a 15 cm atmospheric-to-laboratory channel.

An essential engineering takeaway from this practical deployment concerns the physical limitations of commercial Light Emitting Diodes. The LEDs sourced from local commercial markets are fundamentally optimized for general illumination and visual signaling, not high-speed data communications. Consequently, their internal carrier lifetimes induce heavy capacitive effects at higher frequencies, explaining why the response rapidly degraded beyond 2 kHz.

To scale this design up to megabit or gigabit data rates, specialized communication-grade solid-state sources (such as pure intrinsic Germanium-doped or high-speed gallium nitride quantum-well LEDs) would be imperative. Nevertheless, within the boundaries of available laboratory hardware, the successful integration of our customized TIA, optimized feedback loops, and printed PCBs successfully validated the core mathematical and structural concepts of our UWOC system.

5.6 Experimental Underwater Channel Setup

To rigorously evaluate the point-to-point Underwater Wireless Optical Communication (UWOC) system under realistic channel conditions, a dedicated laboratory aquatic simulator was constructed.

- **Physical Channel Simulator:** The underwater medium was simulated using a custom-engineered glass tank measuring $60 \times 30 \times 30$ cm.
- **Ingress Protection and Isolation:** Operating delicate electronic hardware in proximity to an aquatic environment introduced a critical risk of water damage and short-circuiting. To achieve absolute electrical isolation, two custom watertight acrylic enclosures measuring $20 \times 15 \times 15$ cm with a wall thickness of 5 mm were fabricated. These enclosures housed and sealed the transmitter (TX) and receiver (RX) circuits, allowing optical signals to propagate through the transparent acrylic walls while keeping the PCBs entirely dry.
- **Structural Integration:** To transition the prototype from a loose laboratory assembly to a professional, robust system, custom cases and mounting brackets were modeled and 3D printed. These 3D-printed holders secured the TX circuit rigidly inside its enclosure, ensuring structural stability, professional aesthetics, and precise alignment.
- To overcome this spatial attenuation, a custom optical collimator was designed. Constructed with two coaxial biconvex lenses housed in a narrow, 3D-printed barrel, this collimator focused the wide-angle LED emission into a tight, parallel beam. This dual-lens setup drastically minimized geometric spreading losses, projecting the maximum possible optical power density directly onto the receiver's photodiode.



Figure 5.7: 3D-printed housing components and biconvex lenses+Glass Box +Isolated acrylic.

5.6.1 System Performance & Transmission Results

With the integration of the collimating lens assembly, the system's performance was evaluated across both atmospheric and aquatic mediums at a stable data rate of 2.5 kbps.

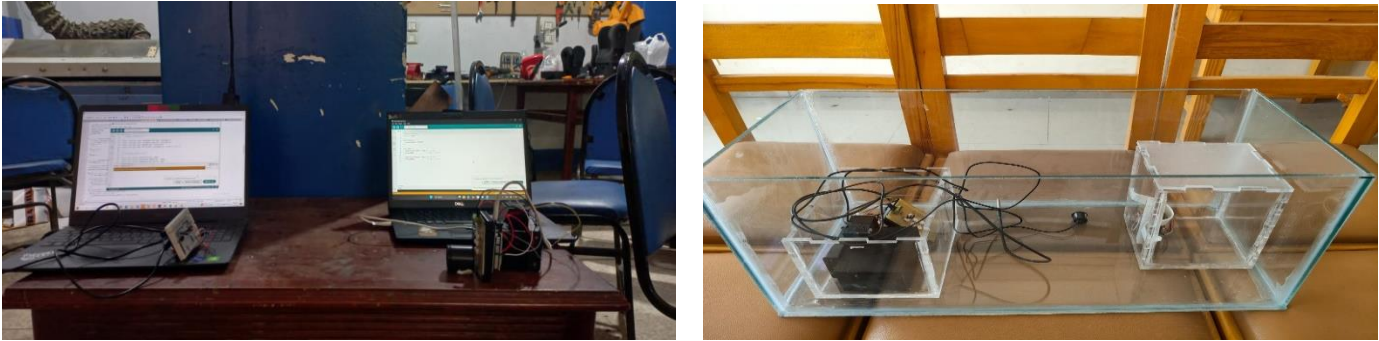


Figure 5.8: Complete experimental UWOC testbed aligned for free-space atmospheric (air) transmission testing and under water.

Table 5.1: Performance comparison of the UWOC system under varying channel mediums and optical configurations.

Medium	Optical Configuration	Maximum Stable Distance
Atmospheric (Air) Un-collimated (Bare LED) 10 cm	Un-collimated (Bare LED)	10cm
Atmospheric (Air) Collimated (Dual-Convex Lens) 50	Collimated (Dual-Convex Lens)	50 cm
Clear water	Collimated	20 cm

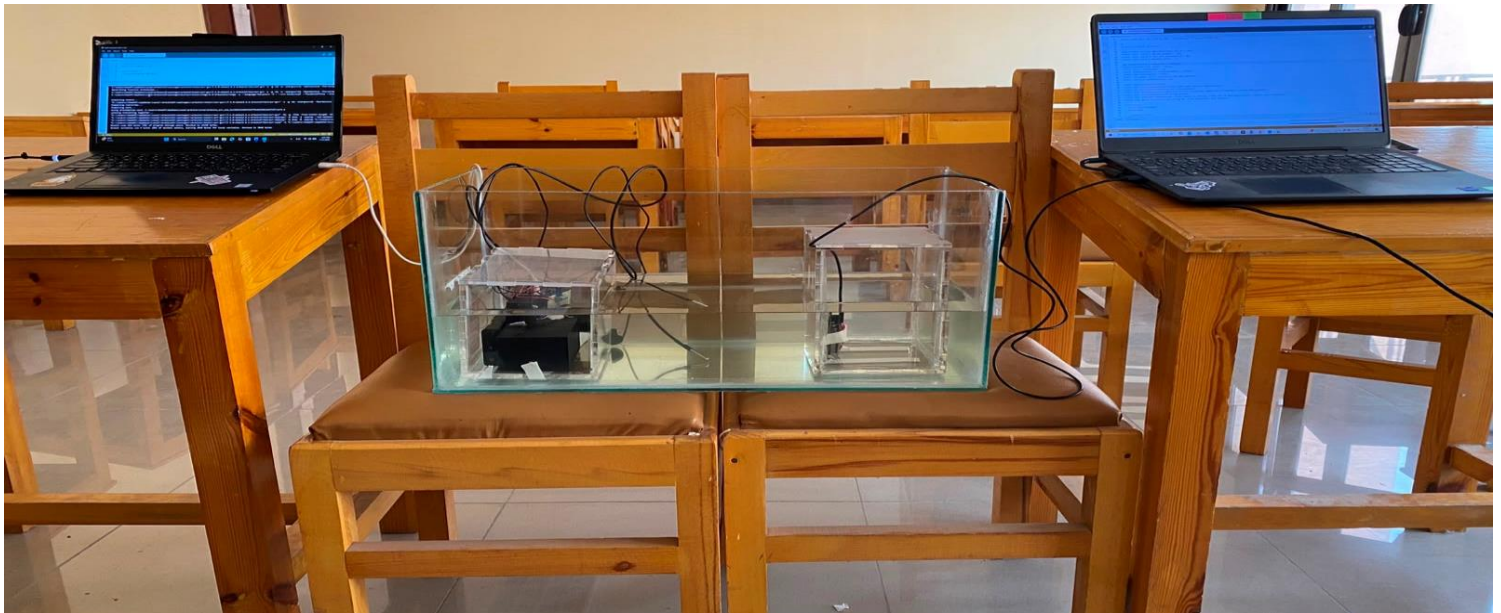


Figure 5.9: Experimental hardware setup for the underwater wireless optical communication (UWOC) link at a 20 cm transmission distance.



Figure 5.10: Graduation project team presenting the final physical setup during the underwater experimental validation.

These experimental outcomes validate the practical design modifications and structural enhancements implemented throughout the development phase. The integration of the custom waterproofing enclosures, 3D-printed structural mounts, and the dual-lens collimator successfully transformed the loose prototyping circuit into a stable, aligned, and resilient system capable of operating in aquatic environments. Although optical attenuation—primarily driven by absorption and scattering—reduced the effective communication range in water relative to free space, the system successfully established a robust, error-free link over a 20 cm clear-water channel at a stable rate of 2.5 kbps. This successful end-to-end hardware validation under realistic environmental constraints confirms the physical viability of our customized transimpedance receiver and collimated transmitter, proving that the system is fully capable of maintaining a reliable short-range underwater optical link.

CHAPTER 6: *Conclusion & Future work*

6.1 *Conclusion*

6.1.1 *The Philosophy of "Design-First" Engineering*

The "Blue-Green OptiCom" project represents a fundamental shift in how we approach Underwater Optical Wireless Communication (UWOC) systems. Rather than relying on high-end, proprietary laboratory equipment or costly FPGA-based DSPs, our primary objective was to demonstrate that sophisticated communication telemetry can be achieved through disciplined, bottom-up analog design. This project was built on the philosophy of "engineering ownership," where every millivolt, every component, and every signal path was manually optimized to overcome the inherent complexities of the underwater environment. By bypassing expensive black-box modules, we were forced to master the underlying physics of our signal chain, which resulted in a more profound understanding of system behavior and greater flexibility in troubleshooting and refinement.

6.1.2 *Simulation as an Analytical Foundation*

Our development cycle was rooted in a high-fidelity simulation workflow that served as the blueprint for our physical realization. Utilizing Optisystem and Proteus, we did not merely visualize the circuit; we performed a deep-dive parametric analysis of the system's performance limits. For instance, by modeling the Voltage-Controlled Current Source (VCCS) stage under a 21V bias, we could pinpoint the exact saturation points of our LED array, preventing the signal clipping that frequently degrades performance in cheaper implementations. This simulation phase was the difference between a prototype that works intermittently and one that is analytically predictable; it allowed us to conduct a comprehensive E/O/E (Electrical-to-Optical-to-Electrical) frequency sweep, which served as our stress test for the system's bandwidth limits before a single piece of hardware was soldered.

6.1.3 *Deep Dive into Hardware Implementation and Optimization*

The core of our achievement lies in the successful transition from simulation to a physical, working prototype. This stage required addressing three major engineering challenges that set our work apart from standard academic exercises:

- ***Custom Driver Circuitry and Signal Integrity:*** *Standard commercial LEDs were never intended for high-speed communication. To mitigate their inherent switching inertia, we engineered a second-order pre-emphasis driver circuit using high-speed MOSFETs. This custom driver allowed us to aggressively manipulate the rise and fall times of the optical pulses, effectively overcoming the sluggish response typical of low-cost LEDs and pushing the system bandwidth to 130 MHz.*

- **Transimpedance Amplifier (TIA) Customization:** Unlike generic pre-built modules, we manually designed and tuned the TIA feedback loops to perfectly match the spectral responsivity of the BPW34 photodiode. This bespoke analog front-end was crucial for maintaining the Signal-to-Noise Ratio (SNR) against the high-frequency ambient noise prevalent in aquatic environments.
- **Hardware-Centric Signal Recovery:** To ensure reliable data transmission, we implemented a real-time hardware-based pulse reconstruction using an LM393 high-speed comparator. This deliberate choice to perform data reconstruction in the hardware layer, rather than relying on software-based algorithms, eliminated computational latency and provided a sharp, deterministic output signal that is far more stable than typical microcontroller-based approaches.

Finally, It has been demonstrated that simple, accessible components—when integrated with comprehensive engineering foresight and custom PCB design—can outperform 'quick-assembly' alternatives in transparency and effectiveness. Every aspect of the system was executed manually: from driver circuits engineered to break frequency constraints, to TIA feedback loops calibrated to the specific aquatic noise of the test environment. This direct implementation enabled a deep engagement with every millivolt in the system, granting a unique capability for troubleshooting and precise fine-tuning that is unattainable for users of off-the-shelf, proprietary systems. Ultimately, the success of this project lies not only in data accuracy but in the realization that the work was not limited to mere 'assembly,' but rather represented a true 'systems engineering' approach where every detail was designed, implemented, and tested personally. This ensures that the project stands as a complete and authentic engineering experience in every sense of the word.

6.2 Future Work

To evolve the current prototype into a resilient, deep-sea industrial solution, future development will focus on the following strategic technical enhancements:

- **High-Power Optoelectronics:** Upgrading to GaN laser diodes and high-sensitivity detectors (APDs/SiPMs) to significantly extend transmission range and data rates (Kaushal & Kaddoum, 2016).
- **AI-Driven Performance Optimization:** Deploying AI models to perform real-time analysis of link quality, allowing for autonomous, dynamic adjustment of hardware parameters to ensure peak transmission efficiency.
- **High-Sensitivity Photodetectors:** Upgrading from standard PIN photodiodes to Avalanche Photodiodes (APDs) or Silicon Photomultipliers (SiPMs) to enhance signal recovery (Arnon, 2010).

- **Advanced Signal Processing:** Implementing OFDM modulation and real-time Forward Error Correction (FEC) to mitigate inter-symbol interference and transmission errors in turbid environments (Oubei et al., 2015).
- **Autonomous Alignment (PAT):** Integrating AI-enhanced computer vision algorithms with gimbal-stabilized transceivers to maintain precise link alignment under dynamic ocean conditions.
- **Robust Marine Packaging:** Utilizing IP68-rated anodized aluminum housings and passive thermal management via copper heat pipes to ensure long-term reliability in extreme deep-sea pressures.
- **Hybrid Communication Architecture:** Developing a failsafe hybrid opto-acoustic network, incorporating AI-based node localization to maintain telemetry continuity during periods of extreme optical attenuation (Islam et al., 2021; Saeed et al., 2018).

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Appendices

Appendix A : system source code and protocol implementation

Listing A.1: Underwater-System

(1)sensor_node_rmt_tx.ino

```
#include <Arduino.h>
#include <string.h>
#include <OneWire.h>
#include <DallasTemperature.h>
#include "driver/rmt_tx.h"
#include "driver/rmt_encoder.h"
#include "esp_err.h"
#include "esp_heap_caps.h"
#include "freertos/FreeRTOS.h"
#include "freertos/task.h"
#include "freertos/queue.h"

// -----
// Hardware configuration
// -----
static const gpio_num_t ULTRASONIC_TRIG_PIN = GPIO_NUM_12;
static const gpio_num_t ULTRASONIC_ECHO_PIN = GPIO_NUM_13;
static const gpio_num_t ONEWIRE_PIN = GPIO_NUM_21;
static const gpio_num_t RMT_TX_PIN = GPIO_NUM_17; // Replaces UART TX

// -----
// Sensor/RMT timing constants
// -----
static const uint32_t LOOP_PERIOD_MS = 2000;
static const uint32_t ULTRASONIC_TIMEOUT_US = 30000;
static const uint32_t MANCHESTER_HALF_BIT_US = 10;
static const uint32_t END_GAP_US = 2400; // End-of-frame idle marker
static const uint8_t PREAMBLE_BYTES = 2;

// -----
// FreeRTOS: queues & task params
// -----
static const UBaseType_t MEASUREMENT_QUEUE_LENGTH = 8;
static const UBaseType_t PAYLOAD_QUEUE_LENGTH = 8;
static const size_t PAYLOAD_BUF_SIZE = 128;

static QueueHandle_t xMeasurementQueue = nullptr;
static QueueHandle_t xPayloadQueue = nullptr;

static const uint8_t SENSOR_TASK_PRIO = 1;
static const uint8_t FORMATTER_TASK_PRIO = 2;
static const uint8_t RMT_TASK_PRIO = 2;

static const uint32_t SENSOR_TASK_STACK = 4096;
static const uint32_t FORMATTER_TASK_STACK = 3072;
static const uint32_t RMT_TASK_STACK = 6144;

struct Measurement {
    float temperatureC;
    float distanceCm;
```

```

};

struct FormattedPayload {
    char text[PAYLOAD_BUF_SIZE];
};

Onewire onewire(ONEWIRE_PIN);
DallasTemperature tempSensor(&onewire);
rmt_channel_handle_t txChannel = nullptr;
rmt_encoder_handle_t copyEncoder = nullptr;

static void initSensors();
static void initRmtTransmitter();
static float readTemperatureC();
static float readDistanceCm();
static Measurement readMeasurements();
static String formatPayload(const Measurement &m);
static void formatPayloadToBuffer(const Measurement &m, FormattedPayload *out);
static size_t manchesterEncodeByte(uint8_t value, rmt_symbol_word_t *symbols, size_t index);
static size_t buildManchesterFrame(const char *payload, size_t payloadLen, rmt_symbol_word_t
*symbols,
                                size_t maxSymbols);
static bool sendManchesterFrame(const char *payload, size_t payloadLen);

static void sensorReaderTask(void *pvParameters);
static void dataFormatterTask(void *pvParameters);
static void rmtTransmitterTask(void *pvParameters);

void setup() {
    Serial.begin(115200);
    delay(300);

    initSensors();
    initRmtTransmitter();

    xMeasurementQueue = xQueueCreate(MEASUREMENT_QUEUE_LENGTH, sizeof(Measurement));
    xPayloadQueue = xQueueCreate(PAYLOAD_QUEUE_LENGTH, sizeof(FormattedPayload));
    if (!xMeasurementQueue || !xPayloadQueue) {
        Serial.println("Fatal: queue creation failed");
        while (true) {
            delay(1000);
        }
    }

    BaseType_t ok = xTaskCreatePinnedToCore(sensorReaderTask, "sensorReader",
    SENSOR_TASK_STACK, nullptr,
                                SENSOR_TASK_PRIO, nullptr, 0);

    if (ok != pdPASS) {
        Serial.println("Fatal: sensorReaderTask create failed");
        while (true) {
            delay(1000);
        }
    }

    ok = xTaskCreatePinnedToCore(dataFormatterTask, "dataFormatter", FORMATTER_TASK_STACK,
    nullptr,
                                FORMATTER_TASK_PRIO, nullptr, 0);

    if (ok != pdPASS) {
        Serial.println("Fatal: dataFormatterTask create failed");
        while (true) {
            delay(1000);
        }
    }
}

```

```

    }
}

ok = xTaskCreatePinnedToCore(rmtTransmitterTask, "rmtTx", RMT_TASK_STACK, nullptr,
RMT_TASK_PRIO, nullptr, 1);
if (ok != pdPASS) {
    Serial.println("Fatal: rmtTransmitterTask create failed");
    while (true) {
        delay(1000);
    }
}

Serial.println("Sensor Node (RMT TX + Manchester, dual-core FreeRTOS) Ready");
}

void loop() {
    vTaskDelay(pdMS_TO_TICKS(1000));
}

static void sensorReaderTask(void *pvParameters) {
    (void)pvParameters;
    for (;;) {
        Measurement m = readMeasurements();
        if (xQueueSend(xMeasurementQueue, &m, portMAX_DELAY) != pdTRUE) {
            // Should not happen with portMAX_DELAY; kept for completeness
        }
        vTaskDelay(pdMS_TO_TICKS(LOOP_PERIOD_MS));
    }
}

static void dataFormatterTask(void *pvParameters) {
    (void)pvParameters;
    Measurement m{};
    for (;;) {
        if (xQueueReceive(xMeasurementQueue, &m, portMAX_DELAY) != pdTRUE) {
            continue;
        }

        FormattedPayload fp{};
        formatPayloadToBuffer(m, &fp);

        Serial.print("TX Payload: ");
        Serial.print(fp.text);
        if (isnan(m.temperatureC)) {
            Serial.println("TX warning: DS18B20 read failed (nan)");
        }
        if (m.distanceCm < 0.0f) {
            Serial.println("TX warning: Ultrasonic timeout (-1)");
        }

        if (xQueueSend(xPayloadQueue, &fp, portMAX_DELAY) != pdTRUE) {
            // Should not happen with portMAX_DELAY
        }
    }
}

static void rmtTransmitterTask(void *pvParameters) {
    (void)pvParameters;
    FormattedPayload fp{};
    for (;;) {
        if (xQueueReceive(xPayloadQueue, &fp, portMAX_DELAY) != pdTRUE) {

```

```

        continue;
    }
    const size_t len = strlen(fp.text, sizeof(fp.text));
    if (!sendManchesterFrame(fp.text, len)) {
        Serial.println("RMT TX failed");
    }
}

static void initSensors() {
    tempSensor.begin();
    Serial.print("DS18B20 devices found: ");
    Serial.println(tempSensor.getDeviceCount());

    pinMode(ULTRASONIC_TRIG_PIN, OUTPUT);
    pinMode(ULTRASONIC_ECHO_PIN, INPUT);
    digitalWrite(ULTRASONIC_TRIG_PIN, LOW);
}

static void initRmtTransmitter() {
    rmt_tx_channel_config_t txCfg = {};
    txCfg.gpio_num = RMT_TX_PIN;
    txCfg.clk_src = RMT_CLK_SRC_DEFAULT;
    txCfg.resolution_hz = 1000000; // 1 tick = 1 us
    txCfg.mem_block_symbols = 256;
    txCfg.trans_queue_depth = 4;

    ESP_ERROR_CHECK(rmt_new_tx_channel(&txCfg, &txChannel));
    rmt_copy_encoder_config_t copyCfg = {};
    ESP_ERROR_CHECK(rmt_new_copy_encoder(&copyCfg, &copyEncoder));
    ESP_ERROR_CHECK(rmt_enable(txChannel));
}

static float readTemperatureC() {
    for (int attempt = 0; attempt < 3; ++attempt) {
        tempSensor.requestTemperatures();
        float t = tempSensor.getTempCByIndex(0);
        if (t != DEVICE_DISCONNECTED_C && t >= -55.0f && t <= 125.0f) {
            return t;
        }
        delay(50);
    }
    return NAN;
}

static float readDistanceCm() {
    digitalWrite(ULTRASONIC_TRIG_PIN, LOW);
    delayMicroseconds(2);
    digitalWrite(ULTRASONIC_TRIG_PIN, HIGH);
    delayMicroseconds(10); // Required 10 us trigger pulse
    digitalWrite(ULTRASONIC_TRIG_PIN, LOW);

    unsigned long duration = pulseIn(ULTRASONIC_ECHO_PIN, HIGH, ULTRASONIC_TIMEOUT_US);
    if (duration == 0) {
        return -1.0f; // Timeout case required by spec
    }

    // Required formula: distance = (duration * 0.1500) / 2
    return (duration * 0.1500f) / 2.0f;
}

```

```

static Measurement readMeasurements() {
    Measurement m;
    m.temperatureC = readTemperatureC();
    m.distanceCm = readDistanceCm();
    return m;
}

static String formatPayload(const Measurement &m) {
    String t = isnan(m.temperatureC) ? String("nan") : String(m.temperatureC, 2);
    String d = String(m.distanceCm, 2);

    String payload = t;
    payload += ",";
    payload += d;
    payload += "\n";
    return payload;
}

static void formatPayloadToBuffer(const Measurement &m, FormattedPayload *out) {
    String payload = formatPayload(m);
    const size_t n = payload.length();
    if (n >= PAYLOAD_BUF_SIZE) {
        out->text[0] = '\0';
        return;
    }
    memcpy(out->text, payload.c_str(), n);
    out->text[n] = '\0';
}

// Manchester mapping per bit:
// bit 1 -> HIGH then LOW
// bit 0 -> LOW then HIGH
static size_t manchesterEncodeByte(uint8_t value, rmt_symbol_word_t *symbols, size_t index) {
    for (int bit = 7; bit >= 0; --bit) {
        bool one = (value >> bit) & 0x01;

        rmt_symbol_word_t sym = {};
        sym.duration0 = MANCHESTER_HALF_BIT_US;
        sym.duration1 = MANCHESTER_HALF_BIT_US;
        sym.level0 = one ? 1 : 0;
        sym.level1 = one ? 0 : 1;

        symbols[index++] = sym;
    }
    return index;
}

static size_t buildManchesterFrame(const char *payload, size_t payloadLen, rmt_symbol_word_t
*symbols,
                                size_t maxSymbols) {
    size_t index = 0;

    // Longer preamble increases chance of synchronization when RX starts mid-frame.
    for (uint8_t i = 0; i < PREAMBLE_BYTES; ++i) {
        if (index + 8 >= maxSymbols) {
            return 0;
        }
        index = manchesterEncodeByte(0x55, symbols, index);
    }

    for (size_t i = 0; i < payloadLen; ++i) {

```



```

        if (index + 8 >= maxSymbols) {
            return 0;
        }
        index = manchesterEncodeByte(static_cast<uint8_t>(payload[i]), symbols, index);
    }

    // End gap helps RX frame separation
    if (index < maxSymbols) {
        rmt_symbol_word_t gap = {};
        gap.duration0 = END_GAP_US;
        gap.level0 = 0;
        gap.duration1 = 0;
        gap.level1 = 0;
        symbols[index++] = gap;
    }

    return index;
}

static bool sendManchesterFrame(const char *payload, size_t payloadLen) {
    const size_t maxSymbols = PREAMBLE_BYTES * 8 + payloadLen * 8 + 1;
    rmt_symbol_word_t *symbols = static_cast<rmt_symbol_word_t *>(
        heap_caps_malloc(maxSymbols * sizeof(rmt_symbol_word_t), MALLOC_CAP_INTERNAL));
    if (!symbols) {
        return false;
    }

    size_t used = buildManchesterFrame(payload, payloadLen, symbols, maxSymbols);
    if (used == 0) {
        heap_caps_free(symbols);
        return false;
    }

    rmt_transmit_config_t txConfig = {};
    txConfig.loop_count = 0;

    esp_err_t err = rmt_transmit(txChannel, copyEncoder, symbols, used *
sizeof(rmt_symbol_word_t), &txConfig);
    if (err == ESP_OK) {
        err = rmt_tx_wait_all_done(txChannel, pdMS_TO_TICKS(100));
    }

    heap_caps_free(symbols);
    return err == ESP_OK;
}

```

(2)receiver_node_rmt_rx.ino

```

#include <Arduino.h>
#include <vector>
#include "driver/rmt_rx.h"
#include "esp_err.h"
#include "freertos/FreeRTOS.h"
#include "freertos/semphr.h"

// -----
// Hardware configuration
// -----
static const gpio_num_t RMT_RX_PIN = GPIO_NUM_16; // Replaces UART RX

// -----

```

```

// Manchester/RMT constants
// -----
static const uint32_t MANCHESTER_HALF_BIT_US = 10;
static const uint16_t RX_MEM_SYMBOLS = 512;
static const uint16_t HALF_BIT_TOLERANCE_US = 80;
static const uint8_t PREAMBLE_BYTES = 2;

rmt_channel_handle_t rxChannel = nullptr;
static volatile bool rxDone = false;
static size_t rxNumSymbols = 0;
static SemaphoreHandle_t rxDoneSemaphore = nullptr;
static bool rxInProgress = false;
static rmt_symbol_word_t rxRawBuffer[RX_MEM_SYMBOLS];
static uint32_t lastRxStatusMs = 0;

static void initRmtReceiver();
static bool receiveSymbols(std::vector<rmt_symbol_word_t> &outSymbols);
static int durationToHalfBits(uint16_t durationUs);
static bool decodeManchesterSymbols(const std::vector<rmt_symbol_word_t> &symbols, String
&decodedText);
static bool parseAndPrint(const String &line);
static bool IRAM_ATTR onRxDone(rmt_channel_handle_t channel, const rmt_rx_done_event_data_t
*edata, void *userData);
static bool parseFloatField(const String &s, float &outValue, bool &isNan);

void setup() {
    Serial.begin(115200);
    delay(300);

    initRmtReceiver();
    Serial.println("Receiver Node (RMT RX + Manchester) Ready");
}

void loop() {
    std::vector<rmt_symbol_word_t> symbols;
    if (!receiveSymbols(symbols)) {
        delay(5);
        return;
    }

    String rxText;
    if (!decodeManchesterSymbols(symbols, rxText)) {
        Serial.println("RX decode error");
        return;
    }

    // Payload can contain one or more '\n' terminated lines.
    int start = 0;
    while (true) {
        int end = rxText.indexOf('\n', start);
        if (end < 0) break;

        String line = rxText.substring(start, end);
        line.trim();
        if (line.length() > 0) {
            parseAndPrint(line);
        }
        start = end + 1;
    }
}

```

```

static void initRmtReceiver() {
    pinMode(static_cast<int>(RMT_RX_PIN), INPUT_PULLDOWN);

    rmt_rx_channel_config_t rxCfg = {};
    rxCfg.gpio_num = RMT_RX_PIN;
    rxCfg.clk_src = RMT_CLK_SRC_DEFAULT;
    rxCfg.resolution_hz = 1000000; // 1 tick = 1 us
    rxCfg.mem_block_symbols = RX_MEM_SYMBOLS;

    ESP_ERROR_CHECK(rmt_new_rx_channel(&rxCfg, &rxChannel));

    rxDoneSemaphore = xSemaphoreCreateBinary();
    rmt_rx_event_callbacks_t cbs = {};
    cbs.on_recv_done = onRxDone;
    ESP_ERROR_CHECK(rmt_rx_register_event_callbacks(rxChannel, &cbs, nullptr));

    ESP_ERROR_CHECK(rmt_enable(rxChannel));
}

static bool receiveSymbols(std::vector<rmt_symbol_word_t> &outSymbols) {
    if (!rxInProgress) {
        rxDone = false;
        rxNumSymbols = 0;

        rmt_receive_config_t cfg = {};
        cfg.signal_range_min_ns = 1000; // Ignore pulses shorter than 1 us
        cfg.signal_range_max_ns = 60000000; // Must stay below driver max (~65.535 ms)

        esp_err_t err = rmt_receive(rxChannel, rxRawBuffer, sizeof(rxRawBuffer), &cfg);
        if (err != ESP_OK) {
            return false;
        }

        rxInProgress = true;
        return false;
    }

    if (xSemaphoreTake(rxDoneSemaphore, pdMS_TO_TICKS(5)) != pdTRUE) {
        if (millis() - lastRxStatusMs > 1000) {
            Serial.println("RX waiting for signal...");
            lastRxStatusMs = millis();
        }
        return false;
    }

    rxInProgress = false;
    if (!rxDone || rxNumSymbols == 0 || rxNumSymbols > RX_MEM_SYMBOLS) {
        rxDone = false;
        rxNumSymbols = 0;
        return false;
    }

    outSymbols.assign(rxRawBuffer, rxRawBuffer + rxNumSymbols);
    Serial.print("RX frame captured, symbols: ");
    Serial.println(static_cast<unsigned int>(rxNumSymbols));
    rxDone = false;
    rxNumSymbols = 0;
    return true;
}

static int durationToHalfBits(uint16_t durationUs) {

```

```

// Convert raw pulse duration to half-bit count with rounding.
int n = (durationUs + (MANCHESTER_HALF_BIT_US / 2)) / MANCHESTER_HALF_BIT_US;
if (n < 1) {
    n = 1;
}
if (n > 32) {
    n = 32;
}
return n;
}

static bool decodeManchesterSymbols(const std::vector<rmt_symbol_word_t> &symbols, String
&decodedText) {
    decodedText = "";

    // Rebuild an idealized half-bit level stream from captured pulse durations.
    std::vector<uint8_t> halfLevels;
    halfLevels.reserve(symbols.size() * 2);

    for (size_t i = 0; i < symbols.size(); ++i) {
        const rmt_symbol_word_t &s = symbols[i];

        if (s.duration0 > 0) {
            int n0 = durationToHalfBits(s.duration0);
            for (int k = 0; k < n0; ++k) {
                halfLevels.push_back(static_cast<uint8_t>(s.level0));
            }
        }
        if (s.duration1 > 0) {
            int n1 = durationToHalfBits(s.duration1);
            for (int k = 0; k < n1; ++k) {
                halfLevels.push_back(static_cast<uint8_t>(s.level1));
            }
        }
    }

    if (halfLevels.size() < 16) {
        return false;
    }

    auto decodeFromPhase = [&](size_t phase, String &outText) -> bool {
        std::vector<uint8_t> bitStream;
        bitStream.reserve(halfLevels.size() / 2);

        // Decode Manchester from pairs of half-bits.
        // 10 => bit 1
        // 01 => bit 0
        for (size_t i = phase; i + 1 < halfLevels.size(); i += 2) {
            uint8_t a = halfLevels[i];
            uint8_t b = halfLevels[i + 1];

            if (a == 1 && b == 0) {
                bitStream.push_back(1);
            } else if (a == 0 && b == 1) {
                bitStream.push_back(0);
            } else {
                // Skip invalid pair to allow loose re-synchronization.
                continue;
            }
        }
    }
}

```

```

auto bitsToTextFrom = [&](size_t startBit, String &txt) {
    txt = "";
    size_t p = startBit;
    while (p + 8 <= bitStream.size()) {
        uint8_t value = 0;
        for (int i = 0; i < 8; ++i) {
            value = (value << 1) | bitStream[p + i];
        }
        txt += static_cast<char>(value);
        p += 8;
    }
};

auto extractCsvLine = [&](const String &txt, String &lineOut) -> bool {
    int nl = txt.indexOf('\n');
    while (nl >= 0) {
        int start = nl;
        while (start > 0 && txt[start - 1] != '\n') {
            --start;
        }
        String line = txt.substring(start, nl);
        int comma = line.indexOf(',');
        if (comma > 0) {
            lineOut = line + "\n";
            return true;
        }
        nl = txt.indexOf('\n', nl + 1);
    }
    return false;
};

// Strategy 1: strict preamble lock then byte decode.
size_t bitPos = 0;
while (bitPos + PREAMBLE_BYTES * 8 <= bitStream.size()) {
    bool syncOk = true;
    for (uint8_t p = 0; p < PREAMBLE_BYTES; ++p) {
        uint8_t b = 0;
        for (int i = 0; i < 8; ++i) {
            b = (b << 1) | bitStream[bitPos + p * 8 + i];
        }
        if (b != 0x55) {
            syncOk = false;
            break;
        }
    }
    if (syncOk) {
        String txt;
        bitsToTextFrom(bitPos + PREAMBLE_BYTES * 8, txt);
        if (extractCsvLine(txt, outText)) {
            return true;
        }
    }
    ++bitPos;
}

// Strategy 2: no preamble lock, test all byte alignments.
for (size_t offset = 0; offset < 8; ++offset) {
    String txt;
    bitsToTextFrom(offset, txt);
    if (extractCsvLine(txt, outText)) {
        return true;
    }
}

```

```

    }
  }
  return false;
};

String candidate;
if (decodeFromPhase(0, candidate) || decodeFromPhase(1, candidate)) {
  decodedText = candidate;
  return true;
}
return false;
}

static bool parseAndPrint(const String &line) {
  String clean = line;
  clean.trim();
  // Remove leading preamble artifacts (e.g., 'U' from 0x55) and other junk
  // until a likely float start is found.
  int start = 0;
  while (start < clean.length()) {
    char c = clean[start];
    bool floatStart = (c >= '0' && c <= '9') || c == '-' || c == '+' || c == '.' || c == 'n'
  || c == 'N';
    if (floatStart) break;
    ++start;
  }
  if (start > 0) {
    clean = clean.substring(start);
  }

  int comma = clean.indexOf(',');
  if (comma < 0) {
    return false;
  }

  String tempStr = clean.substring(0, comma);
  String distStr = clean.substring(comma + 1);
  tempStr.trim();
  distStr.trim();

  float temp = 0.0f;
  float dist = 0.0f;
  bool tempIsNan = false;
  bool distIsNan = false;
  if (!parseFloatField(tempStr, temp, tempIsNan) || !parseFloatField(distStr, dist,
distIsNan)) {
    Serial.print("RX invalid line: ");
    Serial.println(line);
    return false;
  }

  Serial.print("RX raw line: ");
  Serial.println(clean);
  Serial.println("===== DATA RECEIVED =====");
  Serial.print("Water Temperature: ");
  if (tempIsNan) {
    Serial.print("NaN");
  } else {
    Serial.print(temp, 2);
  }
  Serial.println(" C");
}

```

```

Serial.print("Water Distance  : ");
if (distIsNan) {
    Serial.print("NaN");
} else {
    Serial.print(dist, 2);
}
Serial.println(" cm");
Serial.println("=====");
return true;
}

static bool IRAM_ATTR onRxDone(rmt_channel_handle_t channel, const rmt_rx_done_event_data_t
*edata, void *userData) {
    (void)channel;
    (void)userData;

    BaseType_t highTaskWoken = pdFALSE;
    rxNumSymbols = edata->num_symbols;
    rxDone = true;
    xSemaphoreGiveFromISR(rxDoneSemaphore, &highTaskWoken);
    return highTaskWoken == pdTRUE;
}

static bool parseFloatField(const String &s, float &outValue, bool &isNan) {
    String t = s;
    t.trim();
    t.toLowerCase();
    if (t == "nan") {
        isNan = true;
        outValue = NAN;
        return true;
    }

    const char *c = t.c_str();
    char *endPtr = nullptr;
    float v = strtoc(c, &endPtr);
    if (endPtr == c || *endPtr != '\0') {
        return false;
    }

    isNan = false;
    outValue = v;
    return true;
}

```

Appendix B : Components Datasheets

B.1 Transmitting Stage Components

- **ESP32 Microcontroller:**

The ESP32 is a low-cost, low-power system-on-a-chip (SoC) microcontroller with integrated Wi-Fi and dual-mode Bluetooth. In this Underwater Optical Wireless Communication (UWOC) project, the ESP32 serves as the main processing unit (the brain) responsible for acquiring data from the sensors,

performing encoding operations, and controlling the transmitting driver circuit via the ESP-NOW protocol

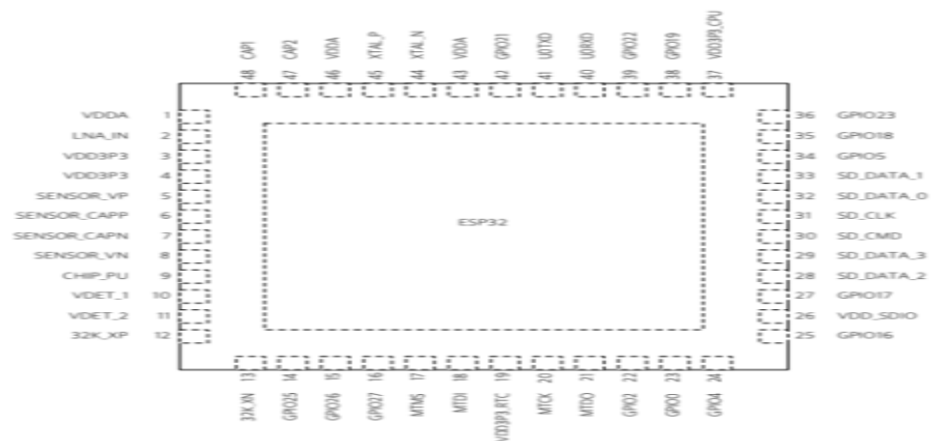


Figure B.1: ESP32 Pin Layout

Table B.1: Pin Description

Name	No.	Type	Function
Analog			
VDDA	1	P	Analog power supply (2.3V ~ 3.6V)
LNA_IN	2	I/O	RF input and output
VDD3P3	3	P	Amplifier power supply (2.3V ~ 3.6V)
VDD3P3	4	P	Amplifier power supply (2.3V ~ 3.6V)
VDD3P3_RTC			
SENSOR_VP	5	I	GPIO36, ADC_PRE_AMP, ADC1_CH0, RTC_GPIO0 Note: Connects 270 pF capacitor from SENSOR_VP to SENSOR_CAPP when used as ADC_PRE_AMP.

Table B.2: Power Consumption by Power Modes

Power mode	Description	Power consumption
Active (RF working)	Wi-Fi Tx packet 13 dBm ~ 21 dBm	160 ~ 260 mA
	Wi-Fi / BT Tx packet 0 dBm	120 mA
	Wi-Fi / BT Rx and listening	80 ~ 90 mA
	Association sleep pattern (by Light-sleep)	0.9 mA@DTIM3, 1.2 mA@DTIM1
Modem-sleep	The CPU is powered on.	Max speed: 20 mA
		Normal speed: 5 ~ 10 mA
		Slow speed: 3 mA
Light-sleep	-	0.8 mA
Deep-sleep	The ULP co-processor is powered on.	0.15 mA
	ULP sensor-monitored pattern	25 μ A @1% duty
	RTC timer + RTC memory	10 μ A
Hibernation	RTC timer only	2.5 μ A

- **2SC5200 Power Transistor:**

The 2SC5200 is a high-power NPN epitaxial silicon transistor designed for high-power audio amplification and general-purpose switching applications. In this project, it is utilized within the transmitting stage as a laser diode/LED driver circuit. Since the ESP32 microcontroller cannot source the high current required to drive the optical transmitter efficiently, the 2SC5200 acts as a current amplifier, switching the optical source based on the encoded Manchester data stream to ensure reliable signal propagation through the underwater channel.

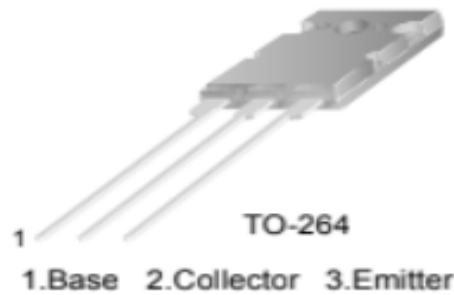


Figure B.2: 2SC5200 Transistor Pin Configuration and Package Outline.

Table B.3: 2SC5200 Absolute Maximum Ratings

Absolute Maximum Ratings* $T_a = 25^\circ\text{C}$ unless otherwise noted

Symbol	Parameter	Ratings	Units
BV_{CBO}	Collector-Base Voltage	250	V
BV_{CEO}	Collector-Emitter Voltage	250	V
BV_{EBO}	Emitter-Base Voltage	5	V
I_C	Collector Current(DC)	17	A
I_B	Base Current	1.5	A
P_D	Total Device Dissipation($T_C=25^\circ\text{C}$) Derate above 25°C	150 1.04	W W/ $^\circ\text{C}$
T_J, T_{STG}	Junction and Storage Temperature	- 50 ~ +150	$^\circ\text{C}$

B.2 Receiving Stage Components

- **BPW34 Silicon PIN Photodiode**

The BPW34 is a high-speed and high-sensitive silicon PIN photodiode enclosed in a miniature flat plastic package. Due to its short response time and low junction capacitance, it is perfectly suited for high-bandwidth optical wireless communication frameworks. In this project, the BPW34 acts as the optical receiver (the detector) deployed at the destination of the underwater channel. It captures the incoming attenuated light pulses (photons) emitted by the transmitter and converts them directly into a proportional, yet extremely weak, analog electric current (photocurrent).

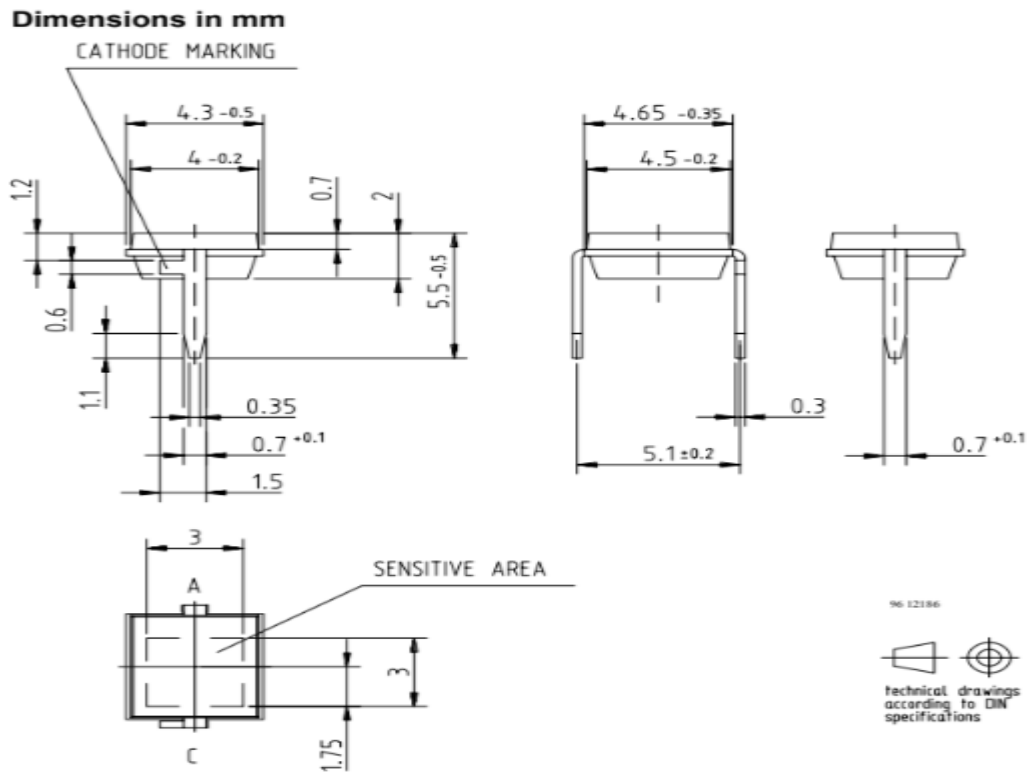


Figure B.3: BPW34 Photodiode Package Dimensions and Pin Configuration.

Table B.4: BPW34 Photodiode Maximum Ratings.

Absolute Maximum Ratings

$T_{amb} = 25^{\circ}\text{C}$

Parameter	Test Conditions	Symbol	Value	Unit
Reverse Voltage		V_R	60	V
Power Dissipation	$T_{amb} \leq 25^{\circ}\text{C}$	P_V	215	mW
Junction Temperature		T_j	100	$^{\circ}\text{C}$
Storage Temperature Range		T_{sig}	-55...+100	$^{\circ}\text{C}$
Soldering Temperature	$t \leq 3\text{ s}$	T_{sd}	260	$^{\circ}\text{C}$
Thermal Resistance Junction/Ambient		R_{thJA}	350	K/W

Table B.5: BPW34 Photodiode Electrical and Optical Characteristics.

Basic Characteristics

$T_{amb} = 25^{\circ}\text{C}$

Parameter	Test Conditions	Symbol	Min	Typ	Max	Unit
Breakdown Voltage	$I_R = 100\ \mu\text{A}, E = 0$	$V_{(BR)}$	60			V
Reverse Dark Current	$V_R = 10\ \text{V}, E = 0$	I_{r0}		2	30	nA
Diode Capacitance	$V_R = 0\ \text{V}, f = 1\ \text{MHz}, E = 0$	C_D		70		pF
	$V_R = 3\ \text{V}, f = 1\ \text{MHz}, E = 0$	C_D		25	40	pF
Open Circuit Voltage	$E_e = 1\ \text{mW}/\text{cm}^2, \lambda = 950\ \text{nm}$	V_o		350		mV
Temp. Coefficient of V_o	$E_e = 1\ \text{mW}/\text{cm}^2, \lambda = 950\ \text{nm}$	TK_{V_o}		-2.6		mV/K
Short Circuit Current	$E_A = 1\ \text{klx}$	I_k		70		μA
	$E_e = 1\ \text{mW}/\text{cm}^2, \lambda = 950\ \text{nm}$	I_k		47		μA
Temp. Coefficient of I_k	$E_e = 1\ \text{mW}/\text{cm}^2, \lambda = 950\ \text{nm}$	TK_{I_k}		0.1		%/K
Reverse Light Current	$E_A = 1\ \text{klx}, V_R = 5\ \text{V}$	I_{ra}		75		μA
	$E_e = 1\ \text{mW}/\text{cm}^2, \lambda = 950\ \text{nm}, V_R = 5\ \text{V}$	I_{ra}	40	50		μA
Angle of Half Sensitivity		ϕ		± 65		deg
Wavelength of Peak Sensitivity		λ_p		900		nm
Range of Spectral Bandwidth		$\lambda_{0.5}$		600...1050		nm
Noise Equivalent Power	$V_R = 10\ \text{V}, \lambda = 950\ \text{nm}$	NEP		4×10^{-14}		W/ $\sqrt{\text{Hz}}$
Rise Time	$V_R = 10\ \text{V}, R_L = 1\ \text{k}\ \Omega, \lambda = 820\ \text{nm}$	t_r		100		ns
Fall Time	$V_R = 10\ \text{V}, R_L = 1\ \text{k}\ \Omega, \lambda = 820\ \text{nm}$	t_f		100		ns

- **OPA381 Transimpedance Amplifier**

The OPA381 is a precision, low-power, high-speed transimpedance amplifier (TIA) specifically designed for photodiode applications. In the receiving architecture of this project, the OPA381 is a critical component used to handle the weak current output from the BPW34 photodiode. It converts the low-level photocurrent (in the microampere range) into a stable, usable voltage signal while providing immense amplification and preserving high signal-to-noise ratio (SNR). This amplified voltage signal is then ready for further filtration and threshold processing.

Table B.6: OPA381 Absolute Maximum Ratings

ABSOLUTE MAXIMUM RATINGS⁽¹⁾

Voltage Supply	+7V
Signal Input Terminals ⁽²⁾ , Voltage	(V-) -0.5V to (V+) + 0.5V
Current	±10mA
Short-Circuit Current ⁽³⁾	Continuous
Operating Temperature Range	-40°C to +125°C
Storage Temperature Range	-65°C to +150°C
Junction Temperature	+150°C
Lead Temperature (soldering, 10s)	+300°C
OPA381 ESD Rating (Human Body Model)	2000V
OPA2381 ESD Rating (Human Body Model)	1500V

- (1) Stresses above these ratings may cause permanent damage. Exposure to absolute maximum conditions for extended periods may degrade device reliability. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those specified is not implied.
- (2) Input terminals are diode clamped to the power-supply rails. Input signals that can swing more than 0.5V beyond the supply rails should be current limited to 10mA or less.
- (3) Short-circuit to ground; one amplifier per package.

PIN ASSIGNMENTS

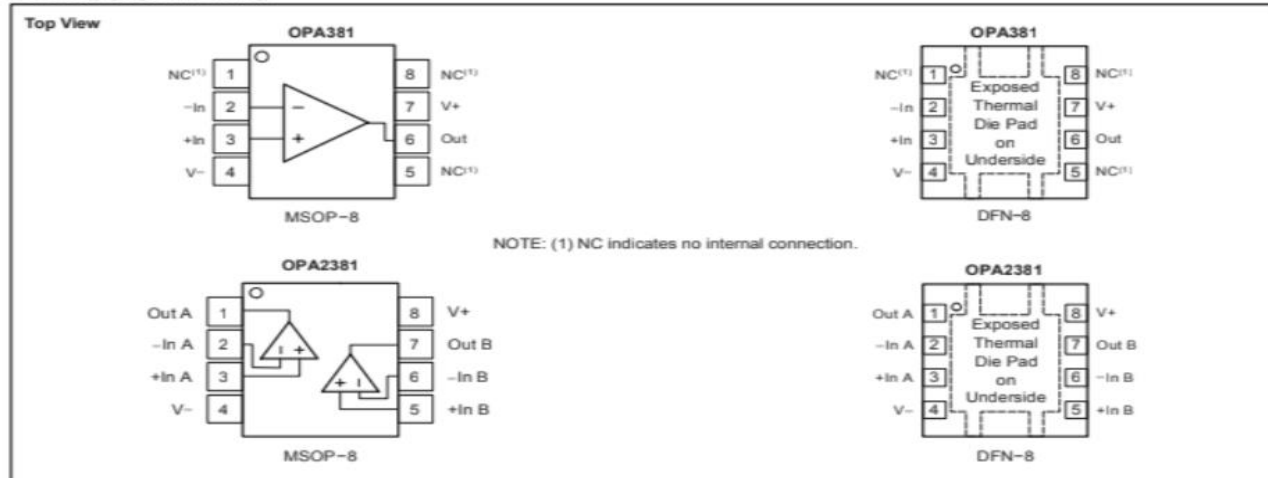


Figure B.4: OPA381 Pin Configuration

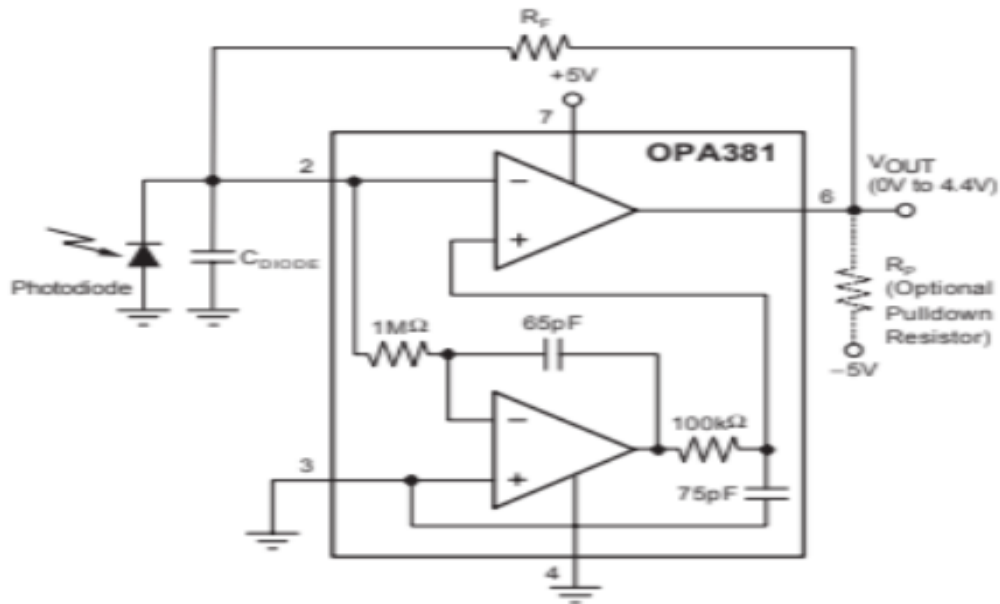


Figure B.5: OPA381 Typical Photodiode Transimpedance Amplifier Circuit.

B.3 Signal Conditioning and Processing

- **LM358N Operational Amplifier**

The LM358N is a next-generation, industry-standard dual operational amplifier (op-amp) housed in an 8-pin plastic dual in-line package (PDIP). In this Underwater Optical Wireless Communication (UWOC) system, the LM358N is deployed in the signal conditioning stage. It functions as a voltage amplifier and an active filter to amplify the analog voltage signal coming from the transimpedance amplifier (TIA) stage. This amplification is crucial to boost the signal's amplitude and filter out environmental optical noise (such as ambient light fluctuations) before the signal undergoes threshold detection.

PIN CONNECTIONS

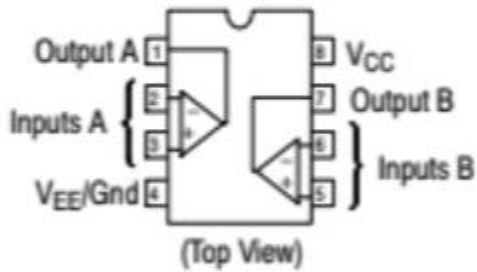


Figure B.6: LM358N Pin Configuration and Internal Schematic.

Table B.7: LM358N Absolute Maximum Ratings

MAXIMUM RATINGS ($T_A = +25^\circ\text{C}$, unless otherwise noted.)

Rating	Symbol	Value	Unit
Power Supply Voltages Single Supply Split Supplies	V_{CC} V_{CC}, V_{EE}	32 ± 16	Vdc
Input Differential Voltage Range (Note 1)	V_{IDR}	± 32	Vdc
Input Common Mode Voltage Range (Note 2)	V_{ICR}	-0.3 to 32	Vdc
Output Short Circuit Duration	t_{SC}	Continuous	
Junction Temperature	T_J	150	$^\circ\text{C}$
Thermal Resistance, Junction-to-Air (Note 3)	$R_{\theta JA}$	Case 846A 238 Case 751 212 Case 626 161	$^\circ\text{C/W}$
Storage Temperature Range	T_{stg}	-65 to +150	$^\circ\text{C}$
ESD Protection at any Pin Human Body Model Machine Model	V_{esd}	2000 200	V
Operating Ambient Temperature Range	T_A	LM258 -25 to +85 LM358, LM358A 0 to +70 LM2904/LM2904A -40 to +105 LM2904V, NCV2904 (Note 4) -40 to +125 NCV2904V (Note 4) -40 to +150	$^\circ\text{C}$

Stresses exceeding Maximum Ratings may damage the device. Maximum Ratings are stress ratings only. Functional operation above the Recommended Operating Conditions is not implied. Extended exposure to stresses above the Recommended Operating Conditions may affect device reliability.

1. Split Power Supplies.
2. For supply voltages less than 32 V the absolute maximum input voltage is equal to the supply voltage.
3. All $R_{\theta JA}$ measurements made on evaluation board with 1 oz. copper traces of minimum pad size. All device outputs were active.
4. NCV2904 and NCV2904V are qualified for automotive use.

Table B.8: LM358N Electrical Characteristics Specifications.

ELECTRICAL CHARACTERISTICS ($V_{CC} = 5.0\text{ V}$, $V_{EE} = \text{GND}$, $T_A = 25^\circ\text{C}$, unless otherwise noted.)

Characteristic	Symbol	LM258			LM358			LM358A			Unit
		Min	Typ	Max	Min	Typ	Max	Min	Typ	Max	
Input Offset Voltage $V_{CC} = 5.0\text{ V}$ to 30 V , $V_{IC} = 0\text{ V}$ to $V_{CC} - 1.7\text{ V}$, $V_O = 1.4\text{ V}$, $R_S = 0\ \Omega$ $T_A = 25^\circ\text{C}$ $T_A = T_{\text{high}}$ (Note 5) $T_A = T_{\text{low}}$ (Note 5)	V_{IO}	–	2.0	5.0	–	2.0	7.0	–	2.0	3.0	mV
Average Temperature Coefficient of Input Offset Voltage $T_A = T_{\text{high}}$ to T_{low} (Note 5)	$\Delta V_{IO}/\Delta T$	–	7.0	–	–	7.0	–	–	7.0	–	$\mu\text{V}/^\circ\text{C}$
Input Offset Current $T_A = T_{\text{high}}$ to T_{low} (Note 5)	I_{IO}	–	3.0	30	–	5.0	50	–	5.0	30	nA
Input Bias Current $T_A = T_{\text{high}}$ to T_{low} (Note 5)	I_B	–	–	100	–	–	150	–	–	75	nA
Average Temperature Coefficient of Input Offset Current $T_A = T_{\text{high}}$ to T_{low} (Note 5)	$\Delta I_{IO}/\Delta T$	–	10	–	–	10	–	–	10	–	$\text{pA}/^\circ\text{C}$
Input Common Mode Voltage Range (Note 6), $V_{CC} = 30\text{ V}$ $V_{CC} = 30\text{ V}$, $T_A = T_{\text{high}}$ to T_{low}	V_{ICR}	0	–	28.3	0	–	28.3	0	–	28.5	V
Differential Input Voltage Range	V_{IDR}	–	–	V_{CC}	–	–	V_{CC}	–	–	V_{CC}	V
Large Signal Open Loop Voltage Gain $R_L = 2.0\text{ k}\Omega$, $V_{CC} = 15\text{ V}$, For Large V_O Swing, $T_A = T_{\text{high}}$ to T_{low} (Note 5)	A_{VOL}	50 25	100 –	– –	25 15	100 –	– –	25 15	100 –	– –	V/mV
Channel Separation 1.0 kHz $\leq f \leq 20\text{ kHz}$, Input Referenced	CS	–	–120	–	–	–120	–	–	–120	–	dB

• LM393N Voltage Comparator

The LM393N is a high-precision, low-power dual voltage comparator integrated circuit enclosed in a standard 8-pin PDIP enclosure. Within the receiver architecture of this project, the LM393N plays a vital role in data recovery. It compares the amplified analog optical signal against a predefined reference voltage (threshold). By performing this real-time comparison, the LM393N converts the continuous, distorted analog waveform back into a clean, sharp digital square wave (binary 0 and 1 pulses). This reconstructed digital bitstream is then routed directly to the ESP32 microcontroller for successful decoding of the transmitted data.

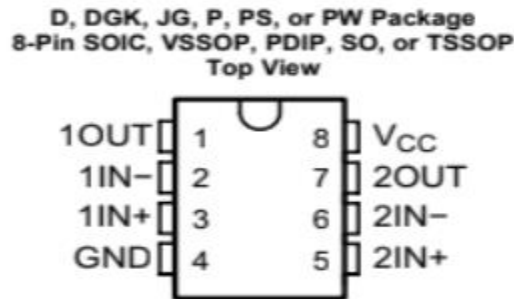


Figure B.7: LM393N Pin Configuration and Internal Schematic.

Table B.9: LM393N Absolute Maximum Rating

over operating free-air temperature range (unless otherwise noted) ⁽¹⁾

		MIN	MAX	UNIT	
V _{CC}	Supply voltage ⁽²⁾	Non-B Versions	–0.3	36	V
		B Versions Only		38	
V _{ID}	Differential input voltage ⁽³⁾	Non-B Versions	–36	36	V
		B Versions Only	–38	38	
V _I	Input voltage (either input)	Non-B Versions	–0.3	36	V
		B Versions Only		38	
I _{IK}	Input current ⁽⁴⁾	–50			mA
V _O	Output voltage	Non-B Versions	–0.3	36	V
		B Versions Only		38	
I _O	Output current	Non-B Versions		20	mA
		B Versions Only		25	
I _{SC}	Duration of output short circuit to ground ⁽⁵⁾	Unlimited			
T _J	Operating virtual-junction temperature	150			°C
T _{stg}	Storage temperature	–65 150			°C

- (1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions* is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability. Production Processing Does Not Necessarily Include Testing of All Parameters.
- (2) All voltage values, except differential voltages, are with respect to network ground.
- (3) Differential voltages are at IN+ with respect to IN–.
- (4) Input current flows thorough parasitic diode to ground and turns on parasitic transistors that increases I_{CC} and may cause output to be incorrect. Normal operation resumes when input current is removed.
- (5) Short circuits from outputs to V_{CC} can cause excessive heating and eventual destruction.

Table B.10: LM393N Electrical Characteristics Specifications.

V_S = 5 V, V_{CM} = (V–) ; T_A = 25°C (unless otherwise noted).

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
V _{IO}	Input offset voltage	V _S = 5 to 36V	–2.5	±0.37	2.5	mV
		V _S = 5 to 36V, T _A = –40°C to +85°C	–4		4	
I _B	Input bias current			–3.5	–25	nA
		T _A = –40°C to +85°C			–50	nA
I _{OS}	Input offset current		–10	±0.5	10	nA
		T _A = –40°C to +85°C	–25		25	nA
V _{CM}	Common mode range	V _S = 3 to 36V	(V–)		(V+) – 1.5	V
V _{CM}	Common mode range	V _S = 3 to 36V, T _A = –40°C to +85°C	(V–)		(V+) – 2.0	V
A _{VD}	Large signal differential voltage amplification	V _S = 15V, V _O = 1.4V to 11.4V; R _L ≥ 15k to (V+)	50	200		V/mV
V _{OL}	Low level output Voltage {swing from (V–)}	I _{SINK} ≤ 4mA, V _{ID} = –1V		110	400	mV
		I _{SINK} ≤ 4mA, V _{ID} = –1V T _A = –40°C to +85°C			550	mV
I _{OH,LKG}	High-level output leakage current	(V+) = V _O = 5 V; V _{ID} = 1V		0.1	20	nA
		(V+) = V _O = 36V; V _{ID} = 1V		0.3	50	nA
I _{OL}	Low level output current	V _{OL} = 1.5V; V _{ID} = –1V; V _S = 5V	6	21		mA
I _O	Quiescent current (all comparators)	V _S = 5 V, no load		400	600	μA
		V _S = 36 V, no load, T _A = –40°C to +85°C		550	800	μA

B.4 Telemetry and Sensory System

- **JSN-SR04 Ultrasonic Distance Sensor**

The JSN-SR04 is an industrial-grade, waterproof ultrasonic distance measurement sensor integrated with a split enclosed probe. It operates by emitting high-frequency sound waves and measuring the time taken for the echo to return after bouncing off an object. In this Underwater Optical Wireless Communication (UWOC) project, the JSN-SR04 is utilized within the sensory subsystem to monitor underwater distance parameters, detect obstacles, or measure depth levels. The sensory data is then processed by the microcontroller to assess the operational environment of the underwater link.

Table B.11: JSN-SR04 Ultrasonic Sensor Electrical Parameters.

	Pulse width output / Serial Output
Operating Voltage	DC 3.0-5.5V
Working current	Less than 8mA
Probe frequency	40KHz
Farthest range	600cm
Recent range	20cm
Distance accuracy	+/- 1cm
Resolution	1mm
Measuring angle	75 degree
Enter the trigger signal	1,10uS above the TTL pulse 2, the serial port to send instructions 0X55
Output the echo signal	Output pulse width level signal, or TTL
Wiring	3-5.5V (power positive) Trig (RX) RX Echo (output) TX GND (power supply negative)
Product Size	L42 * W29 * H12 mm
Operating temperature	-20 ° C to + 70 ° C
Product color	PCB board is blue

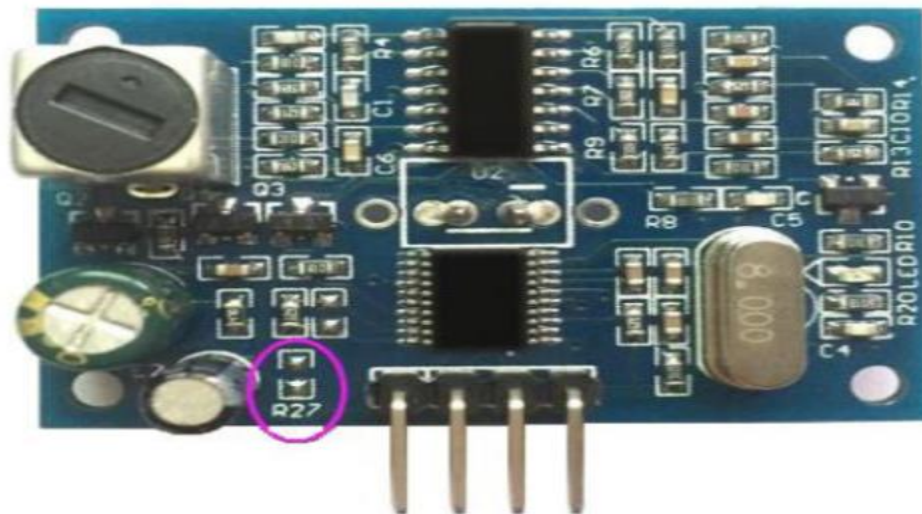


Figure B.8: JSN-SR04 Sensor Dimensions and Pin Configuration.

- **DS18B20 Temperature Sensor**

The DS18B20 is a 1-Wire digital thermometer that provides high-precision temperature measurements encapsulated in a waterproof stainless steel probe. It communicates over a single-wire bus, requiring only one digital pin of the ESP32 microcontroller for data transfer. Within the scope of this project, the DS18B20 is submerged to continuously monitor the water temperature. Tracking temperature fluctuations is essential since thermal gradients directly affect the water density, refractive index, and overall optical attenuation coefficient within the underwater communication channel.

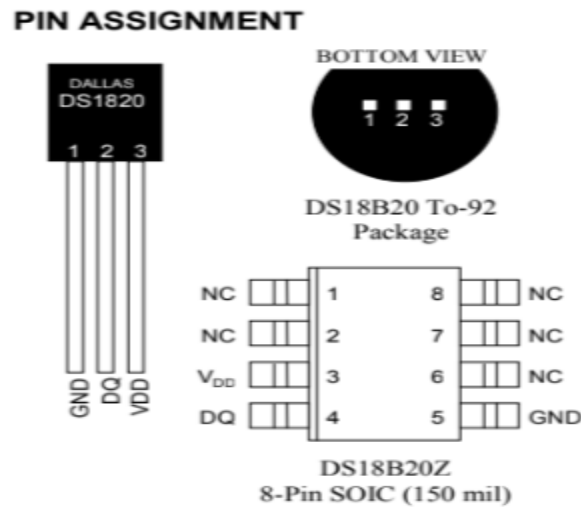


Figure B.9: DS18B20 Pin Configuration and Package Types.

Table B.12: DS18B20 Temperature Sensor Absolute Maximum Ratings

ABSOLUTE MAXIMUM RATINGS*

Voltage on Any Pin Relative to Ground	-0.5V to +6.0V
Operating Temperature	-55°C to +125°C
Storage Temperature	-55°C to +125°C
Soldering Temperature	See J-STD-020A specification

* This is a stress rating only and functional operation of the device at these or any other conditions above those indicated in the operation sections of this specification is not implied. Exposure to absolute maximum rating conditions for extended periods of time may affect reliability.

ملخص المشروع

تُعَد وصلة الضوء الأزرق-الأخضر إحدى التقنيات الرائدة في أنظمة الاتصالات اللاسلكية الضوئية تحت الماء، إذ تتغلب على قيود التوهين الشديدة للموجات الصوتية والترددات الراديوية. وبفضل عملها ضمن نطاق الطول الموجي الأزرق-الأخضر، تُظهر قنوات وأوساط المياه عمومًا فقدًا منخفضًا للإشارة في طيف الضوء المرئي، وتحديدًا في نطاق الطول الموجي 450-570 نانومتر. تُمكن الاتصالات الضوئية تحت الماء من نقل البيانات ذات السعة الكبيرة بسرعة عالية وبشكل آمن عبر مسافات متوسطة وقصيرة، مما يجعلها مجالًا بحثيًا بارزًا في مجال الاتصالات تحت الماء.

يُقدّم هذا المشروع نموذجًا نظريًا وعمليًا وتحليليًا لعدة عوامل تؤثر على حالة وصلة الاتصال الضوئي تحت الماء. ويُحاكي بيئة قناة حقيقية باستخدام طريقة لإرسال واستقبال البيانات بعد مرورها عبر قناة مائية حقيقية، مع حساب قوة الإشارة من البداية إلى النهاية. وأخيرًا، يُحاكي النظام ويتحقق منه باستخدام برامج حاسوبية مثل Optisystem وProteus. ويُعد توليد إشارة اتصال عالية السرعة لجهاز الإرسال تقنية أساسية. في هذا المشروع، تم تنفيذ جهاز إرسال واستقبال يعتمد على مصفوفة من ثنائيات الليدات، ويتكون من عدة ثنائيات ليدات ذات طول موجي مركزي. يستخدم الجهاز نمط تعديل (NRZ-OOK) بمعدل بيانات قياسي في هذا المجال.

شمل ذلك تطوير الكود اللازم لوحدة التحكم الدقيقة وبروتوكول جهاز الإرسال والاستقبال للنموذج التشغيلي المقترح. وفي نهاية المطاف، أُجريت محاكاة وتجربة عملية لوصلة اتصال ضوئي لاسلكي تحت الماء من نقطة إلى نقطة في حوض تجريبي بأبعاد 60 متر طولًا، و30 متر عرضًا، 30 متر عمقًا. وتُظهر النتائج التجريبية أن النظام يعمل بكفاءة

تؤكد النتائج الأولية جدوى هذا النظام لتطبيقات الاتصالات البصرية اللاسلكية عالية السرعة تحت الماء، مثل المركبات البحرية غير المأهولة (UUVs) والمركبات التي يتم تشغيلها عن بُعد (ROVs) والطائرات بدون طيار (UAVs)، وكذلك بين المنصات تحت الماء، أو أثناء استخدام السفن في التطبيقات المدنية والبيئية. كما يؤكد هذا المشروع إمكانية تحسين أداء نظام الاتصالات البصرية اللاسلكية تحت الماء باستخدام مصادر ليزر مختلفة ضمن الطيف المرئي.



جامعة طنطا كلية الهندسة



اتصالات الضوء الأزرق والأخضر : تصميم وتطبيق شبكة اتصالات لاسلكية تحت الماء

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